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Higher Institute Of Technological Studies Rades

End-of-Studies Project Report

Higher Institute Of Technological Studies Rades

by

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MES System

Complete Project Report

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made at

Mavision

Abstract

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List of Abbreviations

Abbreviation	Definition
AL	Application Language (Microsoft Dynamics 365 Business Central)
API	Application Programming Interface
BOM	Bill of Materials
ERP	Enterprise Resource Planning
GUID	Globally Unique Identifier
MES	Manufacturing Execution System
OData	Open Data Protocol
RTL	Right-to-Left
SCRUM	(no acronym — Scrum is a proper noun)
SHA-256	Secure Hash Algorithm 256-bit
UML	Unified Modelling Language

General Introduction

In an industrial context undergoing rapid digital transformation, manufacturing companies face the necessity of optimising their production processes, reducing downtime and ensuring complete traceability of workshop operations. *Manufacturing Execution System* (MES), represent a direct response to these challenges by bridging the gap between decision-making layers *Enterprise Resource Planning* (ERP) and the production floor.

It is within this context that the present end-of-studies project was carried out at **Mavision**. The project consists of the design and development of an integrated MES system, structured around two main components: a business back-end developed in AL language on the Microsoft Dynamics 365 Business Central platform, and a cross-platform mobile/web application developed with the Flutter framework. The system enables machine management, production order tracking, real-time production reporting, as well as user and role-based access management.

This report is structured into several chapters. The first chapter introduces the host organisation and presents the general framework of the project, highlighting the limitations of existing solutions and the relevance of the proposed solution. The second chapter focuses on requirements analysis and specification, as well as the technology choices made. The subsequent chapters trace the various development phases, exploring the conceptual aspects and illustrating the results obtained at the end of each sprint.

Through this approach, we aim to demonstrate how a well-designed digital solution can transform workshop production management by providing visibility, responsiveness and performance, serving both operators and management alike.

Chapter 1

Global Project Framework

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Introduction

This first chapter serves to contextualise our project. It begins with a brief introduction to the host organisation, then describes the objectives to be achieved, provides an in-depth analysis of the existing situation, and finally describes the methodology used to implement this work.

1.1 Presentation of the Host Company

To properly organise this section, we begin with some general details about Mavision before addressing the activities directly relevant to us.

1.1.1 Overview of Mavision

Founded in 2014 and headquartered in Manar II, Tunis, Mavision [1] is a software development company specialising in consulting, auditing and implementing ERP information systems. Mavision supports companies in their IT development projects and helps its clients better organise themselves around their information systems. In addition to its ERP offerings, Mavision develops and integrates reporting solutions to help its clients manage their business activities more effectively. In order to build lasting trust, Mavision places emphasis on its professionalism, commitment and the technical and functional expertise of its team.

Figure 1.1. Mavision Logo [Source: Mavision, https://www.mavision.tn, visited April 2025]

1.1.2 Areas of Expertise

Mavision's activities are structured around three main areas:

ERP Integration: Consulting, implementation and integration of ERP solutions, with a strong focus on Microsoft Dynamics 365 Business Central, as well as project and platform management, architecture and custom business software design.

Business Intelligence: Design and deployment of BI solutions enabling clients to gain strategic visibility over their operations through dashboards, reporting tools and data analysis.

Web & Mobile Technology: Development and integration of mobile phone applications and web platforms that complement ERP systems, enabling clients to manage their business activities with greater agility and efficiency.

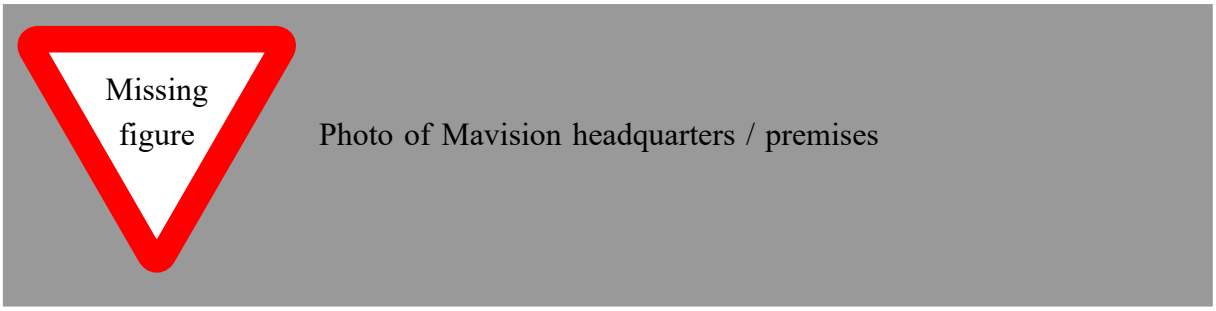

Figure 1.2. Mavision Headquarters

1.2 Project Framework

This project forms part of our end-of-studies internship, an essential step for obtaining the `bachelor's` degree at the `Higher Institute of Technological Studies`. This internship, lasting `four months`, took place from `February 2nd` to `May 30th` at Mavision. During this period, we were assigned to the development team.

This team is responsible for designing and maintaining software solutions that meet the company's business needs, leveraging modern technologies such as Microsoft Dynamics 365 Business Central and Flutter, and agile practices. Within this context, we worked on the development of the MES (*Manufacturing Execution System*) project, a mobile and web solution enabling intelligent management of machines and production operations on the workshop floor.

1.3 State of the Art

1.3.1 Subject Overview

With the rise of Industry 4.0 and the widespread adoption of ERP systems in manufacturing companies, the need to connect decision-making levels with floor operations has become critical. A MES (*Manufacturing Execution System*) is an information system that provides this link in real time: it supervises the execution of production orders, monitors machine status, records production declarations and provides operators and supervisors with a consolidated view of the workshop.

In this context, our project reflects a desire to provide Mavision and its industrial clients with a MES solution natively integrated with Microsoft Dynamics 365 Business Central. The primary objective is to allow operators to start and monitor their production operations from an intuitive mobile/web application, while ensuring data consistency with the ERP.

1.3.2 Problem Statement

The problem of optimising production floor operations arises with particular force in manufacturing companies that rely on an ERP system such as Microsoft Dynamics 365 Business Central. Indeed, while Business Central effectively handles production planning and order management

at the decision-making level, no unified, intelligent and dynamic solution exists to bridge the gap between the ERP and the shop floor in real time.

This gap leads to several consequences: inability to track the progress of production orders as they are being executed, manual entry of production declarations through paper routing sheets or spreadsheet files, lack of visibility for supervisors and managers on machine states and operation statuses, and insufficient traceability of actual shop-floor activity. Thus, a central question arises: how can Mavision provide its industrial clients with a solution that optimises the management of production operations, ensuring effective, accurate and real-time synchronisation between the shop floor and the ERP system?

1.3.3 Study of the Existing Situation

Following an in-depth analysis, it is clear that there is a genuine need within manufacturing companies for a centralised solution to efficiently manage production operations directly from the workshop floor.

Currently, production tracking is often carried out manually, via paper route sheets or spreadsheet files, leading to data entry errors, delays in information reporting and the absence of reliable traceability.

Supervisors and production managers have no real-time visibility into the progress of production orders, nor the current status of machines (running, paused, out of order). There is often no tool to automatically correlate production declarations with orders planned in the ERP.

1.3.4 Critical Analysis of the Existing Situation

In the context of this study, we briefly analysed the MES solution proposed by Delmia Apriso, one of the leading enterprise-grade Manufacturing Execution Systems on the market, whose interface is illustrated in the figure below.

Delmia Apriso is a comprehensive cloud-based MES platform developed by Dassault Systemes, designed for large-scale industrial enterprises. It covers the full spectrum of manufacturing operations including production execution, quality management, labour tracking and supply chain integration. While technically mature and feature-rich, it targets large multinational manufacturers with complex and highly customised deployments, resulting in significant licensing, implementation and maintenance costs that place it out of reach for small and medium-sized manufacturers.

Figure 1.3. Logo of the Delmia Apriso solution [Source: Dassault Systèmes, https://www.3ds.com/products/delmia/apriso, visited April 2025]

From Mavision's perspective as a software provider, the goal is not to use an MES solution internally, but to deliver one to its industrial clients. The solution must therefore be cost-effective to sell, straightforward to deploy and maintain at client sites, and natively integrated into Microsoft Dynamics 365 Business Central, which is already Mavision's core ERP offering. A heavyweight platform such as Delmia Apriso, which requires a separate infrastructure and lengthy implementation cycles, is incompatible with this commercial model. Table 1.1 summarises the key attributes of Delmia Apriso against the specific needs of the project.

Table 1.1. Comparison table with Delmia Apriso

Features	**Project needs**	**Delmia Apriso**
Real-time production order tracking	Yes	Yes
Machine status management	Yes	Yes
Production declaration from mobile	Yes	Partial
Native Business Central integration	Yes	No
Cross-platform mobile interface	Yes	Partial
Role management (Admin / Supervisor / Operator)	Yes	Yes
Secure token-based authentication	Yes	Yes
Adaptability to internal needs	High (custom solution)	Low (proprietary platform)
Ease of deployment at client sites	High	Low
Compatible with SME commercial model	Yes	No

1.3.5 Proposed Solution

The diagnosis of the current situation has highlighted several shortcomings and limitations, as detailed in the previous section. To address these, we propose to design and deploy an innovative MES solution natively integrated with Microsoft Dynamics 365 Business Central.

This solution consists of two complementary parts:

- **An AL back-end (Business Central)**: developed in AL language, it exposes OData V4 endpoints enabling user management, token-based authentication, machine and operation tracking, as well as production declaration.
- **A Flutter application (front-end)**: a cross-platform mobile and web application allowing operators and supervisors to view production orders assigned to their machine, start, pause or complete an operation, declare produced quantities, and visualise progress in real time.

The objective is to optimise every phase of production management by establishing a reliable, traceable and ergonomic digital framework, serving both operators and management.

1.3.6 Objectives

Our MES solution aims to:

- Provide real-time tracking of machine status and production operations directly from the Business Central ERP.
- Allow operators to start, pause and complete operations from an intuitive mobile interface.
- Record production declarations (produced quantities, scrap) and synchronise them with production orders in Business Central.
- Manage users by roles (Administrator, Supervisor, Operator) with secure token-based authentication.
- Offer supervisors a consolidated view of production progress across their work centres.
- Reduce manual data entry and the risk of errors associated with paper-based or non-integrated processes.
- Provide an extensible architecture, ready to accommodate new features (statistics, alerts, QR/label integrations).

1.3.7 Adopted Methodology

The development method corresponds to an ordered and structured sequence of project phases, for efficient and optimal management.

Selection of the Agile Methodology

Agile is an iterative and collaborative working method that takes into account the initial needs of the client and any changes that may arise. Its central ingredient is customer-centred iterative development, which allows a team to gather client feedback on a regular basis and make the necessary adjustments at the right time.

Thus, the client has a better overview of work progress, leading to regular sessions that guarantee the delivery of a functional application at all times throughout the improvement cycles. Consequently, the underlying key idea is to work faster in order to satisfy a client.

Adoption of the Scrum Framework

We adopted the Scrum framework, an agile approach based on short sprints and regular ceremonies. This choice allowed us to deliver functional increments at the end of each sprint, while maintaining smooth communication with the professional and our academic supervisors.

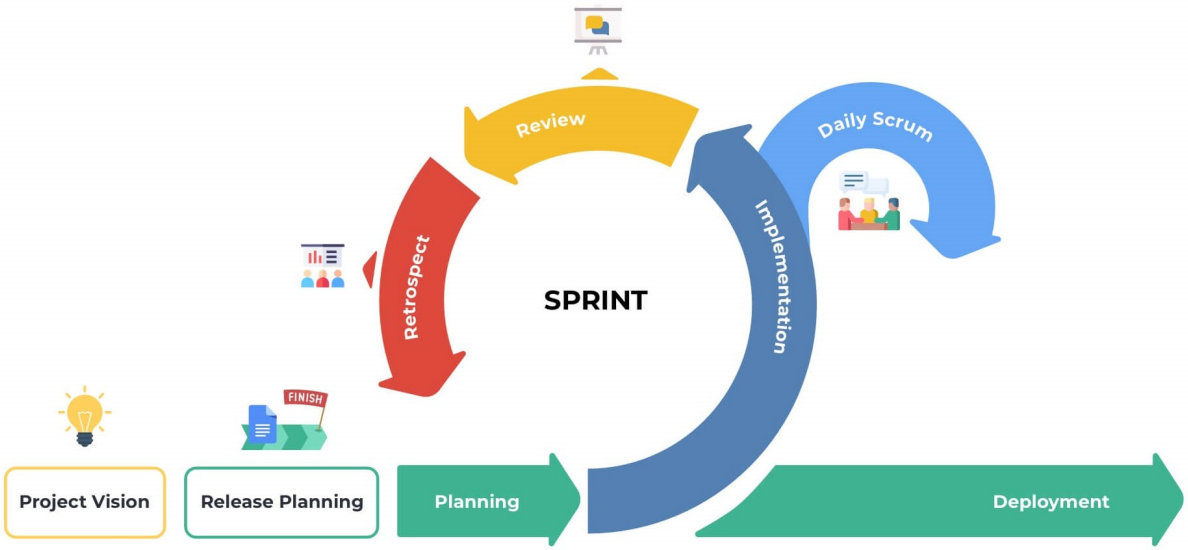

Figure 1.4. Adopted Scrum development cycle

Conclusion

In this chapter, we presented Mavision, the company where the project is being carried out. We then defined the subject context, as well as the objectives of our work and the chosen methodology. In order to ensure the report progresses logically, we will examine the existing solutions used in the company. We will then explain the functional and non-functional requirements of our solution. This will allow us to evaluate existing solutions and identify the weaknesses and strengths for which we should make improvements to our solution. Finally, we will formulate the specific requirements for our solution.

Chapter 2

Requirements Analysis and Specification

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Introduction

Requirements analysis and specification is a crucial phase in the information system development process. It consists, first and foremost, of clarifying the functional and non-functional requirements of the system. This step then allows a complete understanding of the subject in order to position oneself clearly, precisely and without ambiguity.

2.1 Requirements Analysis

This step allows us to understand the overall context of our project, that is to say to identify the main features and determine the actors who will interact with the system.

2.1.1 Identification of Actors

Actors are entities external to the system (people, hardware, systems) that are likely to exchange information with it. The following is a list of the actors who will interact with our system:

Table 2.1. List of MES application users

Actor	Description
Administrator	This is the person responsible for the MES platform. They manage all users (supervisors, operators), passwords and account statuses. They have access to all application features as well as the administration pages in Business Central.
Supervisor	This is an intermediate user. They can monitor the progress of production operations on their work centre, view machine status and track the progress of production orders.
Operator	This is the end user of the MES application. Assigned to a work centre (*Work Center*), they can view the production orders assigned to their machine, start, pause or resume an operation, and declare produced quantities.

2.1.2 Identification of Requirements

At this stage, we will theoretically define the specific requirements of our application, clearly distinguishing functional requirements from non-functional requirements.

Functional Requirements

Each actor of the platform has specific expectations referred to as functional requirements. To define the functional scope, it is necessary that the conceptual solution meets these functional requirements.

- **Authentication**: This feature allows any user to log in to the platform using their Auth ID and password. Upon first login, the user is prompted to change their temporary password.

A secure session token (GUID, 12 h validity) is issued upon each successful login.

- **Two-Factor Authentication (2FA)**: When enabled globally by the administrator, users must complete a second authentication step by scanning their personal badge QR code after entering their credentials. Administrators can view, print, and regenerate badge secrets for any user.
- **Application Setup**: On first launch, the user must configure the middleware host URL (entered manually or obtained by scanning a QR code) so that the application connects to the correct server instance. The host is persisted across sessions and can be changed from the login page.
- **User Management**: This feature allows the administrator to:
  - View the complete list of MES users.
  - Create a new operator or supervisor account by linking it to a Business Central employee and assigning them a work centre.
  - Generate and assign a temporary password.
  - Activate or deactivate an account (automatic revocation of active tokens).
- **Security Policy**: The administrator can configure a password rotation period (in days). A scheduled background job runs every 24 hours and flags users whose password age exceeds the configured period, forcing them to change their password on next login.
- **Machine Management**: This feature allows users to:
  - View the list of machines in a work centre with their current status (Idle, Working).
  - View the production order currently running on each machine.
- **Production Order Management**: This feature allows the operator to:
  - View production orders (Planned, Firm, Released) assigned to their machine.
  - Filter and sort orders by status or date.
  - View the details of an order (item, quantity, planned dates, operation description).
- **Production Operation Management**: This feature is the core of the system. It allows the operator to:
  - Start an operation (validation of prerequisites: released order, free machine, previous operation completed if applicable).
  - Pause or resume an ongoing operation.

  - Declare produced quantities (validation: quantity > 0, no excess of the ordered quantity).
  - View progress in real time (produced quantity, completion percentage, operation status).

- **Progress Dashboard**: Allows the operator and supervisor to view all active operations (Running / Paused) on a given machine, with their respective progress.
- **Password Change**: Allows any logged-in user to change their password by entering their old password and the new one.
- **AI Chat Assistant**: An embedded natural-language assistant allows operators and supervisors to query machine status, production progress, scrap data, BOM, and history without navigating the full interface. The assistant resolves ambiguous machine references, answers fleet-wide questions, surfaces operational anomalies, and provides contextual navigation shortcuts.

Non-Functional Requirements

To define the non-functional scope, it is necessary that the conceptual solution meets these non-functional requirements:

- **Performance**: The platform performs efficiently in terms of handling routine operations (login, order consultation, production declaration). Machine status updates are refreshed every 5 seconds via a streaming mechanism.
- **Security**: Implementation of a token-based authentication system (signed GUID, 12 h expiry) and role management (Admin, Supervisor, Operator) to control access to features. Passwords are stored as SHA-256 hashes with a salt. Optional two-factor authentication enforces badge QR verification as a second factor. A configurable password expiry policy forces periodic credential rotation.
- **Maintainability**: The AL back-end code is organised in layers (Tables, Enums, Codeunits, API Pages) and the Flutter code follows a layered architecture (Data, Domain, Presentation) with the Provider pattern, facilitating updates and the addition of new features.
- **Ergonomics**: The user interface is adaptive (mobile, tablet, PC) thanks to responsive layouts dedicated to each form factor, minimising the learning curve for workshop operators.
- **Portability**: The Flutter application is compiled for Android, iOS, Web, Linux, macOS and Windows from a single codebase. The Business Central back-end is deployable both on-premises and in SaaS (cloud) mode.

2.2 Project Management with Scrum

2.2.1 Backlog Features

Domain 1: Authentication & User Management

ID	User Story	Acceptance Criteria	Priority
US-1.1	As an Administrator, I need to assign access to each user so that users only access features appropriate to their role.	• Ability to assign a user as Operator or Supervisor or Administrator. • Unauthorized actions are blocked with clear error messages.	High
US-1.2	As a User (Operator / Supervisor / Administrator), I need to log into the system with my credentials so that I can access features appropriate to my role.	• User can enter username and password. • System validates credentials against the ERP. • Session token is created upon successful authentication. • User is redirected to the appropriate dashboard based on role.	High
US-1.3	As an Administrator, I need to search users.	• Ability to filter users based on their role. • Ability to search users based on name or email.	High
US-1.4	As an Administrator, I need to see a list of all MES users with their details so that I can manage accounts.	• The user list displays each account's full name, email, role, and work centre. • Clicking a user row opens a password management dialog for that account. • The list refreshes automatically after a new user is created.	High

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ID	User Story	Acceptance Criteria	Priority
US-1.5	As an Administrator, I need to generate and assign a temporary password to a user so that they can log in for the first time.	• Admin can generate a random secure password for any user. • The generated password is sent to the ERP via `AdminSetPassword`. • The target account is flagged *Need To Change Password* after the password is set. • Success or failure is reported to the admin with a clear message.	High
US-1.6	As an Administrator, I need to change the role and work-centre assignment of an existing user so that access rights stay current.	• Admin can select a new role for any user. • Work-centre list updates based on the selected role. • All active tokens for the target user are revoked on save. • The user list refreshes after a successful change.	High
US-1.7	As an Administrator, I need to activate or deactivate a user account so that access can be revoked without deleting the account.	• Admin can toggle the active status of any user. • Deactivation immediately revokes all active tokens. • An admin cannot deactivate their own account. • The user row updates its visual status indicator immediately.	High

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ID	User Story	Acceptance Criteria	Priority
US-1.8	As a User (Operator / Supervisor / Administrator), I need to complete a badge QR code scan after entering my credentials so that my identity is verified by a second factor.	• *2FA* prompt appears only when *TwoFA Enabled* is active in MES Settings. • Camera opens and reads a QR badge code automatically. • Manual entry field available as a fallback. • Session token is persisted only after successful badge verification. • Failed scan shows an error; scanner reactivates. • User can cancel and return to the login page.	High
US-1.9	As an Administrator, I need to view, print, and regenerate a user's badge QR code so that I can distribute it and invalidate a lost or compromised badge.	• Badge QR rendered from the user's *Badge Secret* via a barcode widget. • Print button generates a PDF with landscape/portrait toggle. • Regenerate action shows a confirmation dialog before replacing the secret. • New QR is displayed immediately after regeneration without a page reload. • Actions are available only for active users and to the Admin role.	High

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ID	User Story	Acceptance Criteria	Priority
US-1.10	As an Administrator, I need to globally enable or disable *2FA* and configure a password rotation period so that I can control the organisation's authentication security level.	• *2FA* toggle and password period field available on the *Settings* page. • Changes take effect on the next login attempt. • Settings are persisted in the MES Settings table in Business Central. • A scheduled job (every 24 h) flags users whose password age exceeds the configured period. • Unsaved changes display a persistent banner with *Discard* and *Save* actions.	High
US-1.11	As a first-time user, I need to configure the middleware host URL (or scan a QR code) so that the application connects to the correct server instance.	• *Paring* page shown on first launch when no host is saved. • URL can be entered manually or obtained by scanning a QR code. • URL is validated (no spaces, no illegal characters). • Saved host persists across app restarts. • *Change Host* link on the login page allows re-entering the pairing screen.	High

Domain 2: Machine Monitoring

ID	User Story	Acceptance Criteria	Priority
US-2.1	As an Operator or Supervisor, I need to see the list of machines in my work centre with their current status so that I can see which machines are currently active.	• Machine list is scoped to the authenticated user's work centre. • Each card shows machine name, current status (*Idle*, *Working*), and active production order number. • The list refreshes automatically every 5 seconds.	High

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ID	User Story	Acceptance Criteria	Priority
US-2.2	As an Operator or Supervisor, I need to view the production orders assigned to a machine so that I know what work needs to be done.	• Only *Planned*, *Firm Planned*, or *Released* orders are shown. • Each card displays order number, item description, planned quantity, planned start and end dates, and operation number. • Orders can be searched and filtered by status, and sorted by planned start date.	High
US-2.3	As an Operator or Supervisor, I need to start a production operation so that the system records that the order has begun.	• Only *Released* routing lines may be started. • An operation cannot be started if it is already running. • A machine cannot run two operations simultaneously. • A new MES Operation Status record is created with status *Running*.	High
US-2.4	As a Supervisor or Operator, I need to monitor the current status of all active operations on a machine so that I can follow the progress of each order.	• The Operations Progress tab shows all operations whose latest status record is *Running* or *Paused*. • Each card shows order number, operation number, current status, and last updated timestamp. • Data refreshes automatically every 5 seconds.	High
US-2.5	As an Operator or Supervisor, I need a tabbed view for a machine so that I can navigate between pending orders and active operation progress without leaving the machine context.	• Selecting a machine from the list opens a dedicated machine page. • The page provides at least two tabs: *Orders* and *Progress*. • Switching tabs preserves the state of each tab without reloading data. • The active tab is visually highlighted in the navigation bar.	High

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ID	User Story	Acceptance Criteria	Priority
US-2.6	As an Operator or Supervisor, I need to filter and sort production orders so that I can quickly find the most relevant work.	• Orders can be searched by order number, item description, or date. • Orders can be filtered by status (*All*, *Planned*, *Firm Planned*, *Released*). • Orders can be sorted by planned start date ascending or descending. • On mobile, the status filter is accessible via a compact popup menu.	High
US-2.7	As a Supervisor or Operator, I need to view the history of completed operations for a machine so that I can audit past production.	• The History tab shows all operations with status *Finished* or *Cancelled*. • Each card shows order number, status badge, product name, produced quantity, start time, and end time. • The list supports free-text search and sort by start date. • Tapping a card navigates to the full operation detail page.	High

Domain 3: Production Execution & Traceability

ID	User Story	Acceptance Criteria	Priority
US-3.1	As an Operator or Supervisor, I need to declare the quantity produced in a cycle so that the system records production progress.	• Declared quantity must be positive and not exceed remaining quantity. • Each declaration creates a new progression cycle record. • Cumulative quantity and progress percentage update immediately.	High
US-3.2	As an Operator or Supervisor, I need to pause and resume an active operation so that interruptions are tracked accurately.	• A Running operation can be paused; a Paused one can be resumed. • A machine cannot resume if another operation is already running.	High

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ID	User Story	Acceptance Criteria	Priority
US-3.3	As a Supervisor, I need to finish or cancel an operation so that the machine is released and the order is closed.	• Finishing a complete order (100%) shows a green confirmation dialog. • Cancelling an incomplete order shows a red warning dialog. • Closing sets the execution end time and inserts an Idle machine status.	High
US-3.4	As an Operator or Supervisor, I need a dedicated operation detail page with live data in one place.	• Shows status, order number, product, required and produced quantities. • Live data streams without a manual refresh. • Adapts between mobile and desktop layouts.	High
US-3.5	As an Operator or Supervisor, I need to see all production cycles declared for an operation.	• Cycle table lists operator, quantity, cumulative total, and timestamp. • Records ordered newest-first with pagination. • Refreshes after each new declaration.	High
US-3.6	As an Operator or Supervisor, I need a visual chart of declarations over time to identify performance trends.	• Line chart plots cycle quantity vs. declaration time. • Touch tooltip shows operator, quantity, cumulative total, timestamp. • Auto-scrolls to most recent data point on load and refresh.	Medium
US-3.7	As an Operator or Supervisor, I need to see the Bill of Materials for an operation to know which components are required.	• Components linked to the current routing step are colour-coded by availability (Available / Low Stock / Missing). • Shared components not linked to this step are shown in neutral style. • BOM data refreshes after every consumption event.	High

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ID	User Story	Acceptance Criteria	Priority
US-3.8	As an Operator or Supervisor, I need to report scrapped units with a reason code so that waste is accurately tracked.	• Scrap can only be declared when the operation is Running. • A reason code must be selected before submission. • Each declaration records quantity, code, and operator.	High
US-3.9	As a Supervisor, I need to declare production on behalf of an operator so that output is attributed to the correct person.	• Supervisor role sees an operator selector in the declare dialog. • Selected operator's ID is written to the Declared By field. • If no operator is selected, the supervisor's ID is used.	Medium
US-3.10	As an Operator or Supervisor, I need to scan components into an operation so that consumption is recorded against the BOM.	• Live camera feed reads DataMatrix barcodes. • Internal format parsed locally; external resolved via API. • Duplicate scan increments quantity on the existing row. • Confirming scans inserts consumption records and refreshes the BOM.	High
US-3.11	As an Operator or Supervisor, I need to print a label for a production order so that produced parts are correctly identified.	• Label includes DataMatrix barcode encoding order, machine, item data. • PDF generation supports landscape/portrait toggle. • Print button available only when operation is Finished or progress $\geq$ 100%.	Medium
US-3.12	As an Operator or Supervisor, I need a label printed for each declared unit so that individual pieces carry traceable information.	• One PDF page per declared unit with a label counter. • Same DataMatrix encoding as the production label. • Launched automatically after a successful production declaration.	Medium

Domain 4: Administration & Monitoring

ID	User Story	Acceptance Criteria	Priority
US-4.1	As an Administrator, I need a performance dashboard for all machines so that I can monitor production health across the facility.	• Dashboard shows uptime %, operation count, total produced, and scrap per machine. • Configurable time-range (hours back) filter. • Responsive grid: 1 col mobile, 2 col tablet, 3 col desktop.	High
US-4.2	As an Administrator, I need a filterable, paginated activity log of all system actions so that I can audit operator and machine activity.	• Log entries include type, operator, machine, action, and timestamp. • Filterable by action type and time range. • Paginated at 15 rows per page.	High
US-4.3	As an Administrator, I need a persistent sidebar navigation so that I can move between admin sections without losing context.	• Sidebar has sections for Users & Roles, Machine Dashboard, Activity Logs, and Settings. • All pages kept alive simultaneously via IndexedStack. • Active section is visually highlighted.	High
US-4.4	As any user, I need the application available in English, French, and Arabic so that I can use it in my preferred language.	• Language selector on login screen and in profile page. • All UI strings externalised; no hard-coded text. • Arabic locale activates RTL layout automatically. • Language preference persisted across sessions.	High

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ID	User Story	Acceptance Criteria	Priority
US-4.5	As a first-time user, I need guided coach-mark tutorials so that I can learn the interface without external documentation.	• Tutorials shown exactly once per install per screen. • Coach marks cover: machine list, machine detail tabs, operation detail, and admin dashboard. • Localised skip button text.	Medium

Domain 5: AI Chat Assistant & Analytical Intelligence

ID	User Story	Acceptance Criteria	Priority
US-5.1	As an Operator or Supervisor, I need a natural-language chat assistant so that I can query machine status, production progress, and operational data without navigating the full UI.	• Chat panel opens as a side overlay on wide screens and as a dialog on mobile. • Assistant answers questions about machines, orders, live operations, BOM, history, and scrap. • Responses are scoped to the user's own work centres (access control enforced). • Multi-turn conversation history is maintained within the session. • Conversation can be cleared by the user.	High
US-5.2	As an Operator or Supervisor, I need the assistant to resolve ambiguous machine references (e.g. "Fraiseuse 3") so that I do not need to know exact machine codes.	• Machine names, partial codes, and digit-only references are matched to the correct machine using a five-tier fuzzy resolution strategy. • High-confidence matches ($\geq$ 85%) are resolved silently. • Low-confidence matches present up to 3 candidates for disambiguation. • Machine access control is enforced after resolution.	High

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ID	User Story	Acceptance Criteria	Priority
US-5.3	As an Operator or Supervisor, I need the assistant to answer fleet-wide questions (e.g. "which machines are running?") so that I get a consolidated view without visiting each machine individually.	• Queries about all machines trigger a parallel fan-out across work centres. • Results are filtered by running / idle / overdue / scrap status as requested. • Fan-out is capped at 12 machines to control response time. • Fleet summary includes machine count, utilisation percentage, and active orders.	High
US-5.4	As an Operator or Supervisor, I need the assistant to provide contextual navigation shortcuts so that I can open a machine, operation, or dashboard directly from the chat response.	• Chat responses include up to 4 action buttons when a redirect is relevant. • Supported targets: machine detail, operation detail, machine list, dashboard, and operation history. • Tapping an action navigates to the corresponding screen. • Action buttons appear only on the last assistant message.	Medium
US-5.5	As a Supervisor, I need the assistant to proactively highlight anomalies (overdue orders, scrap spikes, idle operators) so that I can act before problems escalate.	• Scrap rate above 5% is flagged as a warning; above 15% as critical. • Scrap spikes (recent rate $>$ 2.5$\times$ baseline) are explicitly mentioned. • Overdue orders (past planned end) and at-risk orders ($<$ 4 h remaining) are highlighted. • Paused operations exceeding the configured threshold are surfaced.	High

Remaining

ID	User Story	Acceptance Criteria	Priority
US-x.1	As a Supervisor, I need to create a new production order so that I can respond to urgent production needs.	• Form captures: Machine, ProductNo, Planned Quantity, Start Time • Validates product exists in ERP • Validates machine can produce the product • Order is saved to ERP database • Order appears in machine's "Orders to be done" within a short time frame • Audit trail captures order creation	Medium
US-x.2	As an Operator, I need to be able to declare production on another machine of the same type so that I can continue working when my terminal is broken.	• Operator can access machines of same type as their assigned machines • System validates operator is authorized for that machine type • Declaration process identical to US-4.1 • Audit trail shows which machine declaration was made from	Low

2.3 Sprint Plan

The project was organised into five two-week sprints, each delivering a functional increment verified by a sprint review with the professional supervisor.

Table 2.8. MES project sprint plan

Sprint	Domain	Scope
Sprint 1	Authentication & User Management	Secure login, session token management, role-based access control, administrator user creation, password generation and reset, role and work-centre change, account activation/deactivation, two-factor authentication with badge QR code, badge management (view, print, regenerate), password expiry policy configuration, and first-launch host pairing.

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Sprint	Domain	Scope
Sprint 2	Machine Monitoring & Production Order Management	Real-time machine status tracking, production order list per machine, operation start, tabbed machine view with Orders / Progress / History tabs, order filtering and sorting, completed operation history.
Sprint 3	Production Execution & Traceability	Production cycle declarations, pause/resume/finish/cancel operations, operation detail page with live data, production chart, Bill of Materials visibility, component barcode scanning, scrap declaration, supervisor proxy declaration, and label printing.
Sprint 4	Administration, System Configuration	Administrator dashboard and activity log, admin navigation shell, multi-language support (EN/FR/AR) and RTL layout, in-app onboarding tutorials.
Sprint 5	AI Assistant & Analytical Intelligence	Natural-language chat assistant, Node.js reverse proxy middleware, Python AI agent with intent classification, multi-provider LLM support, fuzzy machine resolution, composite fan-out queries, data enrichment, anomaly detection (scrap spikes, overdue orders), and contextual redirect actions.

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2.4 Global Application Architecture

2.4.1 Physical Architecture

The physical architecture of the MES system rests on three main nodes:

- **Business Central Server (on-premises / SaaS)**: hosts the AL extension that exposes OData V4 endpoints and the business REST APIs. In on-premises mode, the server runs on a BC210 instance accessible on port 7048. In SaaS mode, the base URLs follow the schema `api.businesscentral.dynamics.com/v2.0/<tenantId>/<environment>/ODataV4/`.
- **Node.js Middleware**: a reverse-proxy layer running between the Flutter client and Business Central. It handles SSPI/Windows authentication toward BC, exposes a unified REST API consumed by Flutter, and routes AI chat requests to the Python AI agent.
- **Flutter Client (mobile / web / desktop)**: application compiled for Android, iOS, Web or

desktop, communicating with the Node.js middleware via HTTP requests. The middleware proxies these to the BC OData V4 endpoints.

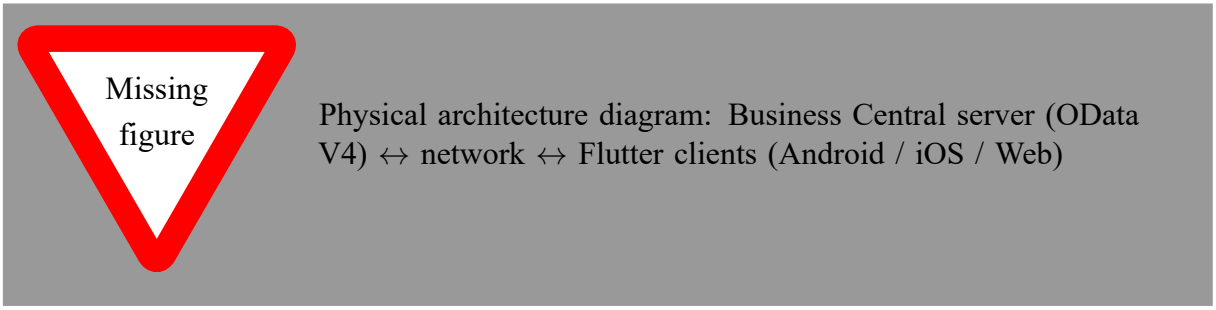

Figure 2.1. MES System Physical Architecture

2.4.2 Software Architecture

The software architecture of the system is organised around four distinct layers.

AL Back-end (Business Central)

The back-end is structured in six layers:

- **Tables (1-Tables)**: define the persistent data model — MES User, MES Auth Token, MES Machine Status, MES Operation Status, MES Operation Progression.
- **Enums (2-Enum)**: define the enumerated types — MES User Role (Operator, Supervisor, Admin), MES Machine Status (Idle, Working), MES Operation Status (Running, Paused, Finished, Cancelled, Interrupted).
- **Codeunits (3-CodeUnit)**: contain the business logic — MES Auth Mgt (authentication, tokens), MES Password Mgt (SHA-256 hashing), MES Machine Fetch and MES Machine Write (machine and operation management), MES Unbound Actions (OData V4 facade).
- **API Pages (4-API)**: expose data for reading/writing via the BC REST API — employees, MES users, work centres.
- **Permission Sets (5-PermissionSet)**: define access rights to AL objects for BC users.
- **Pages (6-Pages)**: administration and debugging interfaces in the BC client (lists, cards, API debug page).

Node.js Middleware

The Node.js middleware is the network bridge between the Flutter client and Business Central. It handles SSPI/Windows authentication for BC OData V4 requests, exposes a unified REST API to the Flutter application, and routes AI chat requests (`POST /api/ai/chat`) to the Python AI agent.

Python AI Agent

The Python AI agent (FastAPI) provides the conversational intelligence layer. It classifies user intent, executes tool chains against Business Central via the Node.js middleware, enriches results with data-analysis helpers, and synthesises natural-language replies using a configurable multi-provider LLM client.

Flutter Front-end

The Flutter front-end follows a three-layer architecture:

- **Data**: contains data models (models) and HTTP services (services) that communicate with BC endpoints via the `http` library.
- **Domain**: contains the providers (Provider pattern) that expose the application state to widgets — AuthProvider, MesUserProvider, MesMachinesProvider, MachineordersProvider, etc.
- **Presentation**: contains the pages and widgets of the user interface, organised by functional domain (auth, admin, machine) with dedicated responsive layouts (mobile, tablet, PC).

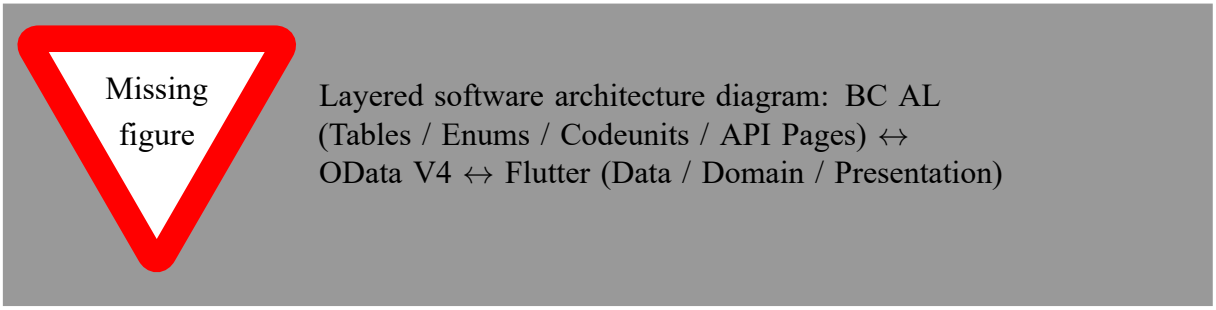

Figure 2.2. MES System Software Architecture

2.5 Work Environment

2.5.1 Software Environment

Table 2.9 summarises the main tools and technologies used in this project.

Table 2.9. MES project software environment

Logo	Tool / Technology	Role in the project
Microsoft Dynamics 365 Business Central	Microsoft Dynamics 365 Business Central	ERP platform hosting the AL back-end and business data

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Logo	Tool / Technology	Role in the project
	AL Language (Application Language)	Development of the BC extension (tables, codeunits, API pages)
	Flutter (Dart)	Development of the cross-platform mobile/web application
	Visual Studio Code	Primary development environment (AL and Flutter)
	Git / GitHub	Version control and collaboration
	Postman	Testing and validation of OData V4 endpoints
	lucid chart	UML and system architecture diagramming

2.5.2 Development Environment

[To be completed: include workstation specifications (OS, RAM, CPU), tool versions (Flutter SDK, Dart SDK, Business Central version, VS Code extensions), and any specific configuration details such as Docker setup or BC on-premises instance parameters.]

Conclusion

In this chapter, we analysed and specified the functional and non-functional requirements of the MES system, identified the actors and their interactions, and presented the overall architecture of the solution. We also described the work environment and the adopted Scrum methodology. The following chapters detail the work carried out sprint by sprint, presenting the conceptual aspects and the results obtained at the end of each iteration.

Chapter 3

Sprint 1 Authentication & User Management

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Introduction

This sprint focuses on delivering a complete authentication and user management module for the *MES* application. The primary goal is to develop a secure authentication system covering *login*, *logout*, and password management, alongside role-based page access control to ensure each user can only reach features appropriate to their role.

In addition, the Administrator is given the ability to add new MES users, generate secure passwords on their behalf, change a user's role and work-centre assignment, and activate or deactivate accounts. The sprint also delivers two-factor authentication via badge QR code scanning, administrator badge management (view, print, regenerate), a configurable password expiry policy, and the first-launch host pairing screen that connects the Flutter application to the correct Node.js middleware instance.

By the end of this sprint, a fully functional *login* screen, *badge scan* flow, *change password* flow, *admin user management dashboard*, *role management dialog*, *settings page*, *paring page*, and *logout* functionality will all be in place.

3.1 Sprint Backlog

This sprint covers **Domain 1:** ***Authentication & User Management***.

Table 3.1. Story Backlog

ID	User Story	Tasks
US-1.1	As an Administrator, I need to assign access to each user so that users only access features appropriate to their role.	*Business Central (AL)* • Implement the functions `AdminCreateUser()` and `AdminSetActive()` in codeunit MES Unbound Actions. • Expose `AdminCreateUser()` and `AdminSetActive()` through codeunit MES Web Service. • Create API page MES User Create API. *Flutter* • Update ApiService to consume the new endpoints. • Implement MesUserService to consume `fetchMesUsers()` and `createMesUser()` endpoints. • Create provider MesUserProvider to manage its state.

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ID	User Story	Tasks
		• Create widget AddUserDialog. • Implement role-based navigation in _AuthGate.
US-1.2	As a User (Operator / Supervisor / Administrator), I need to log into the system with my credentials so that I can access features appropriate to my role.	*Business Central (AL)* • Create the tables MES USER and MES AuthToken. • Create enum MES UserRole. • Implement functions `MakeSalt()`, `HashPassword()`, and `VerifyPassword()` in codeunit MES Password Mgt. • Implement functions `CreateUser()`, `SetPassword()`, `ValidateCredentials()`, `IssueNewToken()`, `ValidateToken()`, `TouchToken()`, `Logout()`, `ValidateChangePassword()`, `ValidateAdminToken()`, `SetActive()`, and `CleanupExpiredTokens()` in codeunit MESAuthMgt. • Implement functions `Login()`, `Logout()`, `Me()`, and `ChangePassword()` in codeunit MES Unbound Actions. • Expose `Login()`, `Logout()`, `Me()`, and `ChangePassword()` in codeunit MES Web Service. *Flutter* • Implement ApiService to consume the new endpoints. • Create provider AuthProvider to manage its state. • Implement authentication routing in _AuthGate. • Create page LoginPage. • Create AppConstants.
US-1.3	As an Administrator, I need to search users.	*Flutter* • Implement user search in AddUserPage. • Implement role filtering in AddUserPage. • Combine role and search filters in AddUserPage.

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ID	User Story	Tasks
US-1.4	As an Administrator, I need to see a list of all MES users with their details so that I can manage accounts.	*Business Central (AL)* • Create the function `fetchAllMESUsers` in codeunit MES Unbound Actions. • Expose `fetchAllMESUsers` through codeunit MES Web Service. *Flutter* • Create model MesUser. • Implement `MesUserService` to consume the new endpoint. • Implement `MesUserProvider` to manage its state and its streams. • Implement user list display in AddUserPage. • Create the widgets GeneratePasswordDialog and EmployeeAvatar.
US-1.5	As an Administrator, I need to generate and assign a temporary password to a user so that they can log in for the first time.	*Business Central (AL)* • Implement function `AdminSetPassword()` in codeunit MES Unbound Actions. • Expose `AdminSetPassword()` through codeunit MES Web Service. *Flutter* • Create widget GeneratePasswordDialog. • Update ApiService to consume `adminSetPassword()`. • Update AuthProvider to account for the new functions in the service.
US-1.6	As an Administrator, I need to change the role and work-centre assignment of an existing user so that access rights stay current.	*Business Central (AL)* • Implement function `AdminChangeUserRole()` in codeunit MES Unbound Actions. • Expose `AdminChangeUserRole()` through codeunit MES Web Service.

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ID	User Story	Tasks
		Flutter • Update MesUserService to consume `changeUserRole()`. • Update MesUserProvider to account for`changeUserRole()`. • Create widget ChangeRoleDialog.
US-1.7	As an Administrator, I need to activate or deactivate a user account so that access can be revoked without deleting the account.	*Flutter* • Update ApiService to account for `toggleUserActiveStatus()`. • Update AuthProvider to account for `toggleUserActiveStatus()`. • Integrate activate and deactivate actions in AddUserPage.
US-1.8	As a User (Operator / Supervisor / Administrator), I need to complete a badge QR code scan after entering my credentials so that my identity is verified by a second factor.	*Business Central (AL)* • Add field `Badge Secret` (Text[64]) to table MES User. • Implement `GenerateBadgeSecret()` in MES User.OnInsert. • Implement `VerifyBadge()`, `GetBadgeSecret()`, and `RegenerateBadgeSecret()` in codeunit MESAuthMgt. • Expose the three procedures through codeunit MES Web Service. • Add field `TwoFA Enabled` (Boolean) to table MES Settings. • Return `twoFAEnabled` flag in the `Login()` response.

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ID	User Story	Tasks
		Flutter • Update AuthProvider.login() to detect `twoFAEnabled` flag and hold pending session state. • Create page/dialog BadgeScanDialog with live camera feed (mobile_scanner) and manual entry fallback. • Implement `verifyBadge()`, `cancelBadgeScan()` in AuthProvider. • Integrate badge verification step into _AuthGate routing logic.
US-1.9	As an Administrator, I need to view, print, and regenerate a user's badge QR code so that I can distribute it and invalidate a compromised badge.	*Flutter* • Create dialog UserBadgeDialog rendering badge QR via barcode_widget package. • Implement `getBadgeSecret()` and `regenerateBadge()` in AuthProvider. • Implement PDF export with landscape/portrait toggle in UserBadgeDialog._exportPdf(). • Add *View Badge* action to UserActionMenu (active users only).
US-1.10	As an Administrator, I need to globally enable or disable *2FA* and configure a password rotation period so that I can control the organisation's authentication security level.	*Business Central (AL)* • Add field `PW change period` (Duration) to table MES Settings. • Implement `GetMESSettings()` and `UpdateMESSettings()` in codeunit MES Settings Functions. • Expose both procedures through codeunit MES Web Service. • Implement codeunit MES Password Expiry Worker (Job Queue, runs every 24 h) that flags users whose password age exceeds the configured period. • Register the worker in MES Setup.RegisterPasswordExpiryWorker().

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ID	User Story	Tasks
		Flutter • Create model SettingModel with `pwChangePeriod` and `twoFAEnabled` fields. • Create SettingService consuming `fetchSettings` and `updateSettings` endpoints. • Create MesSettingsProvider managing fetch, dirty-state detection, and save. • Create page MesSettingsPage with toggle and numeric input; animated dirty banner with *Discard* / *Save* actions. • Add *Settings* section to AdminPage sidebar.
US-1.11	As a first-time user, I need to configure the middleware host URL (or scan a QR code) so that the application connects to the correct server instance.	*Flutter* • Create page ParingPage shown on first launch when `AppConstants.host` is empty. • Implement URL validation (no spaces, no backslash, no illegal characters). • Implement QR scanning via QrScannerDialog to populate the host field. • Persist accepted host via `AppConstants.changeHost()` (SharedPreferences). • Add *Change Host* link on LoginPage for reconfiguration. • Add `AppConstants.loadHost()` call to the app initialisation sequence in `main()`.

3.2 Design

3.2.1 Use Case Diagram

The UML Use Case Diagram (Figure 3.3) shows the three primary actors (Operator, Supervisor, Administrator) and their interactions with the *MES* authentication and user management system. In addition to the core login, logout, change password, and user creation flows, the diagram includes the *Change Role / Work Centre* and *Activate / Deactivate Account* use cases available exclusively to the Administrator, as well as the -*Badge QR Scan* two-factor verification step and the -*Paring / Host Configuration* first-launch flow.

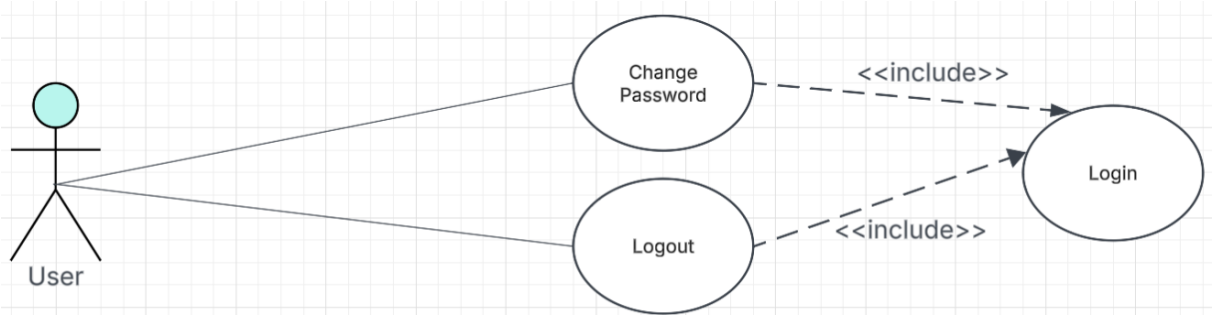

Figure 3.1. UML Use Case Diagram.

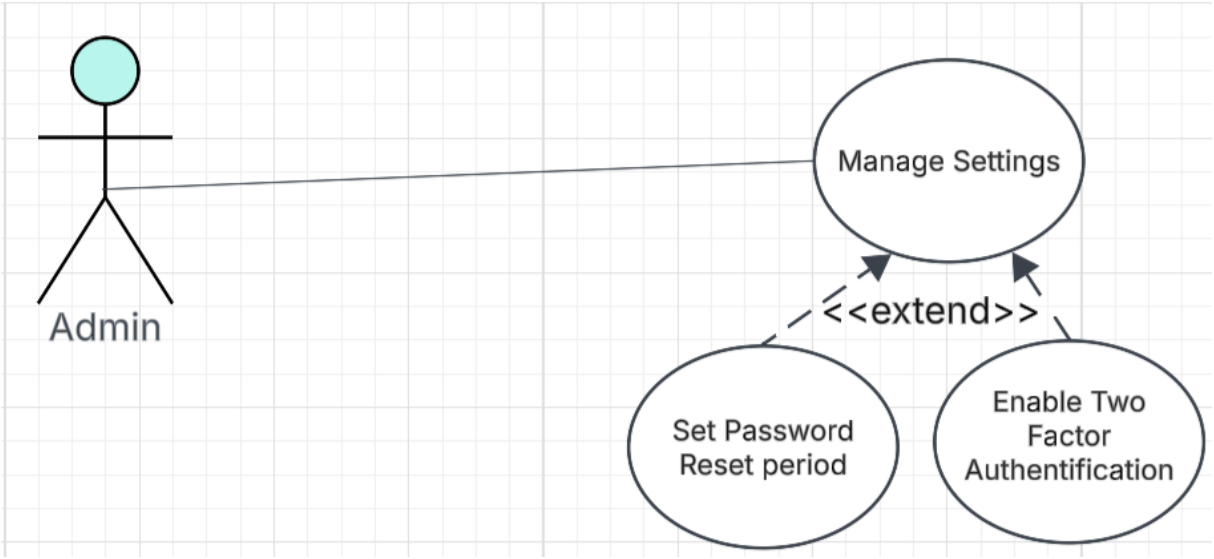

Figure 3.2. UML Use Case Diagram.

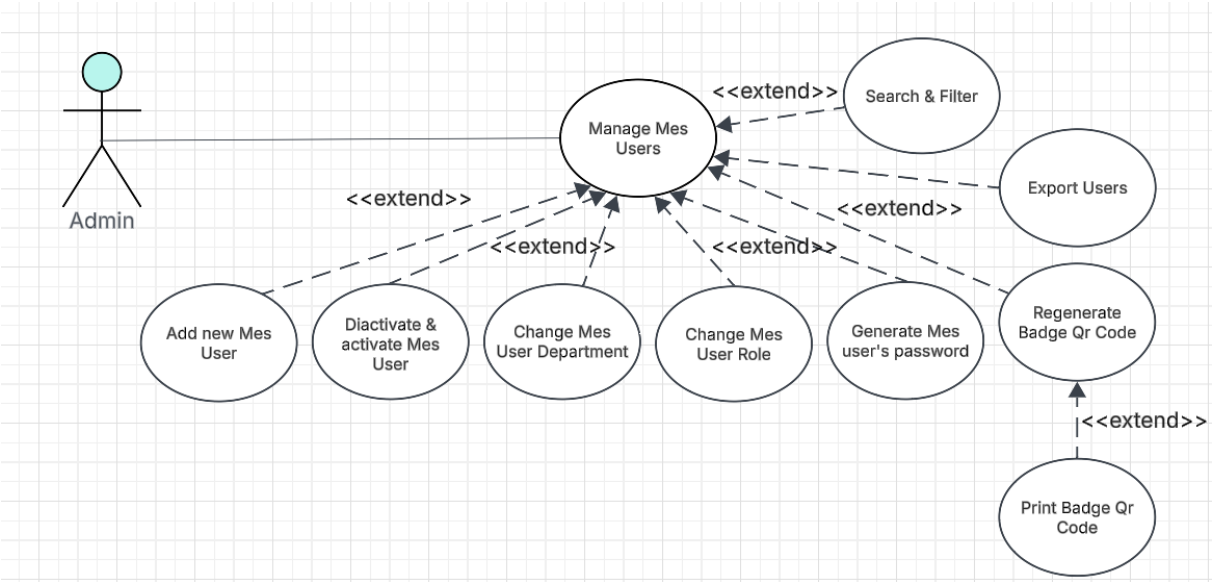

Figure 3.3. UML Use Case Diagram.

3.2.2 Textual Use Case Description

Use Case Name	Login
Primary Actors	Operator, Supervisor, Administrator
Preconditions	• The user account exists in the ERP system. • The user has an assigned role and department. • The account must have a password (there is a possibility that there is an account without a password). • The user account is active.
Postconditions	• If the credentials are valid, the user is successfully authenticated. • If *2FA* is enabled, the badge QR scan step is completed before a session token is issued. • A session token is generated and stored securely. • The user is redirected to the appropriate dashboard based on their role (Admin or MES interface).
	• If the *need change password* flag is set to true, they are redirected to the *Change Password* page. • If the credentials are invalid, access is denied and an error message is displayed.

Continued on next page

Main Scenario	1. The user launches the application. 2. If no host URL is configured, the system displays the *Paring* page; the user enters or scans the middleware URL. 3. The system displays the *login screen*. 4. The user enters *Auth ID* and *password*. 5. The system automatically detects the device ID. 6. The user clicks the *Login* button. 7. The system validates the credentials against the ERP database. 8. The system generates a secure session token. 9. If *2FA* is enabled, the system opens the *Badge Scan* dialog; the user scans their badge QR or enters the code manually. 10. The system verifies the badge secret and generates a secure session token. 11. The system checks the user role. 12. The system redirects the user to the appropriate dashboard (Admin or MES).

Table 3.2. Textual Use Case Description — Login

3.2.3 Sequence Diagrams

The following sequence diagrams illustrate the communication flow between the Flutter frontend, the Business Central OData layer, and the AL business logic for each authentication flow.

Figure 3.4 depicts the login flow, in which the user submits credentials, the AL layer validates them against the Business Central user store, and the application either returns a structured error or issues a session token (or, when *2FA* is enabled, holds a pending state until the badge scan is verified) and redirects the user to the appropriate page based on their role and password-change flag.

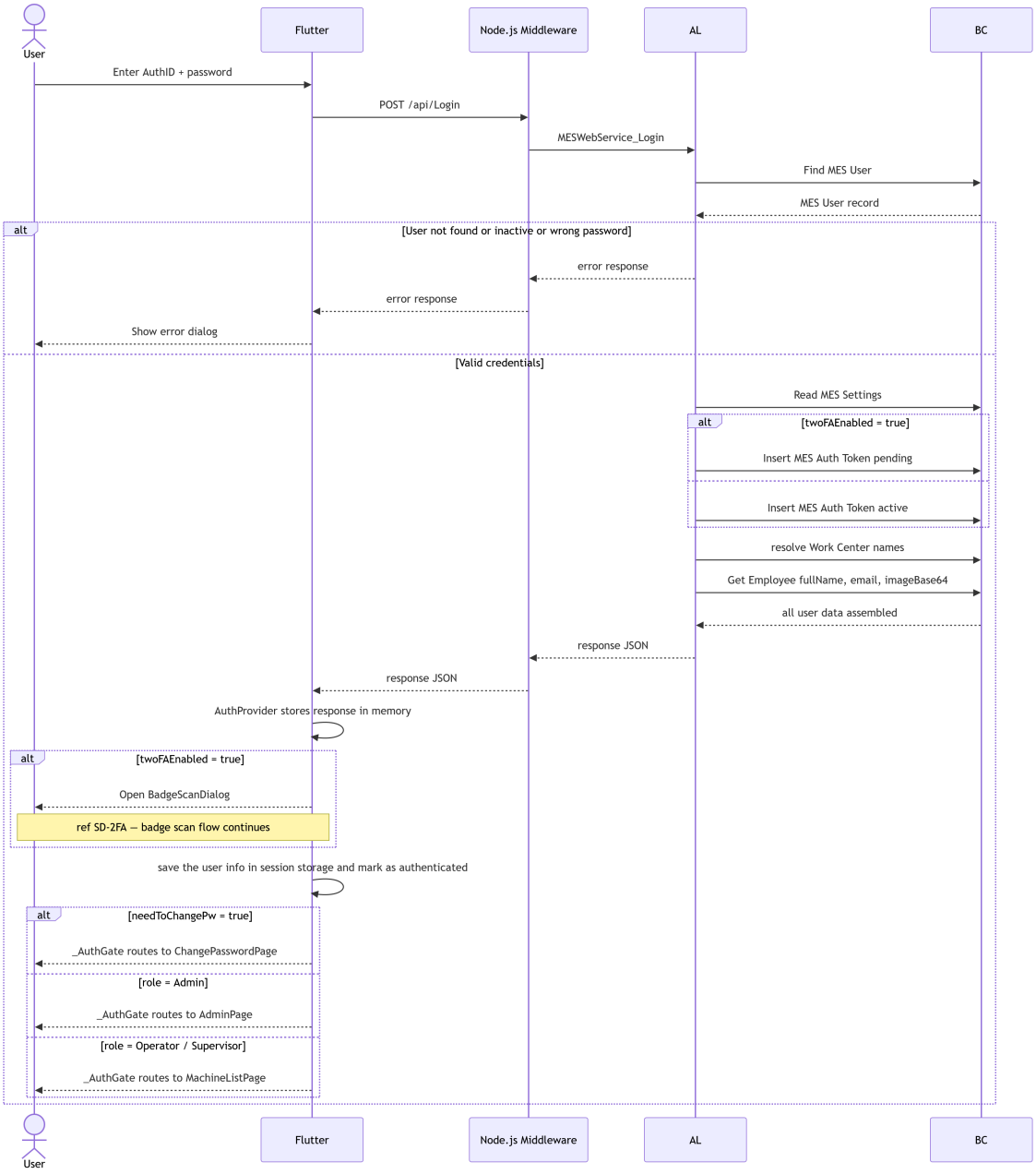

Figure 3.4. SD-01: Login.

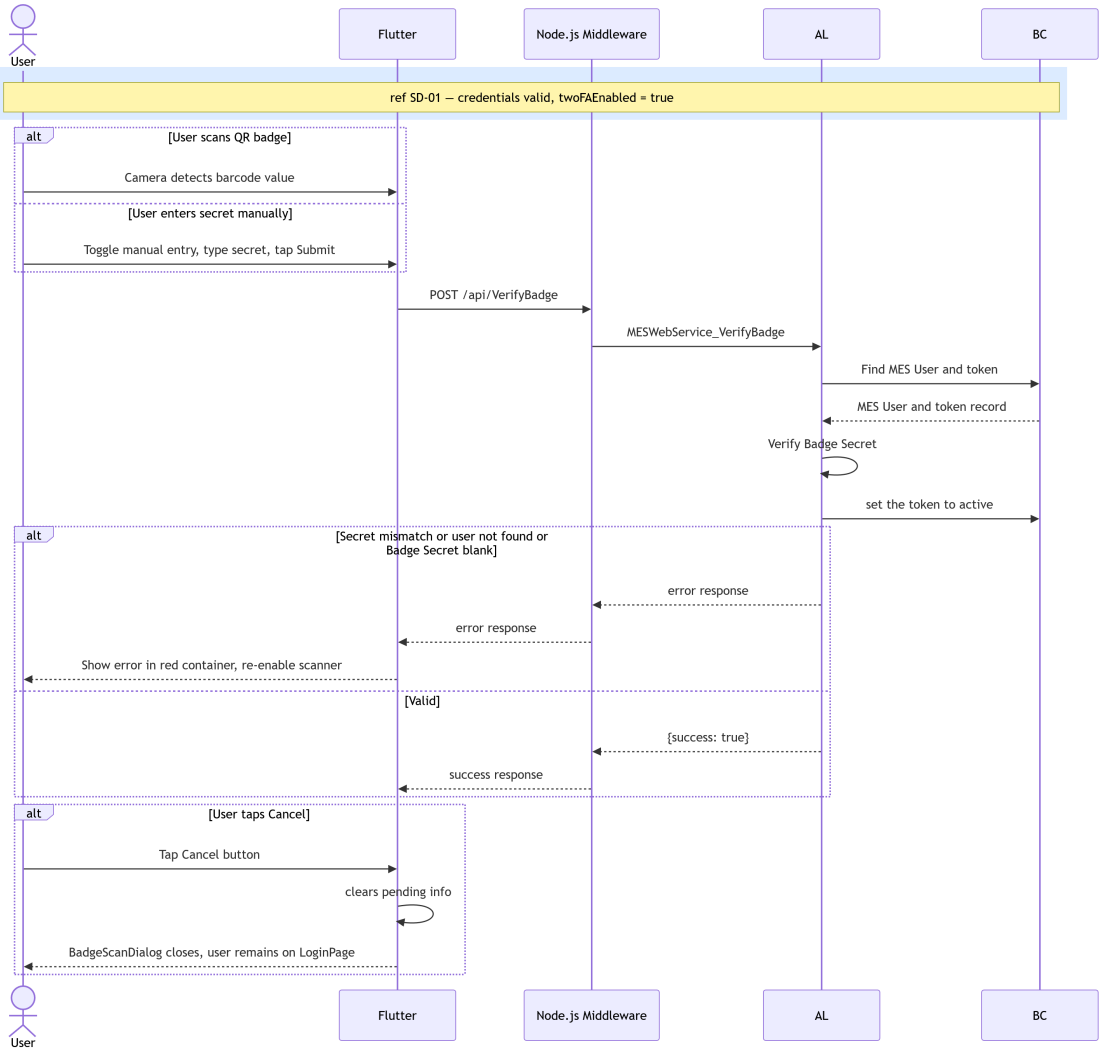

Figure 3.5. SD-2FA: Badge Scan.

Figure 3.6 presents the token validation flow, which is invoked by the AL backend before processing any authenticated request. The backend retrieves the token record from Business Central, verifies its validity and expiry, confirms that the associated user account is still active, and either updates the `LastSeenAt` timestamp or returns an error that forces the Flutter client to redirect the user to the login page.

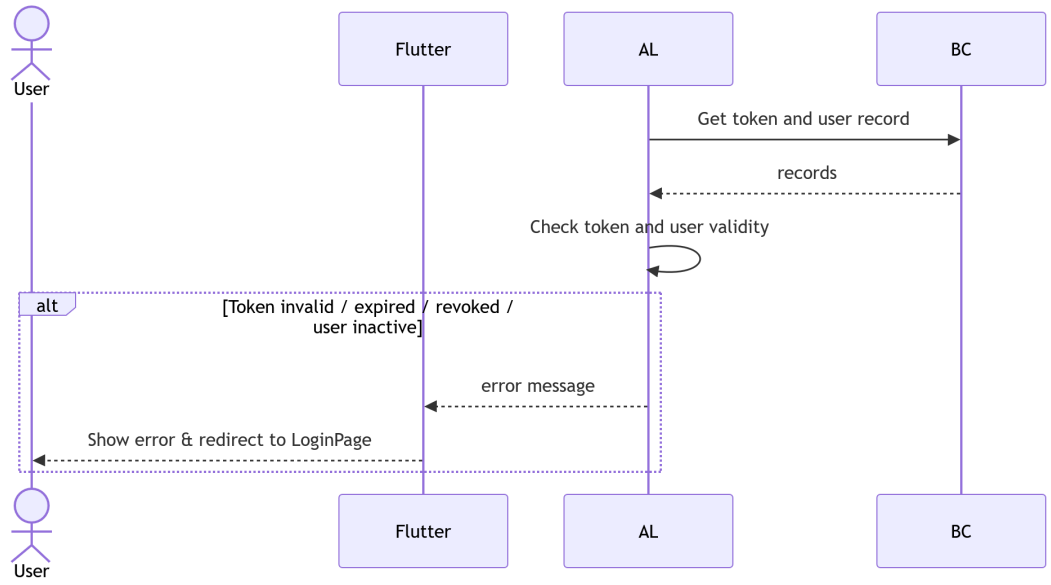

Figure 3.6. SD-02: Token Validation.

Figure 3.7 illustrates the logout flow, in which the Flutter client sends a logout request, the AL layer retrieves and revokes the current session token in Business Central, and the client clears its local AuthProvider state before navigating back to the login page.

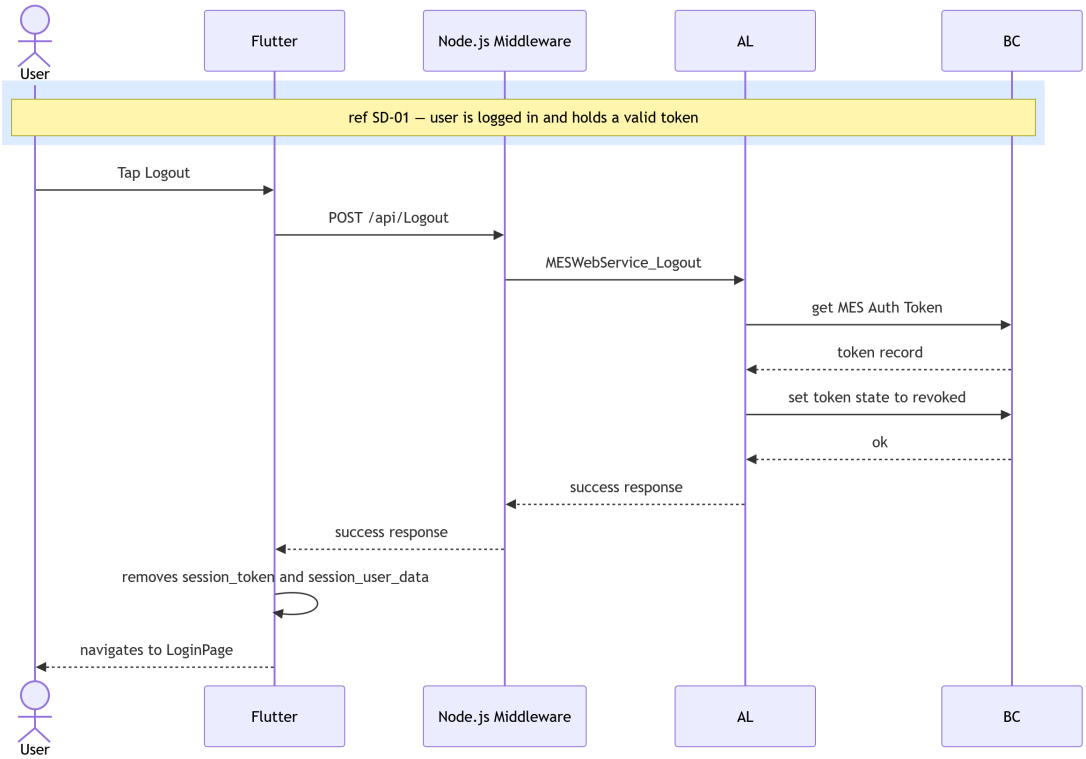

Figure 3.7. SD-03: Logout

Figure 3.8 describes the change-password flow. The Flutter client first validates locally that the two password fields match and satisfy the strength rules, then forwards the request to the AL

layer, which re-validates the token (SD-02), verifies the current password hash, checks the new password strength, and — on success — updates the credential record in Business Central and revokes all existing tokens for the account.

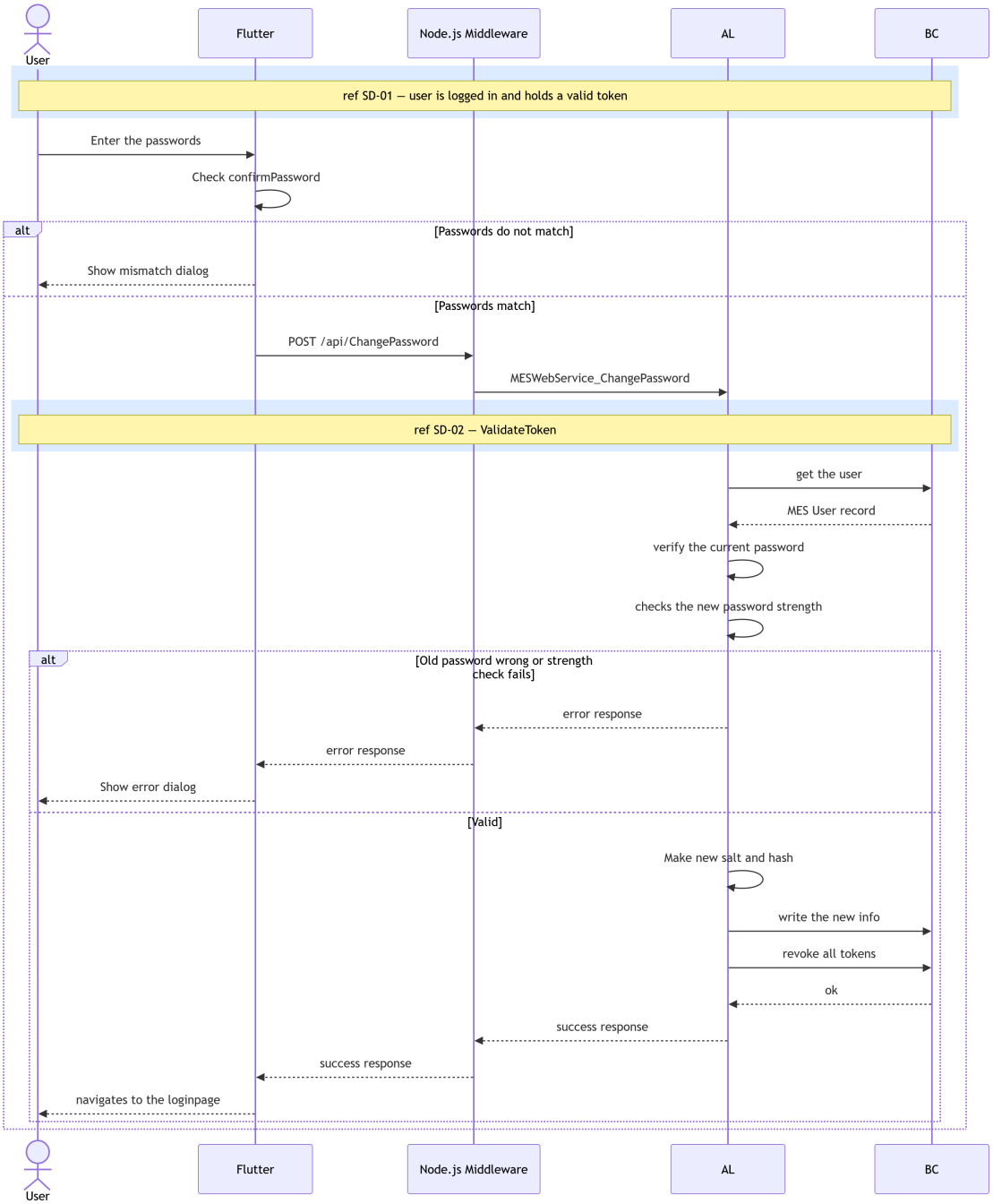

Figure 3.8. SD-03: Change Password

Figures 3.9 through 3.12 cover the four administrator user-management operations. All four follow the same pattern: the Flutter client sends an admin-scoped request, the AL layer validates the token and confirms that the caller holds the Admin role (SD-02), executes the requested Business Central write operations, and triggers a reactive refresh so that the updated user list is reflected immediately in the UI.

Figure 3.9 shows the user creation flow, in which the AL layer inserts a new *MES User* record with `IsActive` and `NeedToChangePw` set to `true`, then iterates over the supplied work-centre list to create the corresponding assignment records.

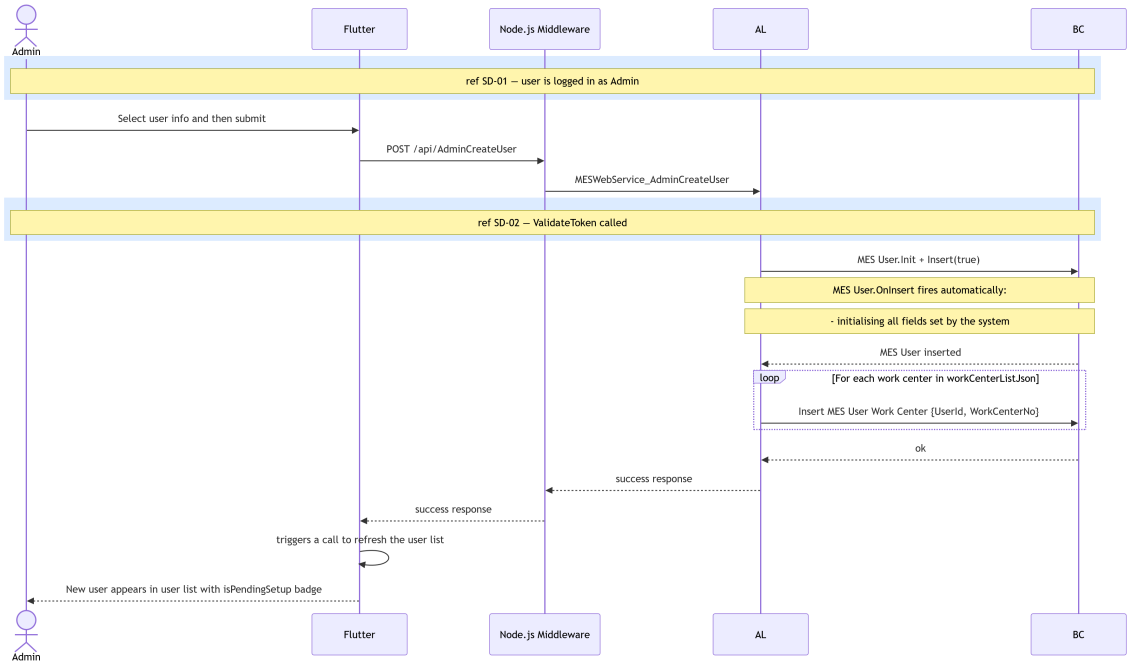

Figure 3.9. SD-05: Admin - Create User.

Figure 3.10 depicts the admin set-password flow. The administrator generates a temporary credential, the AL layer marks the target account as requiring a password change on next login, revokes all active tokens for that account, and returns the new credential to the Flutter client as copyable text.

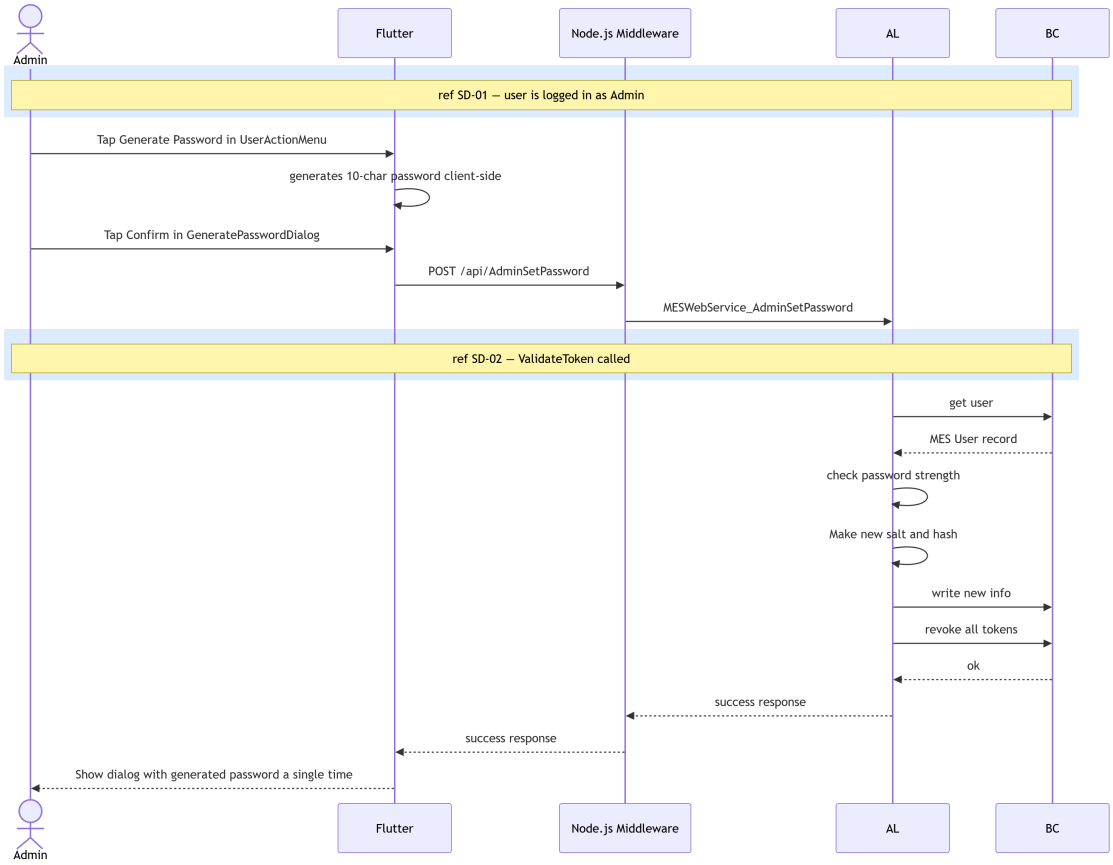

Figure 3.10. SD-06: Admin - Set Password.

Figure 3.10 depicts the admin set-password flow. The administrator generates a temporary credential, the AL layer marks the target account as requiring a password change on next login, revokes all active tokens for that account, and returns the new credential to the Flutter client as copyable text.

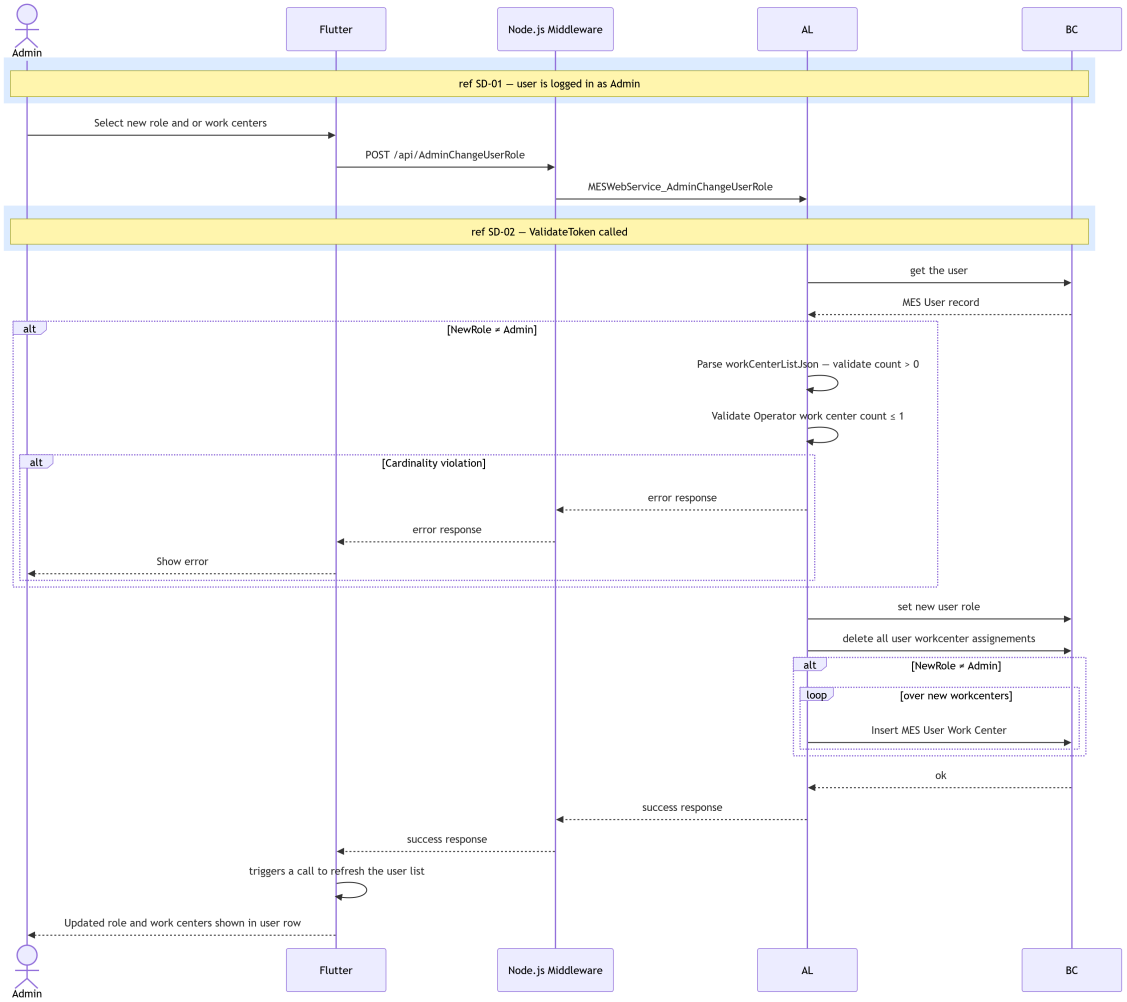

Figure 3.11. SD-07: Admin - Change Role / Work Centre.

Figure 3.12 presents the account activation and deactivation flow. When deactivating an account, the AL layer additionally revokes all active session tokens for that user, ensuring that any currently authenticated sessions are immediately invalidated.

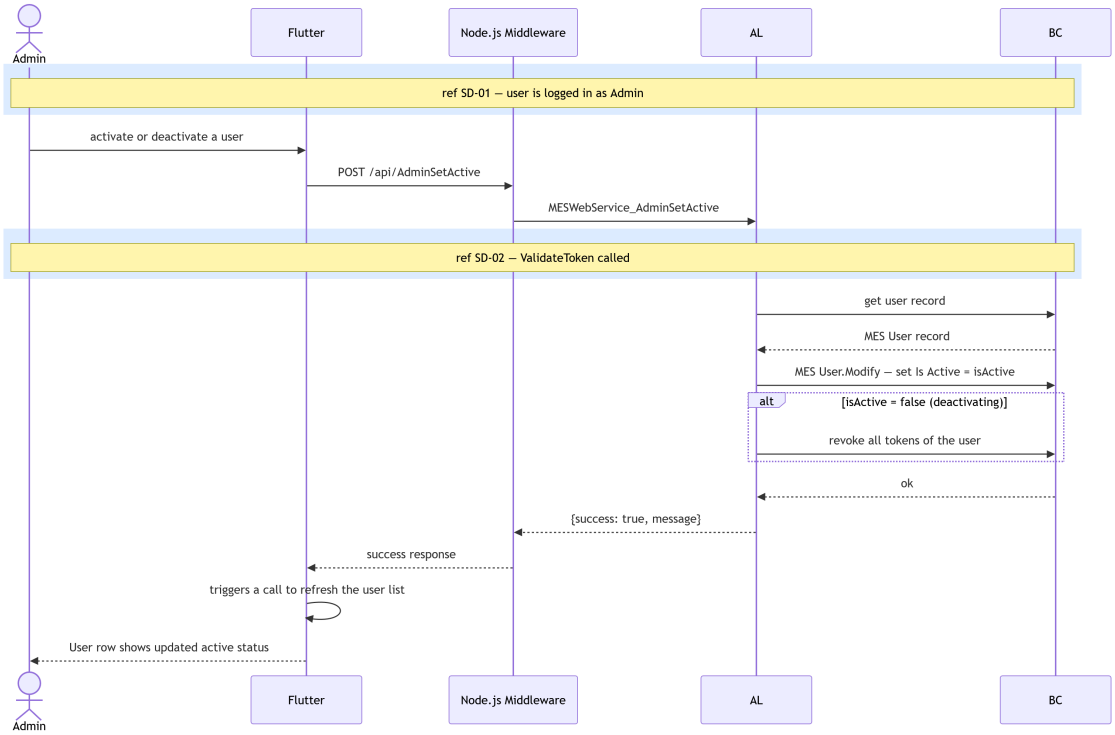

Figure 3.12. SD-08: Admin - Toggle Account Active Status.

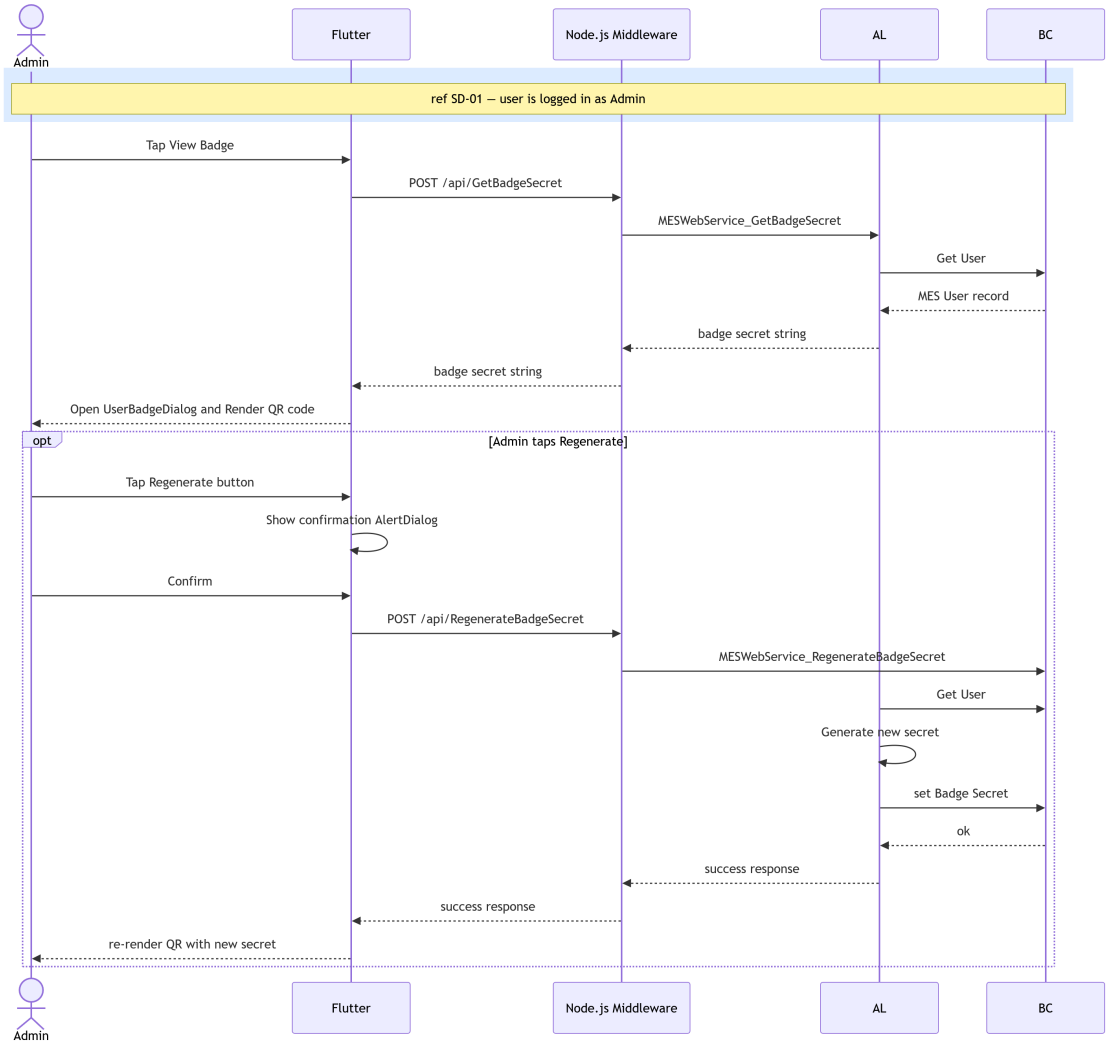

Figure 3.13. SD-ADMIN-BADGE: Admin - regenerate 2FA user badge.

3.2.4 Class Diagram

The UML Class Diagram of the *MES* authentication domain, shown in Figure 3.14, depicts the `MesUser`, `AuthToken`, and `PasswordPolicy` classes with their attributes and associations. The `BadgeSecret` field on `MesUser` and the `MESSettings` singleton are also shown, reflecting the 2FA and password expiry extensions introduced in this sprint.

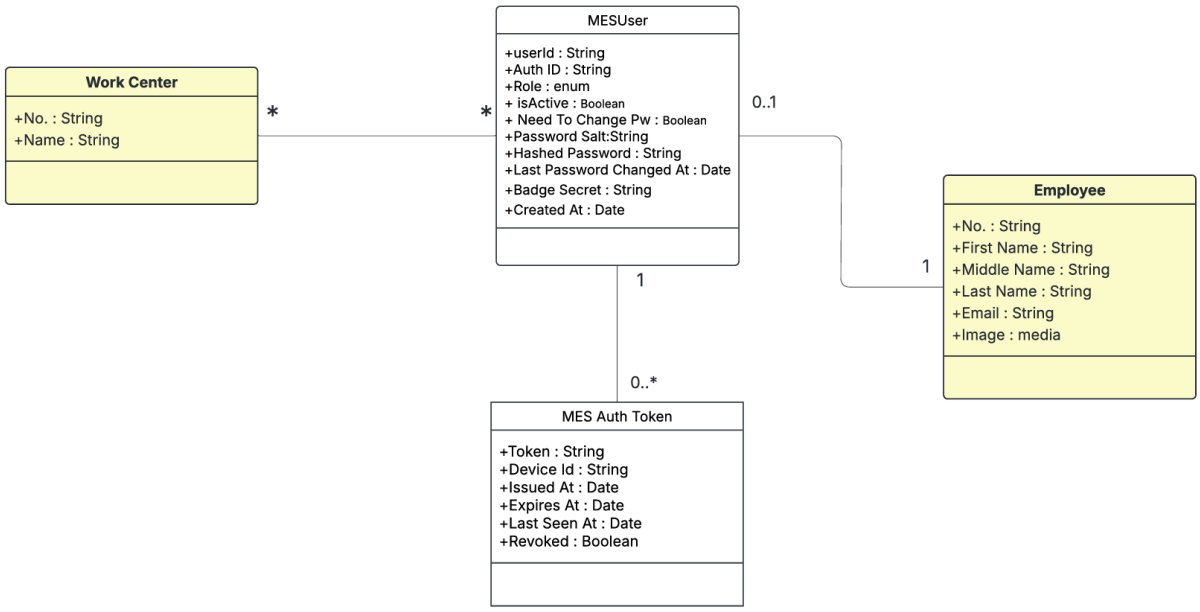

Figure 3.14. UML Class Diagram.

3.3 Implementation

3.3.1 Backend Implementation

Backend Structure Overview

The backend project structure is illustrated in Figure 3.15.

```
AL-backend/
          src/
                1-Tables/
                      NEW/
                          MES_AuthToken.al
                          MES_USER.al
                2-Enum/
                      NEW/
                          MES_UserRole.al
                3-CodeUnit/
                      NEW/
                          MESAuthMgt.al
                          MESPasswordMgt.al
                          MESSetup.al
                4-API/
                      NEW/
                          MESAuthActions.al
                          MESAuthApi.al
                5-PermissionSet/
                      NEW/
                          MESAuthApi.permissionset.al
                6-Pages/
                    NEW/
                        MESApiDebug.Page.al
                        MESUserCard.Page.al
                        MESUserList.Page.al
```

Figure 3.15. Backend project structure.

MES User Table Implementation

The `MES User` table is implemented to store user information, as described in Table 3.3.

Name	Type	Description
User Id	Code[50]	Primary key. Login identifier used at the MES login prompt (e.g., `JDOE`, `OP001`). If blank on insert, a GUID-derived value is assigned automatically.
Employee ID	Code[50]	Foreign key to `Employee."No."`. Required for every MES account and enforced as unique (one Employee maps to one MES User).
Auth ID	Text[100]	Auto-generated external identifier in the form `AUTH-XXXXXXXX`. Used as the login identifier at the MES login prompt. Uniqueness enforced.
Role	Enum (`MES User Role`)	Authorization role for the MES application (e.g., Operator / Supervisor / Admin). Used to control permissions and access.

Continued on next page

Name	Type	Description
Is Active	Boolean	Account enable/disable flag. If false, authentication is blocked and existing tokens are revoked when deactivated.
Need To Change Pw	Boolean	If true, the user must change their password before using the MES application (set at creation and on forced password resets).
Password Salt	Text[64]	Random hex salt used in password hashing so identical passwords produce different hashes across users.
Hashed Password	Text[128]	Hashed password value (hex string). Passwords are never stored in plaintext; hashing uses the password plus salt.
Created At	DateTime	UTC timestamp set on insert only (audit trail/reporting).
Last Password Changed At	DateTime	Updated whenever the password is modified. Used by the password expiry worker to determine whether a rotation is due.
Badge Secret	Text[64]	SHA-256-derived secret generated on user creation. Used to verify badge QR code scans during the two-factor authentication step.

Table 3.3. MES User (Table 50101) — Field Dictionary

Authentication Token Table

The `MES Auth Token` table stores session information as described in Table 3.4.

Name	Type	Description
Token	Guid	Primary key. Raw GUID token presented by the client as a Bearer credential. Generated at issue time (never reused). Used for all direct token lookups (validation/logout).
User Id	Code[50]	Foreign key to `MES User."User Id"`. Identifies the MES user who owns this session/token. Enforces referential integrity via `TableRelation`.
Device Id	Text[100]	Optional identifier for the device that requested the token (e.g., `tablet-floor-02`, `scanner-01`). Stored for traceability/audit only; not used for validation.

Continued on next page

Name	Type	Description
Issued At	DateTime	UTC timestamp set when the token is created. Useful for audit trail and token age inspection.
Expires At	DateTime	UTC timestamp after which the token must be rejected even if not revoked. Set to `Issued At + TTL`. Token validation enforces `Expires At > CurrentDateTime()`.
Last Seen At	DateTime	Updated on every successful token validation. Supports session monitoring (activity/idle detection). Not used for expiry enforcement (use `Expires At`).
Revoked	Boolean	If true, token is explicitly invalidated and must be rejected even if not expired. Set on logout and during bulk revocation (e.g., password change, account deactivation).

Table 3.4. MES Auth Token (Table 50106) — Field Dictionary

Key points of the token design include a unique *GUID* token, auto-expiry after 12 hours, last activity tracking, and soft revocation.

MES Settings Table

The `MES Settings` singleton table (Table 50100) stores the two global security parameters configurable from the *Settings* page.

Name	Type	Description
PW change period	Duration	Password rotation interval stored in milliseconds. A value of 0 does not disable the feature; only negative values do.
TwoFA Enabled	Boolean	Global flag. When true, all users must complete a badge QR code scan after entering their credentials.

Table 3.5. MES Settings (Table 50100) — Field Dictionary

Password Management and Encryption

The password management codeunit ensures that plaintext is never persisted. A unique random *salt* (64-character SHA-256 hex string) is generated per user and the password is hashed using single-pass *SHA-256* before storage. Data classification attributes mark sensitive fields as *Customer Content* or *System Metadata*.

Key points include: passwords are never stored in plaintext; a unique salt per user (64 characters) is generated; *SHA-256* hashing is applied; and data is classified as *Customer Content* or *System Metadata*.

Authentication Logic — Login, Logout, and Change Password

Login

The login validates the user credentials against the `MES User` table. If the credentials are correct and the account is active, and *2FA* is disabled in `MES Settings`, a token is generated and stored in the `MES Auth Token` table. When *2FA* is enabled, the backend returns the `twoFAEnabled` flag and the Flutter client suspends the session until the badge QR code is verified via the `VerifyBadge` endpoint.

Logout

On logout, the user's session authentication token is revoked. The token record matching the supplied session token is located in the `MES Auth Token` table and its `Revoked` flag is set to `true`, immediately invalidating the session (soft revocation).

Change Password

Changing password requires a valid authentication token, a correct current password, and a new password that complies with security rules.

Password Expiry Worker

The `MES Password Expiry Worker` (Codeunit 50129) is a scheduled Business Central Job Queue entry that runs every 24 hours. On each run it reads the `PW change period` value from `MES Settings` and computes a cutoff datetime equal to `CurrentDateTime() minus the period`. It then iterates all active MES Users whose `Last Password Changed At` falls before the cutoff (or is zero, meaning the password has never been changed) and sets their `Need To Change Pw` flag to `true`. Affected users are prompted to change their password on their next login.

3.3.2 REST API and Web Services Implementation

To enable communication between Flutter and Microsoft Dynamics Business Central, a set of *RESTful APIs* were implemented using AL.

API Pages for MES User Management

The *API Page* definition for the `/mesUsers` endpoint exposes the `MES User` table as a read-only *OData* feed.

The *API Page* definition for the `/createMesUser` endpoint differs from the read endpoint by setting validation triggers. `DelayedInsert = true` buffers all field values until the complete request body has been received before committing the new `MES User` record to the database.

Key point: the difference between these two APIs is the *Delayed Insert* flag. When set to `true`, it delays insertion until all fields are received, which is required for `POST` operations. `GET` is not affected.

OData Functions and Web Service Configuration

Exposed these functions in the MES Web Service codeunit (50126). Configured WebServices.xml to publish MES Web Service as an ODataV4 web service under the service name `MESWebService`.

3.3.3 Frontend Implementation

The frontend of this system was developed using Flutter. The application is structured into four main layers: a ***UI Layer*** consisting of Screens and Widgets; a ***State Management Layer*** built on Providers; a ***Service Layer*** for API communication; and a ***Model Layer*** for Data Models.

The data flow follows a unidirectional pattern from the *UI Layer* through the *State Management Layer* (`Provider` pattern) and *Service Layer* to the Business Central REST API backend, as illustrated in Figure 3.16.

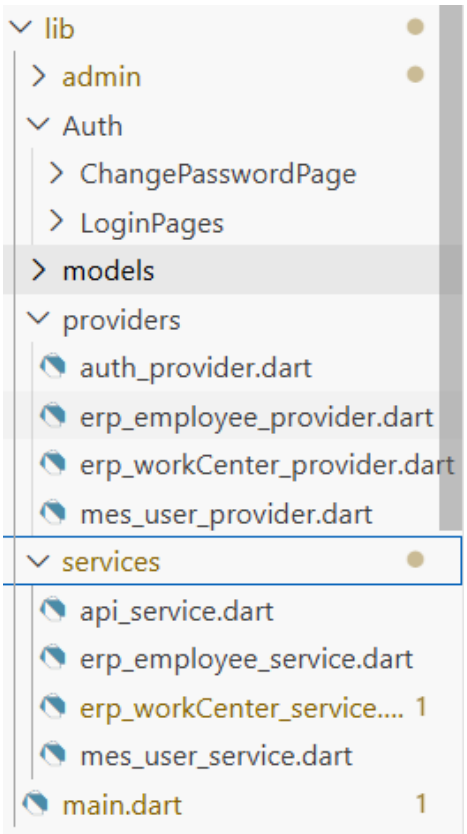

Figure 3.16. Flutter layered architecture.

Service Layer (API Communication)

The *Service Layer* is responsible for handling all *HTTP* communication between Flutter and the Business Central backend.

Two services were implemented: the `api_service` manages authentication requests such as *login*, *logout*, and *password change*, while the `mes_user_service` handles MES user operations including fetching and creating new users. A dedicated `setting_service` handles fetching and updating the MES global settings.

`api_service.dart`

The class declaration and constructor show the base URL configuration and *HTTP* client initialisation used by all authentication endpoints. The *login* method sends user credentials and device ID to the `/login` endpoint and parses the returned session token, role, *needChangePassword* flag, and *twoFAEnabled* flag. When *2FA* is enabled, the token is held in memory rather than persisted until badge verification succeeds. The *logout* method sends the active session token to the `/logout` endpoint to trigger server-side token revocation.

The *change password* method submits the current password, new password, and session token to

the `/changePassword` endpoint. Shared error-handling and response-parsing utilities are used by all methods in the service class to propagate structured exceptions to the *State Management Layer*.

`mes_user_service.dart`

The class declaration and *fetch users* method send an authenticated *HTTP GET* request to the `/mesUsers` endpoint and deserialise the response into a list of `MesUser` model objects. The *create user* method sends a *HTTP POST* request to the `/createMesUser` endpoint with the new user's role and employee reference in the request body. The *generate password* method and associated response deserialisers handle the password generation endpoint and map the *JSON* result into the `GeneratedPassword` model.

State Management Layer using Provider

The *Provider* pattern is implemented to ensure reactive UI updates and centralised data handling. Three providers were implemented: `AuthProvider`, which manages authentication state and the 2FA pending flow; `MesUserProvider`, which handles MES user data; and `MesSettingsProvider`, which manages the global settings fetch, dirty-state detection, and save operations. Each provider wraps the corresponding service layer calls and exposes reactive state (authenticated user, user list, settings, loading flag, error message) consumed by the *UI Layer* via `context.watch`.

User Interface

The *Paring Page*, shown in Figure 3.17, is the very first screen presented when the application has no saved host URL. The user enters the middleware URL manually or taps *Scan QR Code* to open the QrScannerDialog and populate the field automatically. On submission the URL is validated and persisted via `AppConstants.changeHost()`.

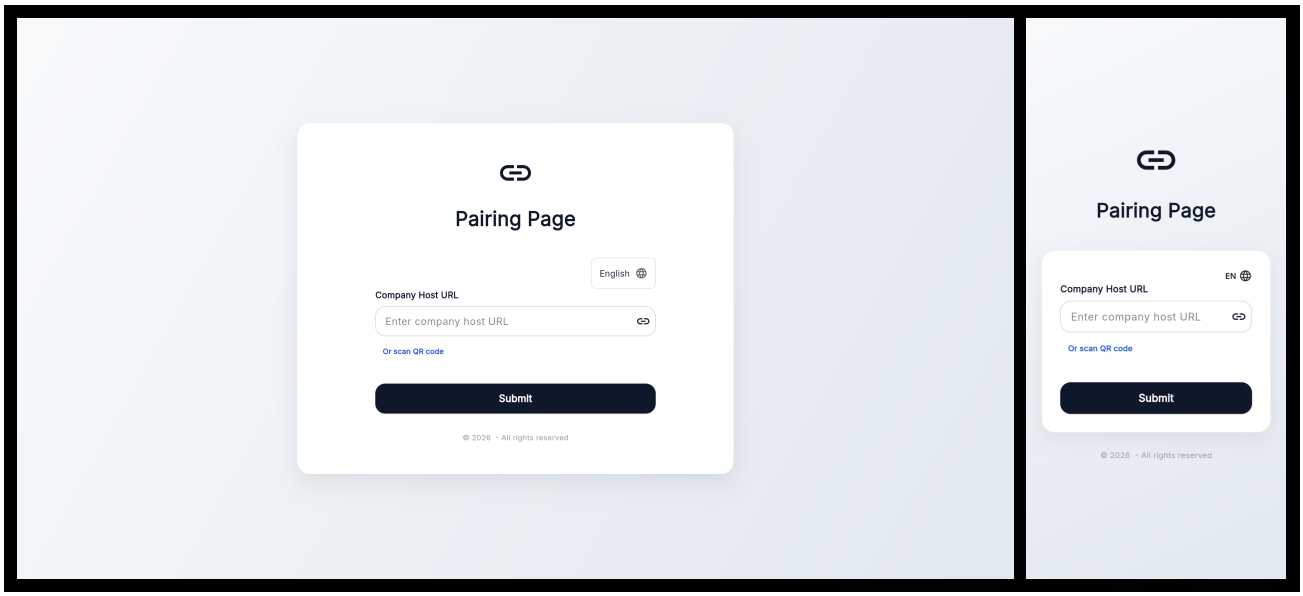

Figure 3.17. *Paring Page* — desktop and mobile views.

The *Login Page*, shown in Figure 3.18, is the application entry screen presented to all users on launch. It contains *User ID* and *Password* input fields and a *Login* button. Successful authentication redirects the user to the role-appropriate dashboard, or to the *Badge Scan Dialog* if *2FA* is enabled.

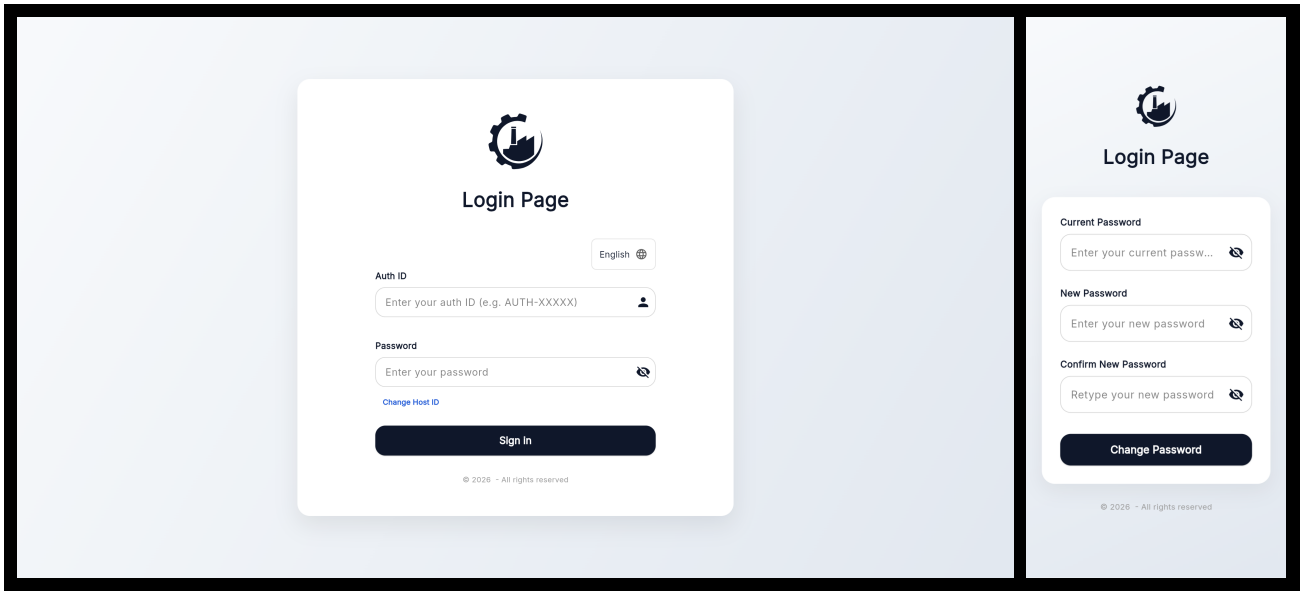

Figure 3.18. *Login Page* — desktop and mobile views.

The *Badge Scan Dialog*, shown in Figure 3.19, is displayed after a successful first-factor login when *2FA* is enabled. A live camera viewport reads the user's badge QR code; a manual entry field is available as a fallback. The dialog displays any verification error and re-enables the scanner on failure. The user can cancel at any time, which clears the pending session state and returns to the login page.

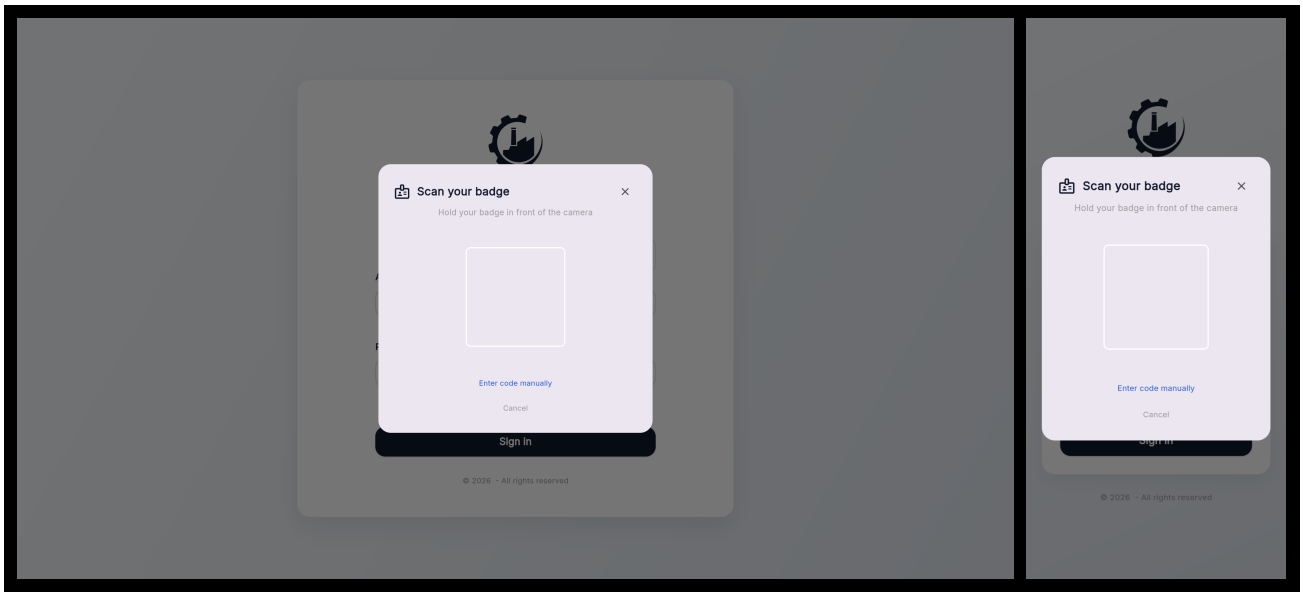

Figure 3.19. *Badge Scan Dialog* — desktop and mobile views.

The *MES User List Page*, shown in Figure 3.20, is the Administrator dashboard screen listing all registered *MES* users with their assigned roles and active status, providing action buttons to assign roles, generate passwords, view or regenerate badges, and activate/deactivate each account.

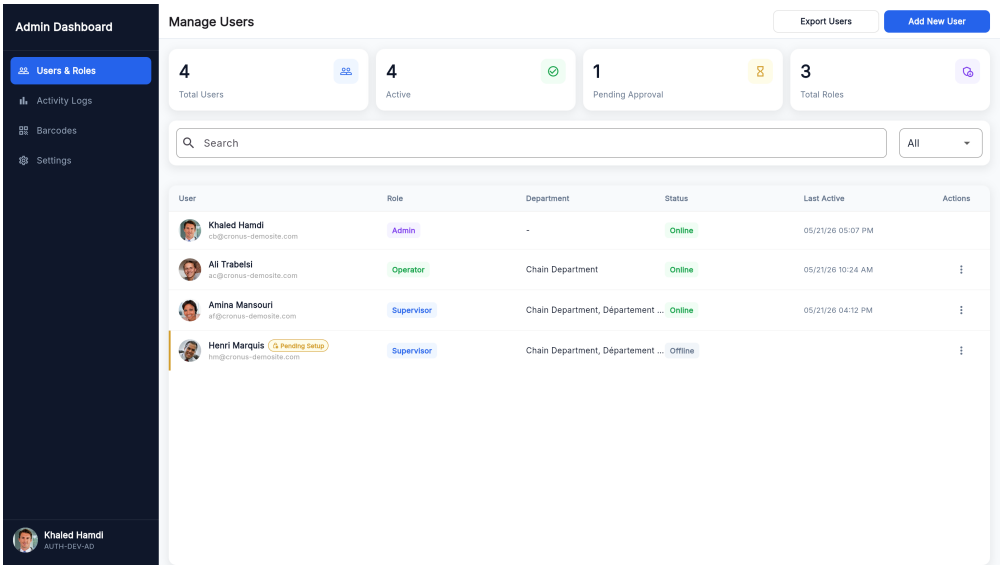

Figure 3.20. *MES User List Page.*

The *Assign MES Role Dialog*, shown in Figure 3.21, is a modal dialog allowing the Administrator to select a Business Central employee record and assign them an *MES* role (Operator or Supervisor), creating a new `MES User` record via the `/createMesUser` endpoint.

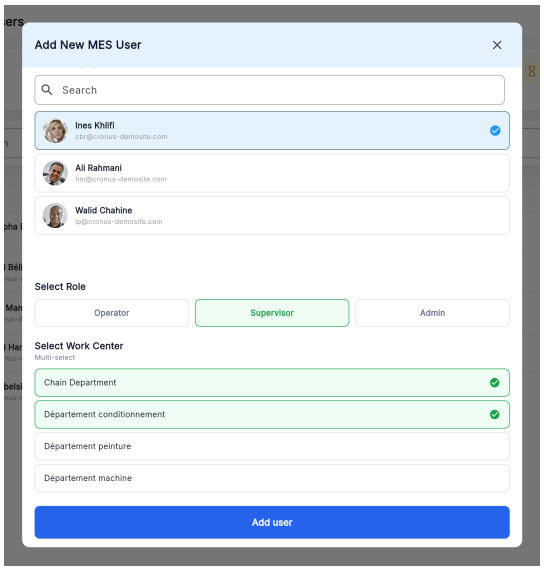

Figure 3.21. *Assign MES Role Dialog.*

The *Generate Password Dialog*, shown in Figure 3.22, is a modal dialog through which the Administrator triggers server-side secure password generation for a selected *MES* user. The generated password is displayed once and must be communicated to the user out-of-band.

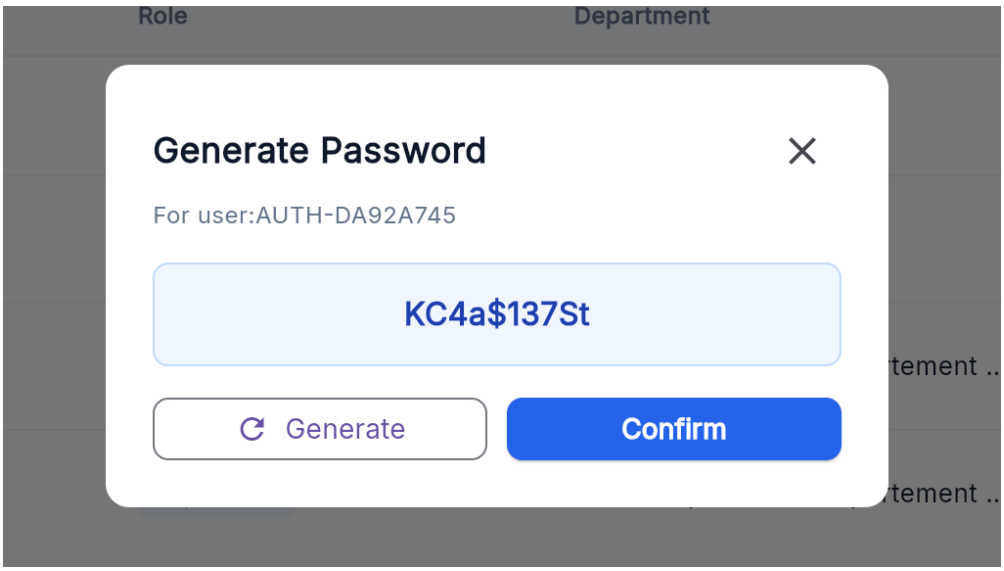

Figure 3.22. *Generate Password Dialog.*

The *Password Generated Successfully Dialog*, shown in Figure 3.23, is a confirmation dialog showing the plaintext password to the Administrator for one-time transmission to the user. The password is not stored in plaintext on the server after this point.

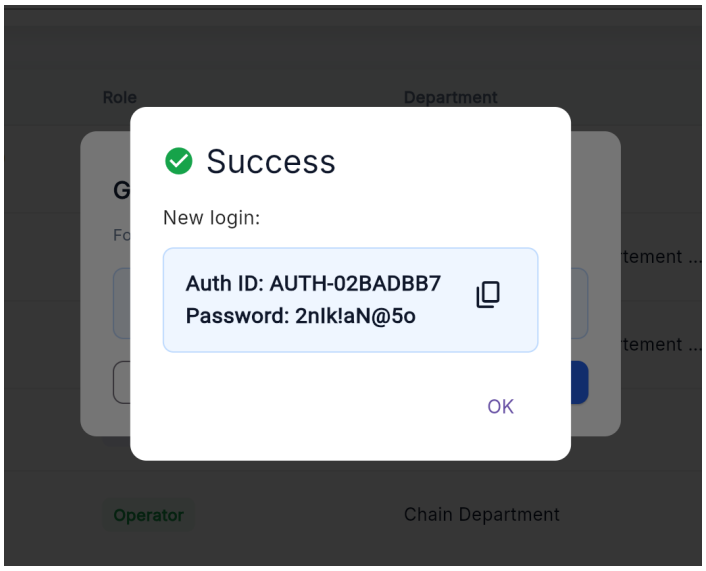

Figure 3.23. *Password Generated Successfully Dialog*.

The *Change Password Page*, shown in Figure 3.24, is presented to users whose *needChangePassword* flag is `true` after login, requiring them to enter their current password and choose a new password before accessing the main application.

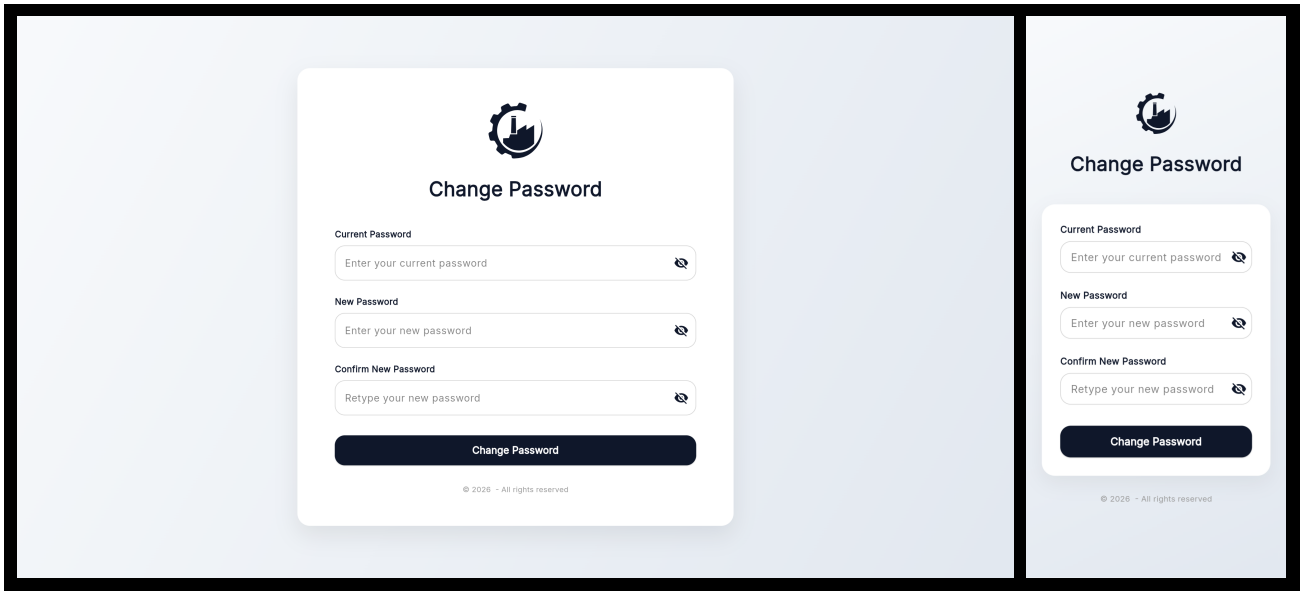

Figure 3.24. *Change Password Page* — desktop and mobile views.

The *Change Role Dialog*, shown in Figure **??**, is a modal dialog through which the Administrator updates the role and work-centre assignment of an existing *MES* user. The dialog pre-populates the current role and presents a three-button row (Operator / Supervisor / Admin); selecting Admin hides the work-centre list, selecting Operator shows a single-select list, and selecting Supervisor shows a multi-select list. An inline error banner displays any server-side validation failure. Tapping *Save Changes* submits the update and revokes all active tokens for the affected user.

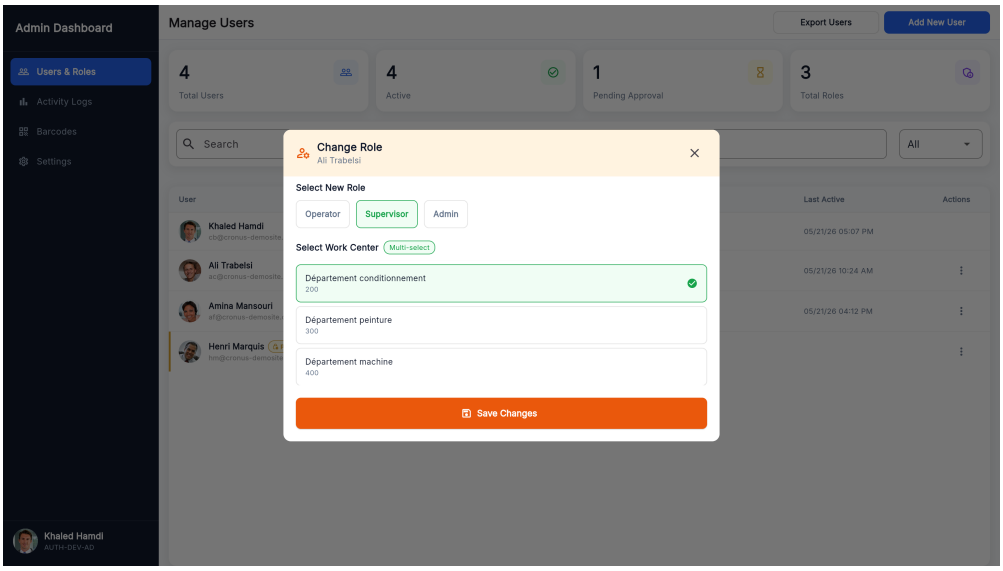

Figure 3.25. *Change Role Dialog.*

The *User Badge Dialog*, shown in Figure 3.26, allows the Administrator to view a user's badge QR code, print it as a PDF (with landscape/portrait toggle), and regenerate the underlying secret after confirmation. It is accessible via the *View Badge* action in the user row action menu and is available only for active accounts.

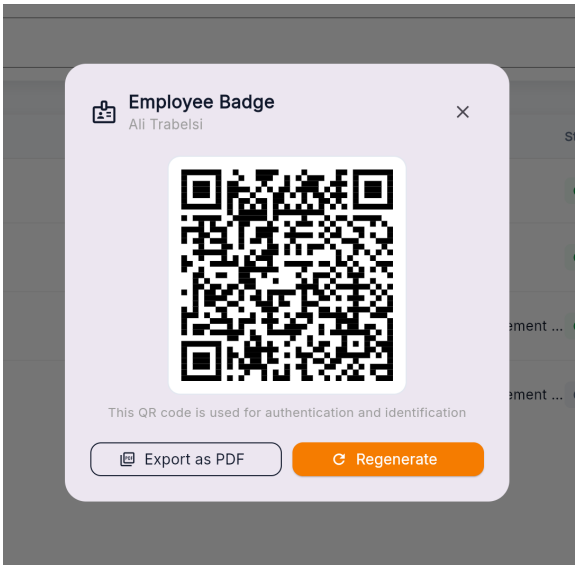

Figure 3.26. *User Badge Dialog.*

The *Settings Page*, shown in Figure 3.27, is accessible from the Administrator sidebar. It exposes two controls: a numeric field for the password rotation period (in days) and a toggle for global *2FA*. An animated action banner appears at the bottom of the page whenever the values differ from the last saved state, offering *Discard* and *Save* actions.

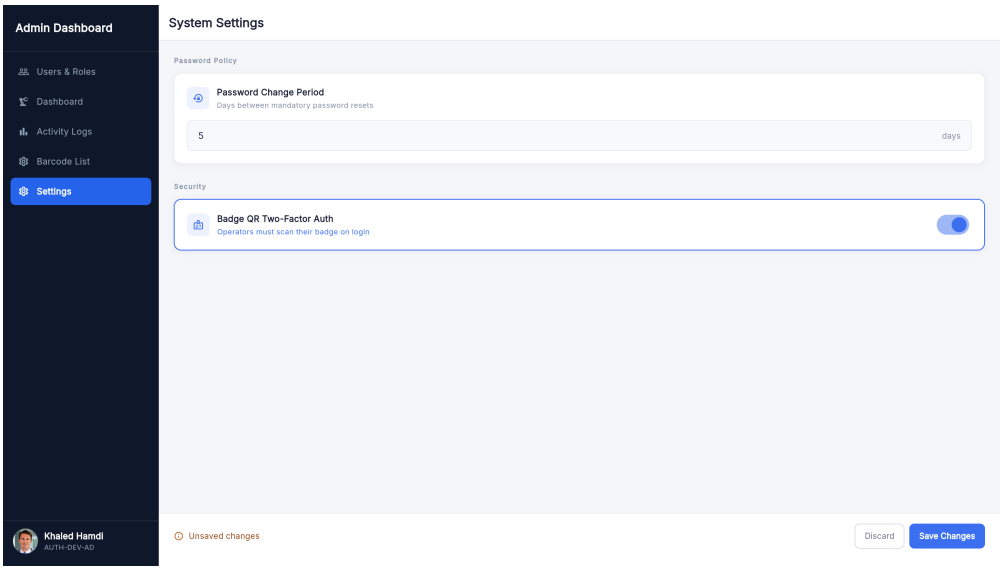

Figure 3.27. *Settings Page.*

3.4 Testing and Validation

3.4.1 Integration testing

For integration testing, we chose Postman, a powerful and user-friendly tool for API evaluation. Figure 3.28 shows a test case for the `/mesUsers` endpoint.

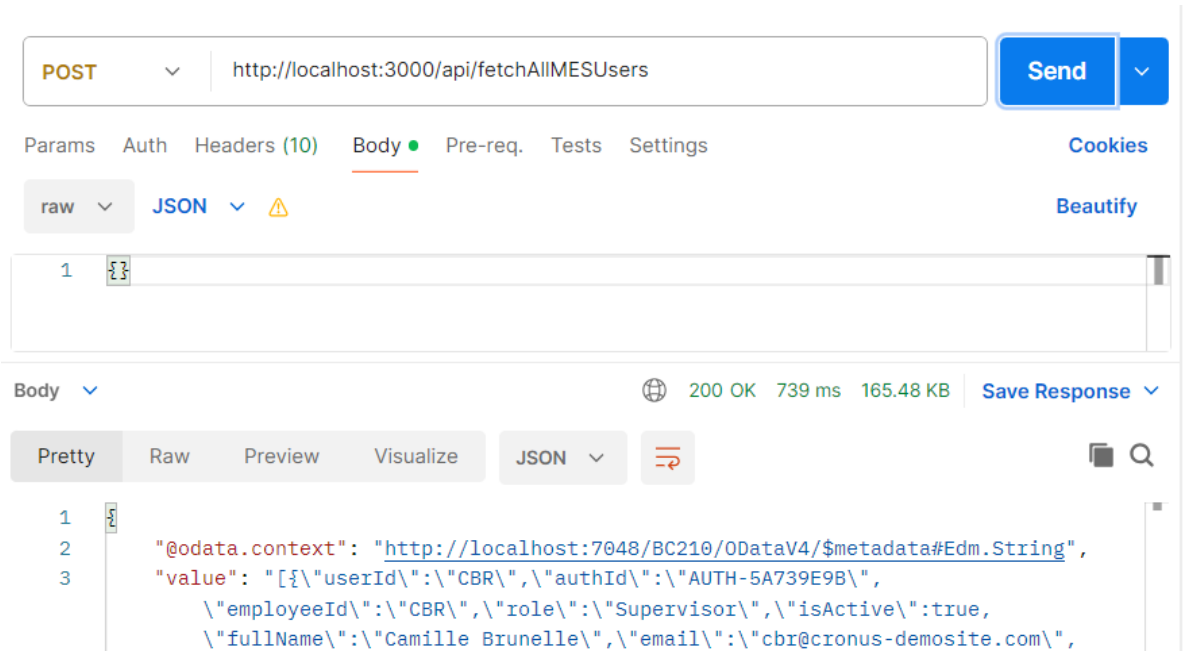

Figure 3.28. Postman endpoint test — fetchAllMesUsers.

Conclusion

This sprint successfully delivered a complete authentication and user management foundation for the *MES* application. The backend, built in Microsoft Dynamics Business Central using

AL, implements secure credential validation, *SHA-256* password hashing, session token management with automatic expiry, optional two-factor authentication via badge QR code scanning, a configurable password expiry policy enforced by a scheduled background job, and a full set of *RESTful* API endpoints exposed through MES Web Service (codeunit 50126). The Flutter frontend provides a clean, role-aware user experience covering first-launch host pairing, login, badge QR verification, logout, forced password change, administrator-level user creation, badge management (view, print, regenerate), role and work-centre management, account activation control, and security policy configuration.

Chapter 4

Sprint 2 Machine & Production Management

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Introduction

This sprint covers **Machine Monitoring, Production Order Management, and Operation History** in the *MES* application, giving Operators and Supervisors real-time visibility into machine states and a full audit trail of completed operations.

It addresses three related sub-domains: live machine status tracking from Business Central to the Flutter frontend, production order management with operation lifecycle initiation, and a history view showing all completed and cancelled operations per machine.

4.1 Sprint Backlog

This sprint covers **Domain 2:** ***Machine Monitoring, Production Order Management & History***.

Table 4.1. Sprint Backlog

ID	User Story	Tasks
US-2.1	As an Operator or Supervisor, I need to see the list of machines in my work centre with their current status so that I can see which machines are currently active.	*Business Central (AL):* • Create table MES MachineStatus. • Create enum MES Machine Status. • Create function `FetchMachines()` in codeunit MES Machine Fetch. • Expose `FetchMachines()` through codeunit MES Web Service. • Create page MES Machine List. *Flutter:* • Create model MachineModel. • Create service MESMachineListService to consume the `FetchMachines()` API. • Create provider MesMachinesProvider. • Create page MachineListPage. • Create widget MachineCard. • Implement automatic machine list refresh in MESMachineListService.
US-2.2	As an Operator or Supervisor, I need to view the production orders assigned to a machine so that I know what work needs to be done.	*Business Central (AL):* • Create function `getMachineOrders()` in codeunit MES Machine Fetch. • Expose `getMachineOrders()` in codeunit MES Web Service.

Continued on next page

ID	User Story	Tasks
		Flutter: • Create model MachineOrderModel. • Create service ErpMachineOrdersService to consume the `getMachineOrders()` API. • Create provider MachineordersProvider. • Create page MachineOrderPage. • Create widget OrderCard. • Create model BadgeStyle.
US-2.3	As an Operator or Supervisor, I need to start a production operation so that the system records that the order has begun.	*Business Central (AL):* • Create table MES Operation state. • Create enum MES Operation Status. • Create function `TryStartOperation()` in codeunit MES Machine Validation. • Create function `InsertMESOperation()` in codeunit MES Machine Write. • Expose `InsertMESOperation()` through codeunit MES Web Service. • Create page MES Operations. *Flutter:* • Implement ErpMachineOrdersService that Consumes the `startOperation` API. • Implement MachineordersProvider to manage its state. • Implement start operation action handling in ActionButtons. • Implement automatic navigation to OrdersProgressionPage
US-2.4	As a Supervisor or Operator, I need to monitor the current status of all active operations on a machine so that I can follow the progress of each order.	*Business Central (AL):* • Create function `fetchMachineOperationStatus()` in codeunit MES Machine Fetch. • Expose `fetchMachineOperationStatus()` through codeunit MES Web Service.

Continued on next page

ID	User Story	Tasks
		Flutter: • Update ErpMachineOrdersServiceto Consume the `fetchMachineOperationStatus`. • Create model OperationStatusStyle. • Create page OrdersProgressionPage. • Create the widgets OperationCard, OperationStatusBadge, widget OperationInfoGrid, and OperationProgressBar. • Implement automatic refresh for operation status monitoring.
US-2.5	As an Operator or Supervisor, I need a tabbed view for a machine so that I can navigate between pending orders and active operation progress without leaving the machine context.	*Flutter:* • Create page MachineMainPage. • Create the widgets TopNavigationBar and NavButton. • Implement tab navigation in MachineMainPage. • Implement navigation from MachineCard to MachineMainPage.
US-2.6	As an Operator or Supervisor, I need to filter and sort production orders so that I can quickly find the most relevant work.	*Flutter:* • Create widget GlobalSearchBar. • Implement order search, filtering, and sorting in MachineOrderPage.
US-2.7	As a Supervisor or Operator, I need to view the history of completed operations for a machine so that I can audit past production.	*Business Central (AL):* • Create function `fetchOperationsStatusAndProgress()` in codeunit MES Machine Fetch and Expose it in codeunit MES Web Service.

Continued on next page

ID	User Story	Tasks
		Flutter: • Update ErpMachineOrdersService to consume the `fetchMachineOperationStatusAndProgress` API. • Update MachineordersProvider to manage its state. • Create page MachineHistoryPage. • Create the widgets HistoryCard, HistoryBadgeAndId, and HistoryInfoGrid. • Implement history search and sorting in MachineHistoryPage. • Implement navigation from HistoryCard to OperationDetailPage.

4.2 Design

4.2.1 Use Case Diagram

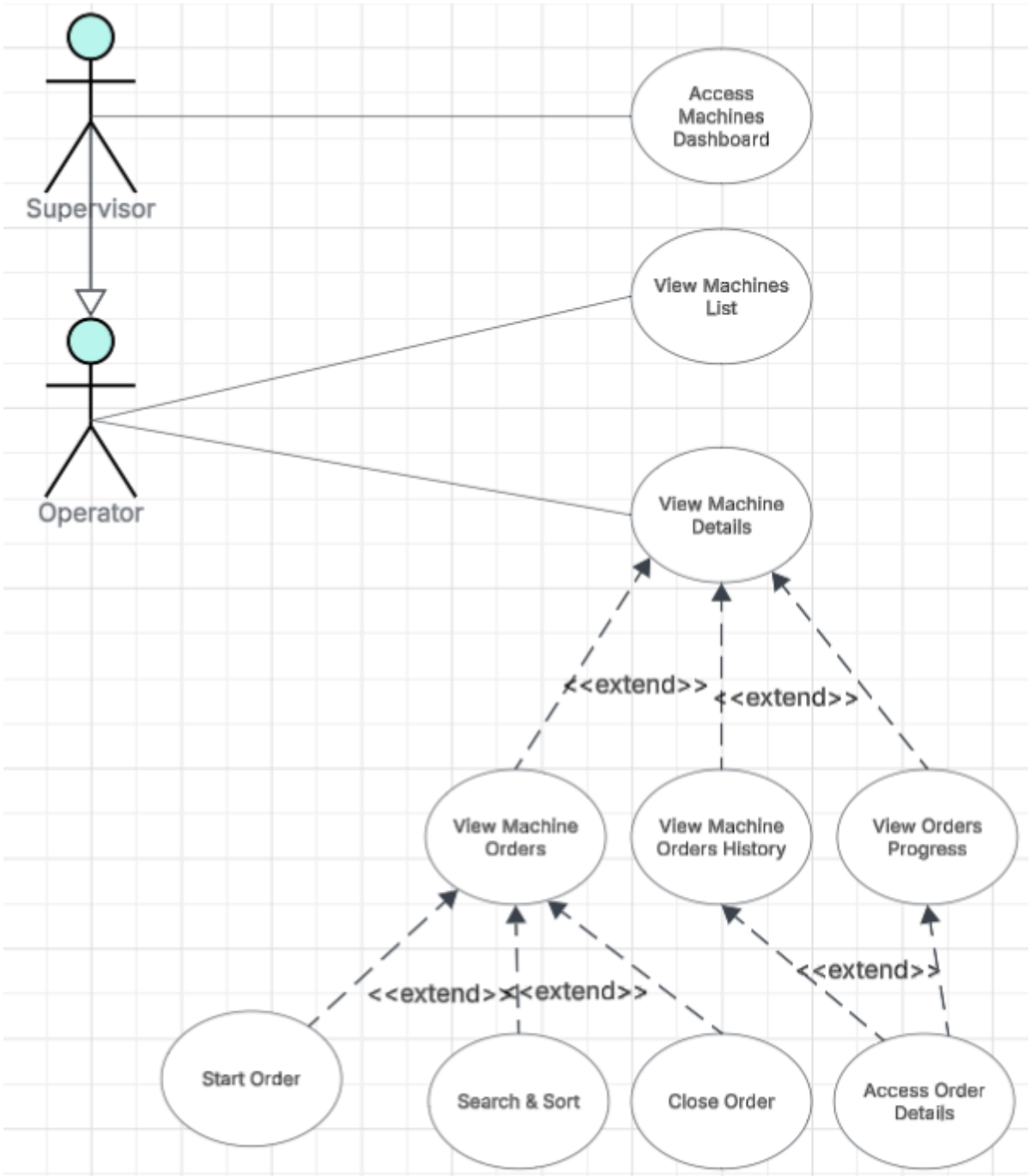

Figure 4.1. UML Use Case Diagram — Machine & Production Management.

The UML Use Case Diagram (Figure 4.1) shows the two primary actors (Operator and Supervisor) and their interactions with the machine monitoring, production order management, and history subsystem. In addition to viewing machines, browsing orders, and managing operation lifecycle transitions, the diagram includes the *View Operation History* use case, which allows both actors to access the completed and cancelled operation log for any machine in their work

centre.

4.2.2 Textual Use Case Description

The following table (Table 4.2) provides the detailed textual description of the *Start Operation* use case.

Table 4.2. Textual Use Case Description — Start Operation

Use Case Name	Start Operation
Primary Actor	Operator
Preconditions	• The Operator is authenticated and has a valid session token. • The target routing line has status *Released*. • No other operation is currently *Running* on the same machine. • The operation has not already been started (no existing *Running* record for this order/operation pair).
Postconditions	• A new `MES Operation Status` record is inserted with status *Running* and the current *UTC* timestamp. • The *Operations Progress* tab reflects the new running operation within the next polling cycle. • If any precondition fails, a descriptive error dialog is shown to the operator and no record is inserted.
Main Scenario	1. The Operator navigates to the *Machine List Page* and selects a machine. 2. The system displays the *Machine Orders Page* listing all applicable production orders. 3. The Operator locates a *Released* order and taps *Start Order*. 4. The frontend sends a *POST* request to `MESWebService_startOperation`. 5. The backend validates all preconditions. 6. On success, a `MES Operation Status` record is inserted. 7. The frontend navigates to the *Orders Progression Page*.

4.2.3 Sequence Diagrams

The following sequence diagrams illustrate the communication flow between the Flutter frontend, the Business Central OData layer, and the AL business logic for the machine-listing and order-management flows introduced in Sprint 2.

Figure 4.2 depicts the machine-list polling flow. The Flutter client issues a `FetchMachines` request for each work centre resolved from the authenticated user's profile every five seconds. For every machine returned by Business Central, the AL layer fetches the latest *MES Machine Status* row and the work-centre name before assembling the response, which the client uses to rebuild the *MachineCard* grid.

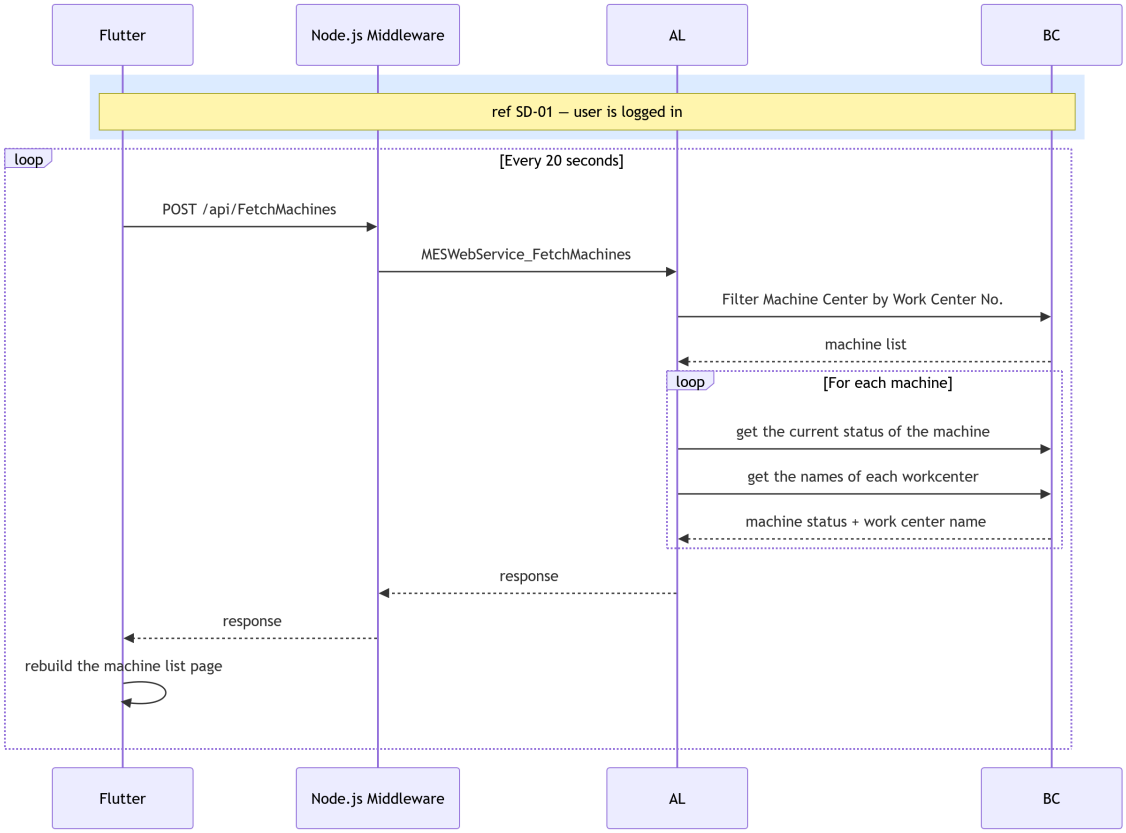

Figure 4.2. SD-09: Fetch Machines.

Figure 4.3 presents the machine-orders retrieval flow, triggered when the user opens the *Orders* tab. The AL layer filters *ProdOrderRoutingLine* records for the selected machine, skips any operation that has already been started, and enriches each remaining routing line with the corresponding *ProdOrderLine* details before returning the list to the Flutter client as a set of *OrderCard* entries.

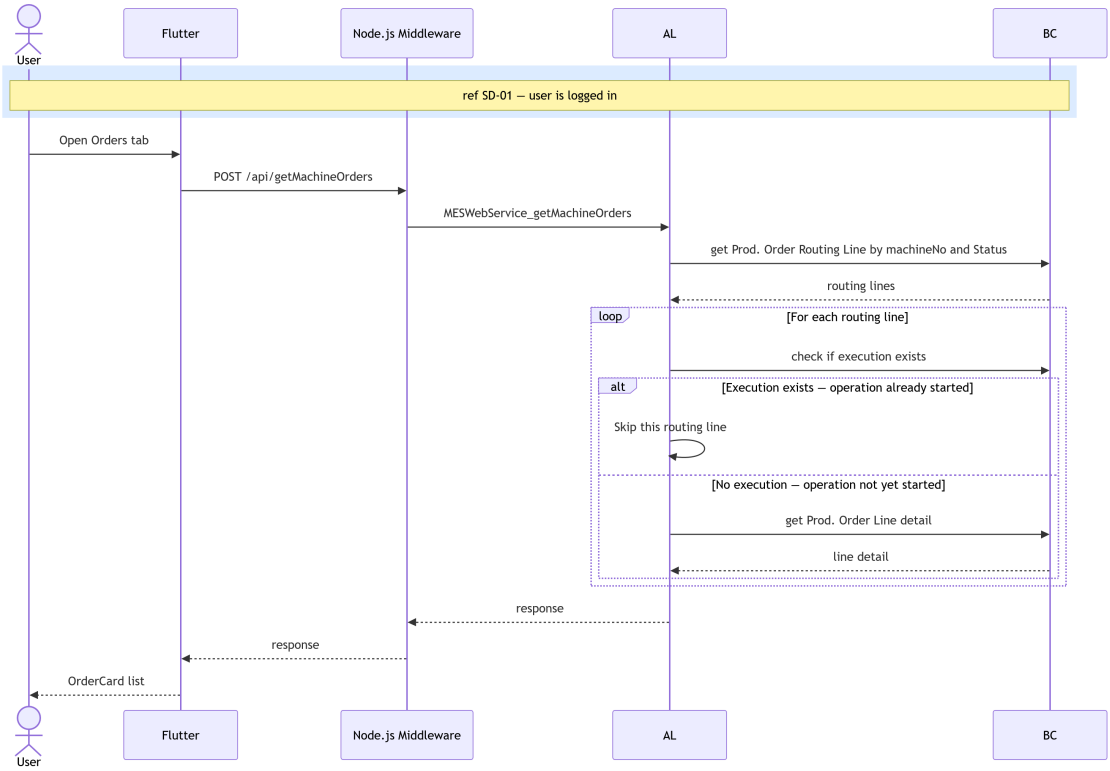

Figure 4.3. SD-10: Get Machine Orders.

Figure 4.4 illustrates the operation start flow. After the user selects a released order and taps *Start*, the AL layer confirms that the order is in the *Released* state and that neither the operation nor the machine is currently running. On success, it creates the *MES Operation Execution*, *MES Operation State*, and *MES Operation Progression* records, marks the machine as working, and records the initial user interaction before redirecting the Flutter client to the *Progress* tab.

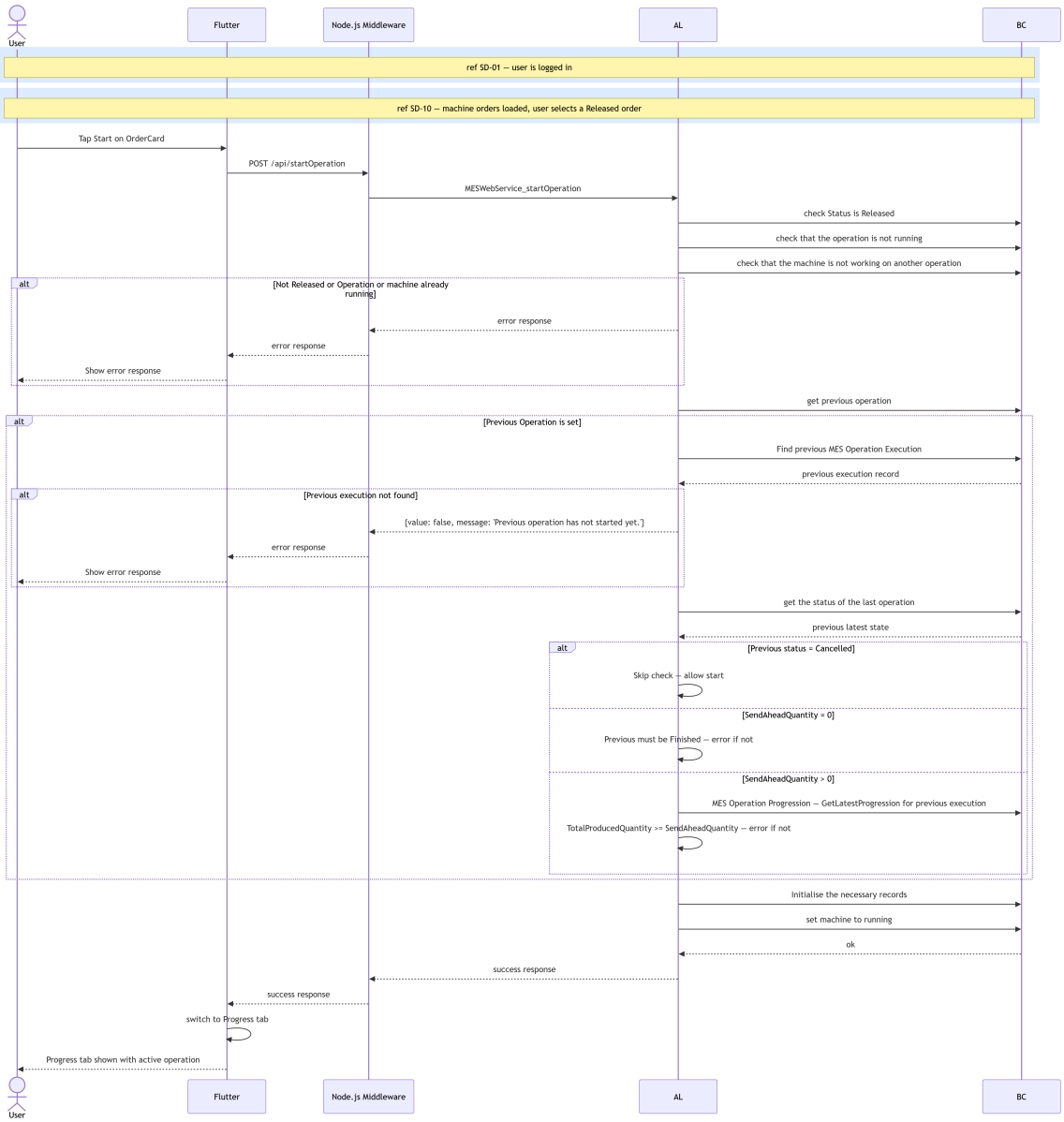

Figure 4.4. SD-11: Start Operation.

Figures 4.5 and 4.6 present the two polling flows that supply real-time and historical data to the *MachineDetailPage*. Both are triggered either on demand or on a fixed five-second interval once an operation is active.

Figure 4.5 describes the ongoing-operations state polling flow. The AL layer retrieves all active operations for the machine, resolves their current state and cumulative produced quantity from Business Central, and returns the aggregated data to the Flutter client.

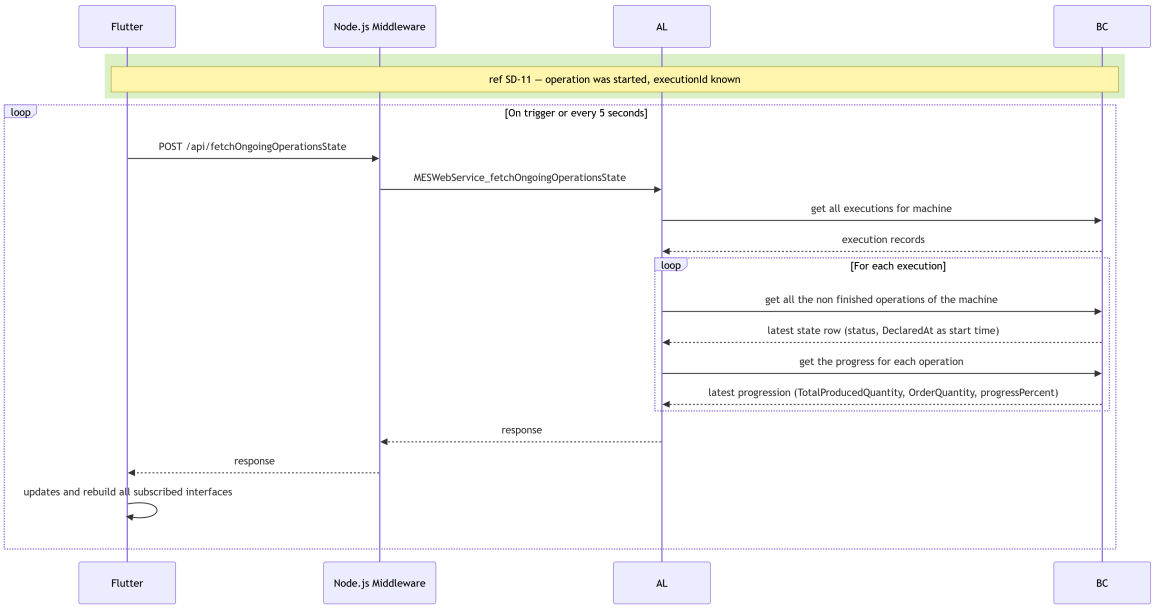

Figure 4.5. SD-23: Fetch Operation progress.

Figure 4.6 presents the operation history retrieval flow, triggered when the user opens the *MachineDetailPage*. The AL layer queries Business Central for finished and cancelled operations associated with the machine, enriches each record with production quantities and start and end times, and returns the assembled list so the Flutter client can display the operations history.

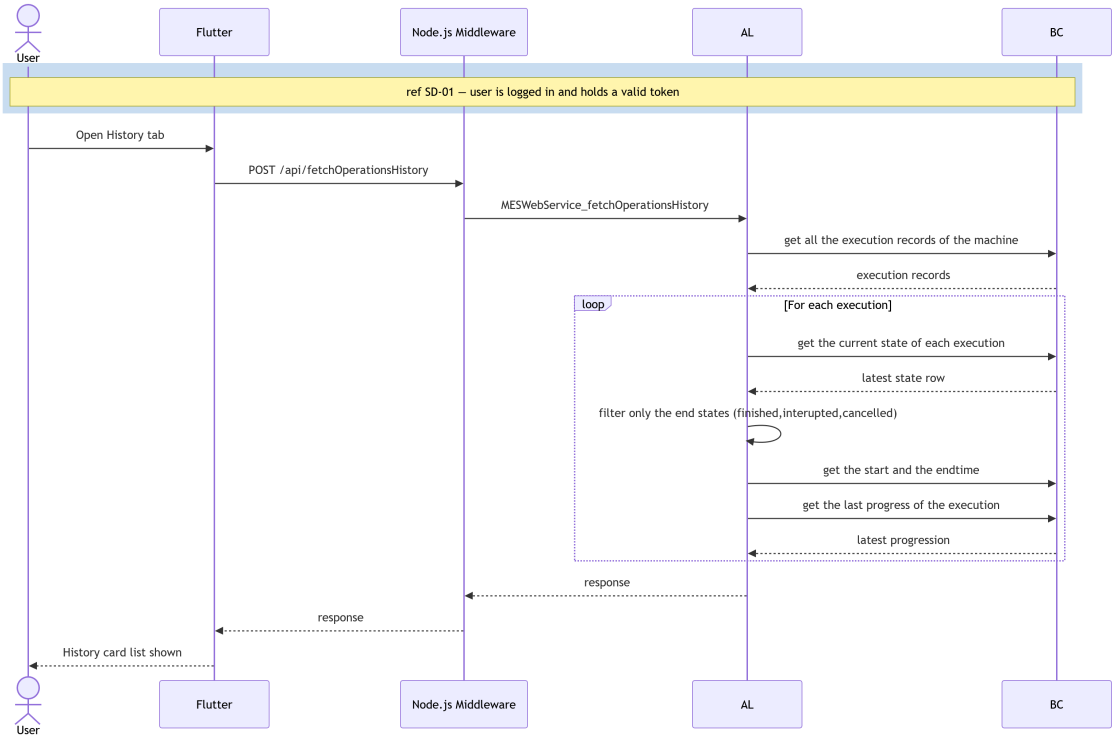

Figure 4.6. SD-24: Fetch Operation History

4.2.4 Class Diagram

The UML Class Diagram (Figure 4.7) of the machine and production management domain shows the `MES Machine Status` and `MES Operation Status` tables and their relationships with the Business Central standard tables `Machine Center` and `Prod. Order Routing Line`.

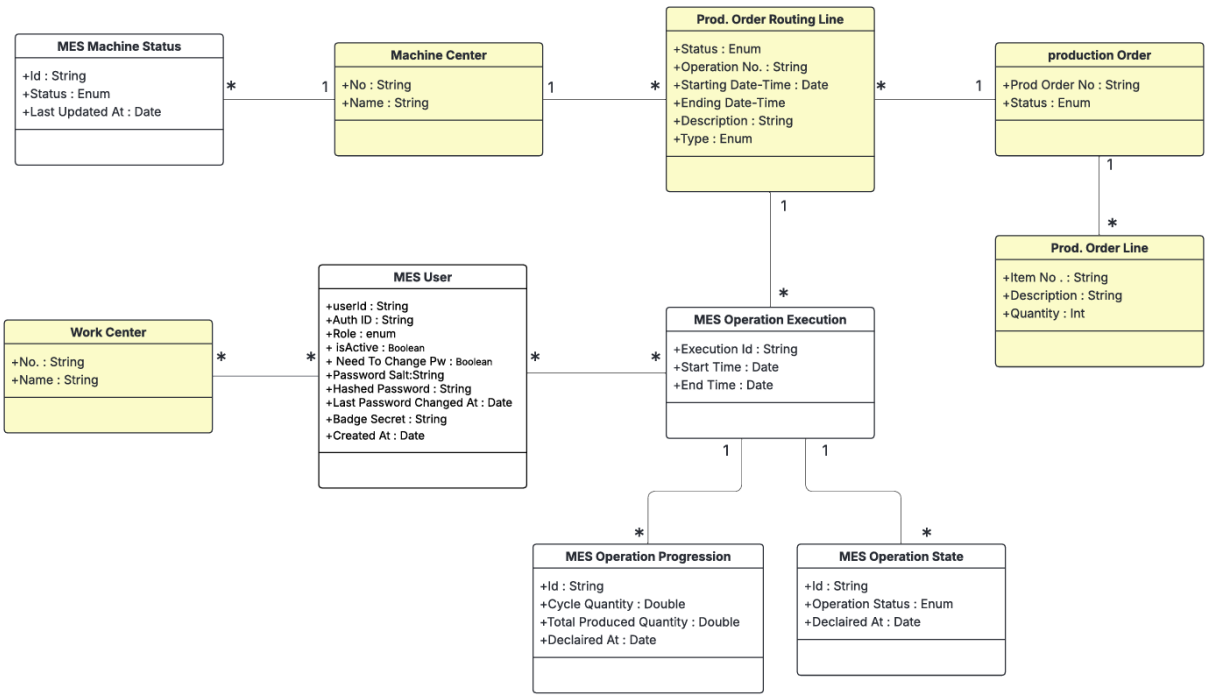

Figure 4.7. UML Class Diagram — Machine & Production Management.

4.3 Implementation

4.3.1 Backend Implementation

Backend Structure Overview

The machines domain follows the same layered structure established in Sprint 1. Two new subfolders were added under `src/1-Tables/machines/` and `src/2-Enum/machines/`, and a new codeunit was placed under `src/3-CodeUnit/machines/`. The OData web service is published through the existing `WebServices.xml` mechanism under a new service name.

MES Machine Status Table

The `MES Machine Status` table (50107) stores the full history of machine state changes as an append-only log. Each insert creates a new record; the most recent record per machine represents the current state. Table 4.3 lists the fields and their descriptions.

Table 4.3. MES Machine Status (Table 50107) — Field Dictionary

Name	Type	Description
Id	Code[50]	Primary key. GUID-derived value assigned automatically on insert if blank.
Machine No.	Code[20]	Foreign key to `Machine Center."No."`. Identifies the machine this status record belongs to.
Status	Enum (`MES Machine Status`)	Current machine state: *Idle* or *Working*.
Current Prod. Order No.	Code[20]	Foreign key to `Prod. Order Routing Line."Prod. Order No."`. Records which production order the machine is currently executing, if any.
Updated At	DateTime	UTC timestamp set automatically on insert. Used to find the most recent status record for a given machine.

MES Operation Status Table

The `MES Operation Status` table (50108) implements the same append-only pattern for operation lifecycle events. Every state transition (start, pause, resume, finish) inserts a new record; queries always sort by `Last Updated At` descending to obtain the current state. Table 4.4 provides the full field dictionary.

Table 4.4. MES Operation State (Table 50108) — Field Dictionary

Name	Type	Description
Id	Code[50]	Primary key. GUID-derived identifier assigned automatically on insert.
Execution Id	Code[50]	Foreign key to `MESOperationExecution`.
Operator Id	Code[50]	Foreign key to `MES User."User Id"`. The operator who triggered the state transition.
Operation Status	Enum (`MES Operation Status`)	Lifecycle state at the time of this record: *Running*, *Paused*, *Finished*, *Cancelled*, or *Interrupted*.

Continued on next page

Name	Type	Description
Declared At	DateTime	UTC timestamp set automatically on insert. Together with the execution index, this field is used to retrieve the current status via descending sort.

Key points:

- Both tables use an append-only design: no `Modify` is called on existing records.
- The *secondary index* `ExecutionTimeline("Execution Id", "Last Updated At")` on `MES Operation State` supports efficient retrieval of the latest event per group.

Enumerations

Two enumerations support the domain. `MES Machine Status` (Enum 50101) defines two values: *Idle* and *Working*. `MES Operation Status` (Enum 50102) defines five values: *Running*, *Paused*, *Finished*, *Cancelled*, and *Interrupted*. Both enumerations are declared `Extensible = true` to allow future additions without modifying the base objects.

Machine Actions Business Logic

Machine and operation business logic is encapsulated across two codeunits: `MES Machine Fetch` (50131) for read operations and `MES Machine Write` (50132) for write operations. Their public procedures are called directly by the OData web service layer.

FetchMachines. Accepts a *JSON array of work centre numbers* (`workCenterNoJson`) and returns a *JSON* array of machine objects. For each `Machine Center` record scoped to the given work centre, the procedure looks up the latest `MES Machine Status` record via `FindLast()` and inlines the current status and active production order number into the response object. If no status record exists, the machine is reported as *Idle*.

getMachineOrders. Accepts a machine number and returns a *JSON* array of production routing lines whose status is *Planned*, *Firm Planned*, or *Released* and whose type is *Machine Center*. Routing lines that already have an existing `MES Operation Execution` record are excluded from the result, ensuring that only work not yet started is shown in the order list. For each qualifying line, the corresponding `Prod. Order Line` is joined to supply item description and order quantity.

startOperation. Accepts a production order number, operation number, and machine number. The procedure delegates precondition checking to a `[TryFunction]` (`TryStartOperation`) that verifies the routing line is *Released*, confirms no *Running* record exists for the same order/operation, and checks that the machine is not already busy with another operation. If all checks pass, a new `MES Operation Status` record with status *Running* is inserted by

`InsertMESOperation`. On failure, the last error text is captured and returned as a structured *JSON* error response.

fetchOngoingOperationsState. Returns the current active operations for a machine as a *JSON* array. The procedure sets a descending composite key on (Machine No, Prod Order No, Operation No, Last Updated At) and iterates the result set, emitting at most one entry per (order, operation) pair — the one with the most recent timestamp. Only entries whose latest status is *Running* or *Paused* are included in the output.

4.3.2 REST API and Web Services Implementation

Communication between Flutter and Business Central for the machines domain uses the same *OData V4 Unbound Actions* pattern established in Sprint 1.The procedures are exposed through the existing `MESWebService` (codeunit 50126) registered in `WebServices.xml`. After deployment, the four procedures are reachable at the following endpoints:

- `POST .../ODataV4/MESWebService_FetchMachines`
- `POST .../ODataV4/MESWebService_getMachineOrders`
- `POST .../ODataV4/MESWebService_startOperation`
- `POST .../ODataV4/MESWebService_fetchOngoingOperationsState`

All endpoints follow the same *JSON* request/response convention used by the authentication layer: the request body carries named parameters as top-level *JSON* fields, and the response wraps the AL return value inside a `"value"` field which the Flutter service layer double-decodes.

4.3.3 Frontend Implementation

The machines domain retains the same four-layer call stack introduced in Sprint 1 (*Model* → *Service* → *Provider* → *UI*), but Sprint 2 introduced a significant reorganisation of the *UI layer* file structure, making the codebase easier to maintain and facilitating code reuse across components.

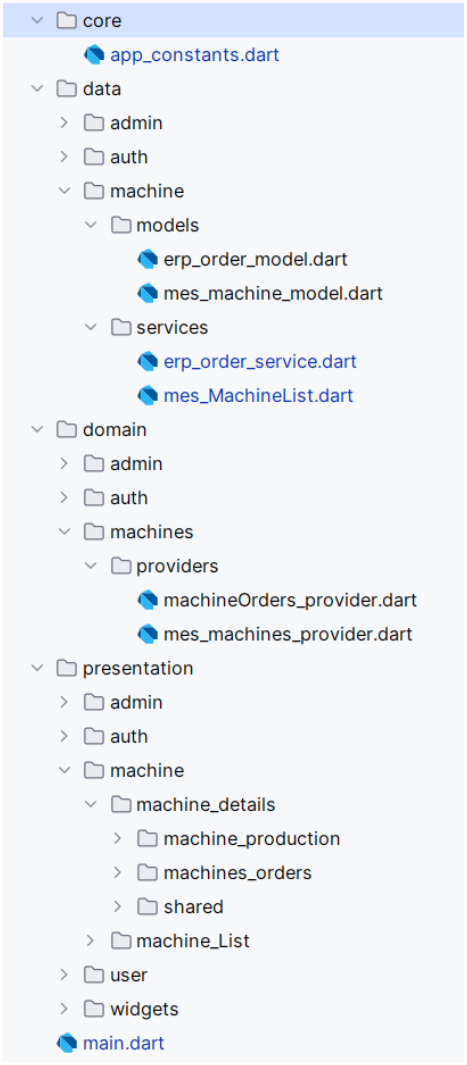

Figure 4.8. Flutter machines domain — layered architecture.

Model Layer

Two models were introduced. `MachineModel` (`mes_machine_model.dart`) holds the machine number, name, current status string, and active order number, populated from the `FetchMachines` response. `MachineOrderModel` (`erp_order_model.dart`) holds the full set of production order fields returned by `getMachineOrders`, including order number, status, operation number, planned start and end *DateTime* values, item number, item description, order quantity, and operation description.

Service Layer

Two service classes handle all network communication for the machines domain. `MESMachineListService` (`mes_MachineList.dart`) exposes a `fetchMachines` method that posts the work centre number and parses the double-encoded *JSON* array into a list of `MachineModel` objects. It also exposes a `streamMachines` method that wraps `fetchMachines` in an `async*` generator loop with a 20-second `Future.delayed` delay, emitting an updated list on every cycle and yielding an empty list on error to prevent the *StreamBuilder* from entering an error state.

`ErpMachineOrdersService` (`erp_order_service.dart`) exposes three methods: `getMachineOrders` for fetching the order list, `getStartOperationValidation` for triggering the start operation endpoint and parsing the success/failure result, and `fetchMachineOperationStatus` for retrieving active operation records. A `streamMachines` generator on this service applies the same 5-second polling pattern for the operations progress view.

State Management Layer

Two providers were introduced. `MesMachinesProvider` (`mes_machines_provider.dart`) is a plain (non-`ChangeNotifier`) provider that delegates directly to the `MESMachineListService` stream. Because all state is carried by the stream itself, no `notifyListeners` mechanism is needed and the *StreamBuilder* in the UI subscribes directly to the emitted values.

`MachineordersProvider` (`machineOrders_provider.dart`) wraps both the order list fetching and the operation start flow with the standard `ChangeNotifier` pattern. It exposes `getMachineOrders` which populates a `machineOrders` list, and `startOrder` which calls the service and returns a boolean success flag. It also exposes `getMachineOperationsStatusStream` which passes the polling stream through directly to the UI layer.

User Interface

The *Machine List Page* is the entry point for operators after login. It displays a live-updating grid (tablet/desktop) or list (mobile) of machine cards scoped to the authenticated work centre. Each card shows the machine name, its current status with a colour-coded badge, and the active production order number. The layout adapts across breakpoints: a single-column list below 600 dp, a two-column grid below 1024 dp, and a four-column grid above. As illustrated in Figure 4.9, the interface is consistent across both desktop and mobile platforms.

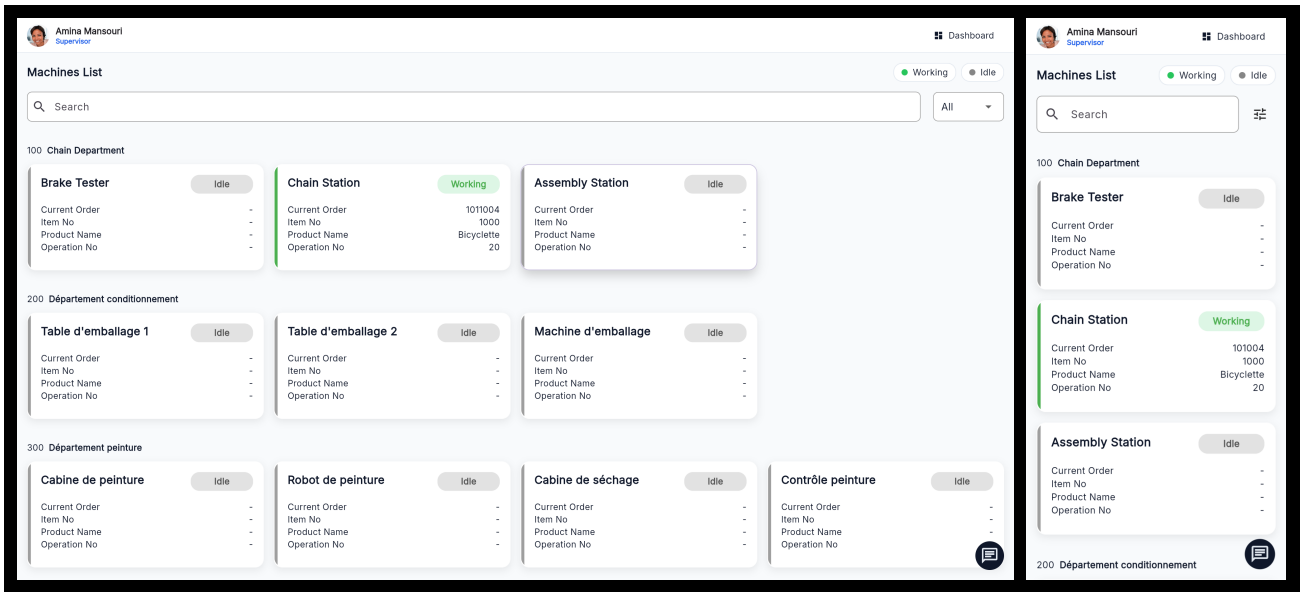

Figure 4.9. *Machine List Page* — desktop and mobile views.

The user interface for the machines domain is split across two top-level pages accessible via a shared *Top Navigation Bar* with tabs for *Orders*, *Progress*, *Consumable*, and *History*.

The *Machine Orders Page* displays the production routing lines assigned to the selected machine. A *Global Search Bar* allows filtering by free text (order number, item description, or date), by

status, and sorting by planned start date. Order cards use a responsive layout that switches between a stacked *narrow layout* and a side-by-side *wide layout* at a 520 dp breakpoint. Only *Released* orders display an active *Start Order* button; *Planned* and *Firm Planned* orders show it disabled. Figure 4.10 illustrates both the desktop and mobile views.

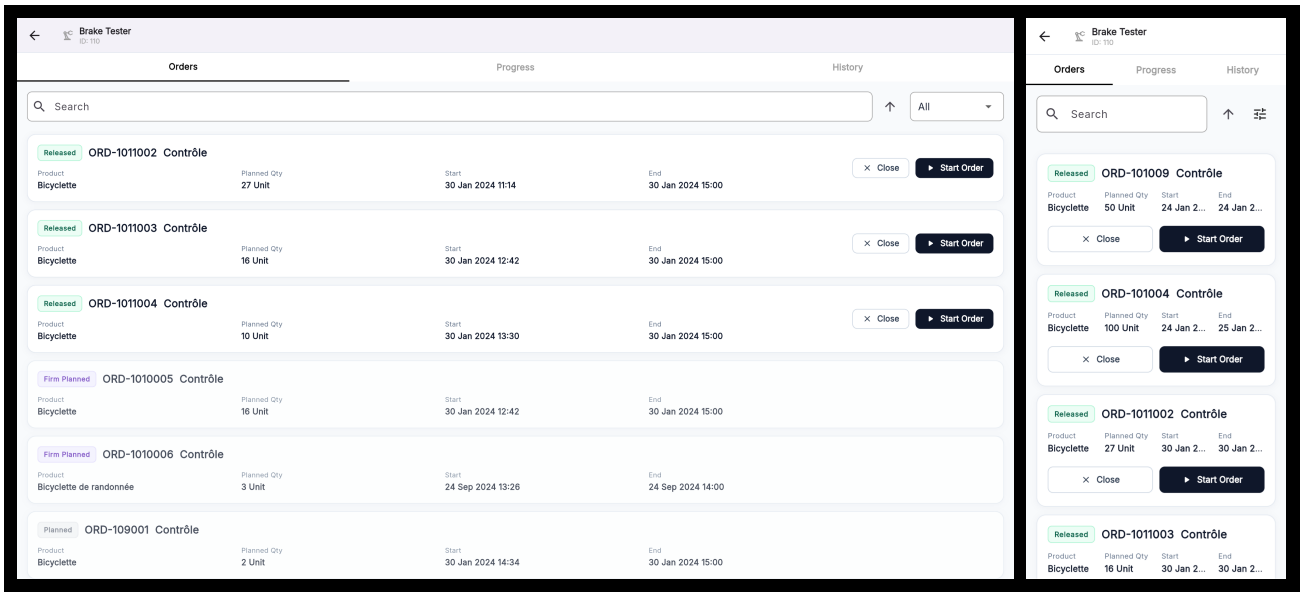

Figure 4.10. *Machine Orders Page* — desktop and mobile views.

The *Orders Progression Page* displays the currently active operations on a machine, polling every 5 seconds. Each operation card shows a colour-coded status badge (green for *Running*, amber for *Paused*), the order and operation identifiers, the last updated timestamp, and a progress bar. The card layout also switches between narrow and wide variants at the 520 dp breakpoint. *Pause* and *Resume* action buttons adapt their colour and label to the current operation status, as shown in Figure 4.11.

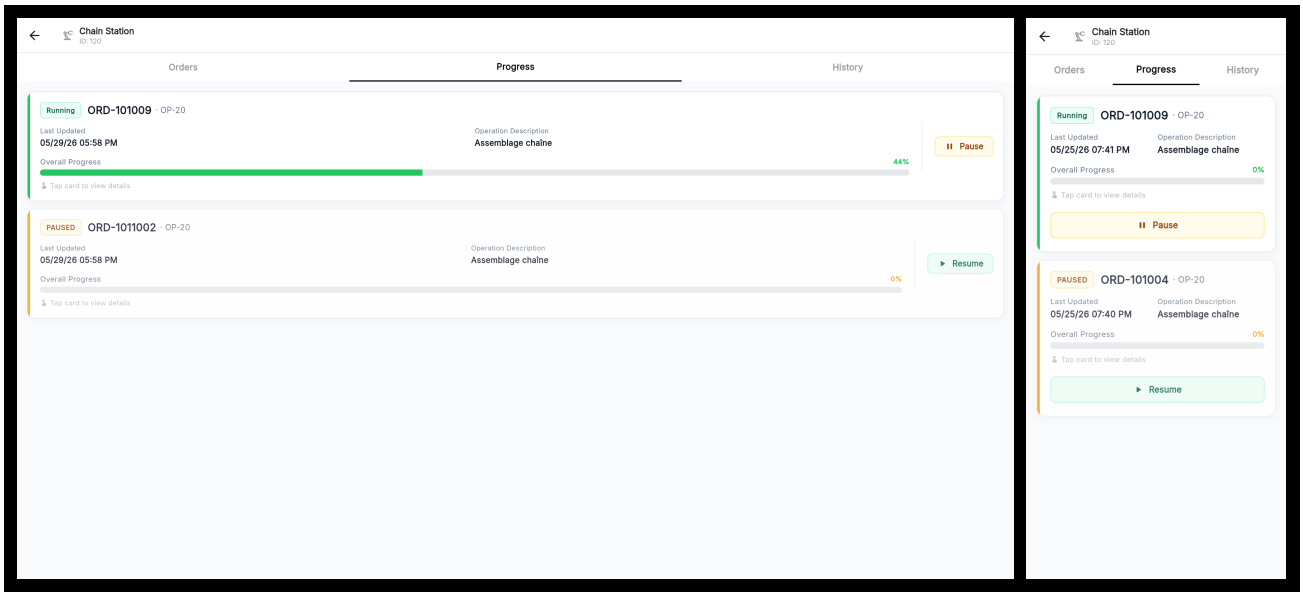

Figure 4.11. *Orders Progression Page* — desktop and mobile views.

The *Machine History Page* (Figure 4.12) displays completed operations assigned to the selected machine, limited to *Finished* and *Cancelled* statuses. It shares the same layout and filtering behaviour as the *Machine Orders Page* — including free-text search, status filtering, and sort controls — but omits the *Start Order* button. Tapping an operation card navigates to its *Operation Detail Page* for review. The *Machine History Page* is accessible via the *History* tab in the TopNavigationBar. Figure 4.12 shows the desktop and mobile views.

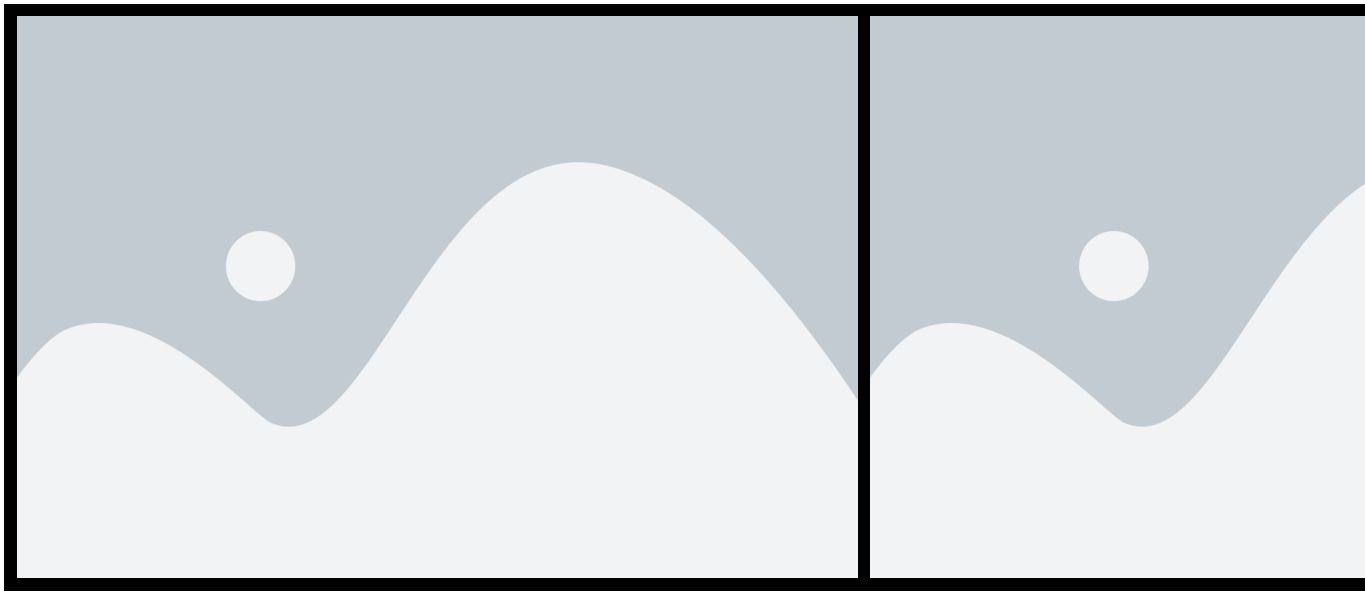

Figure 4.12. *Machine History Page* — desktop and mobile views.

4.4 Testing and Validation

4.4.1 Integrations tests

For integration testing, we chose Postman, a powerful and user-friendly tool for API evaluation. Figure 4.13 shows a test case for the `/fetchmachines` endpoint.

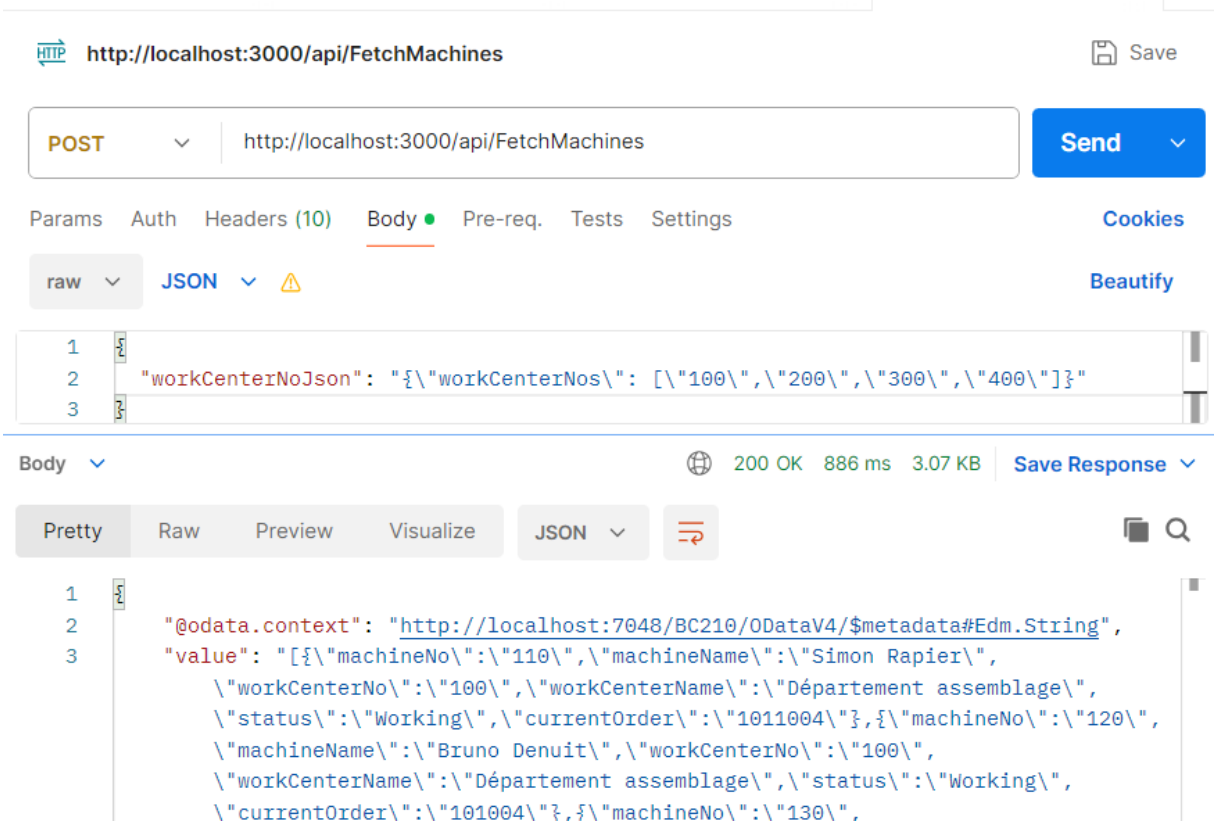

Figure 4.13. Postman endpoint test — fetchMachines.

Figure 4.14 shows a test case for the `/getMachinesOrders` endpoint.

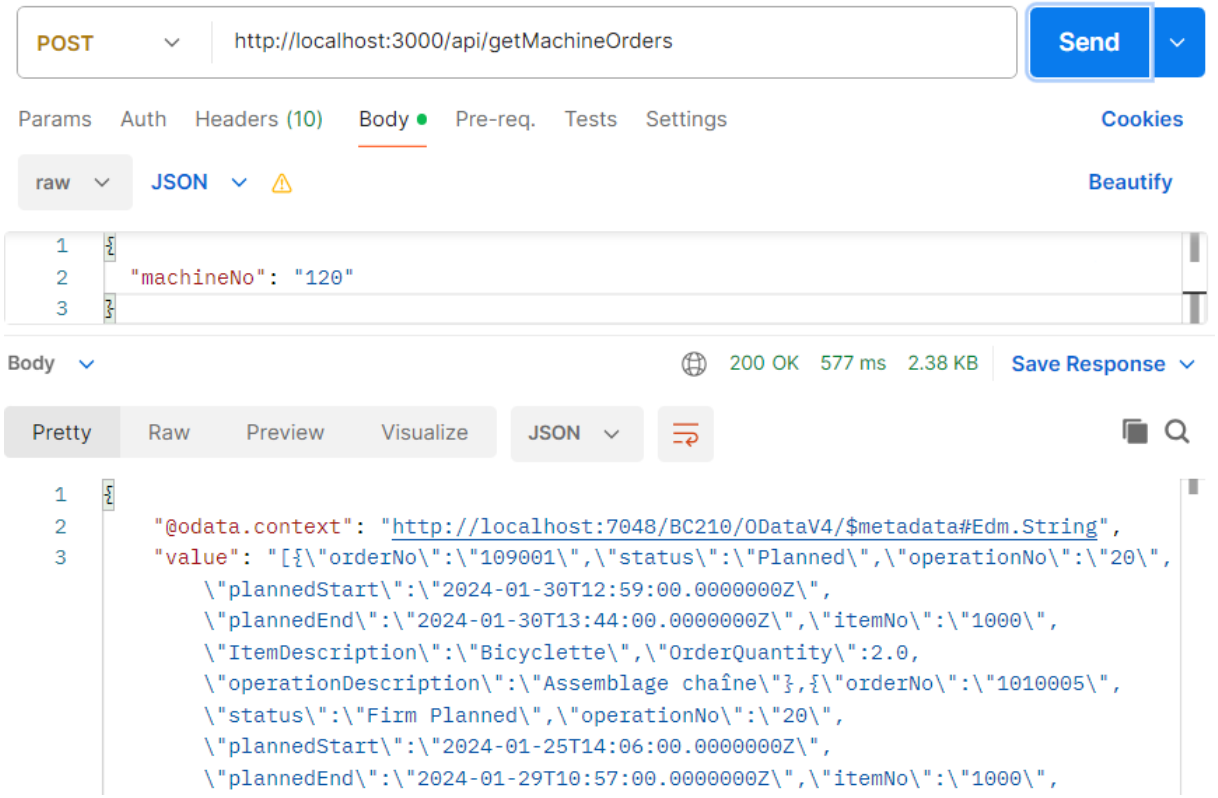

Figure 4.14. Postman endpoint test — getMachinesOrders.

Conclusion

This sprint successfully delivered the machine monitoring and production order management foundation of the *MES* application.On the backend, two new append-only tables (`MES Machine Status` and `MES Operation Status`) and dedicated codeunits (`MES Machine Fetch` and `MES Machine Write`) were implemented in Business Central AL, exposing four *OData V4* endpoints that cover the full machine and operation lifecycle. The append-only design ensures a complete, tamper-evident audit trail of every state transition without requiring any record modification.

Chapter 5

Sprint 3 Production Execution & Traceability

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Introduction

This sprint report covers **Domain 3: Production Execution & Traceability**. Building upon the authentication and machine monitoring infrastructure established in the previous sprints, this sprint delivers the core production execution features: declaring production output, pausing and resuming operations, finishing or cancelling orders, tracking production cycles over time, monitoring Bill of Materials consumption, component barcode scanning, scrap declaration, supervisor proxy declaration, and label printing. The result is a fully operational shop-floor execution loop from order assignment through to order closure, material traceability, and printed part identification.

5.1 Sprint Backlog

This sprint covers **Domain 3:** ***Production Execution & Traceability***.

Table 5.1. US-3.1 — Declare Production

ID	User Story	Acceptance Criteria
US-3.1	As an Operator or Supervisor, I need to declare the quantity of parts produced in a cycle so that the system records my production progress.	*Business Central (AL):* • Create table MES Operation Progression. • Create the functions: – `TryDeclareProduction()` in codeunit MESMachineValidation. – `InsertNewProgressionCycle()` in codeunit MESMachineInsert. – `declareProduction()` in codeunit MESMachineWrite. • Expose `declareProduction()` through the codeunit MES Web Service. • Create page MES Operation Progression list page. *Flutter:* • Create ErpMachineOrdersService to consume the `declareProduction` API and MachineordersProvider to manage it. • Create widget DeclareProductionDialog. • Add the `Declare Production` action in ActionButtonsContainer.

Continued on next page

ID	User Story	Acceptance Criteria
US-3.2	As an Operator or Supervisor, I need to pause and resume an active operation so that production interruptions are tracked accurately.	*Business Central (AL):* • Create the functions: – `TryPauseOperation()` and `TryResumeOperation()` in codeunit MESMachineValidation. – `ExecuteOperationTransition()`, `pauseOperation()`, and `resumeOperation()` in codeunit MESMachineWrite. • Expose `pauseOperation()` and `resumeOperation()` throught codeunit MES Web Service. *Flutter:* • Implement ErpMachineOrdersService to consume the new endpoints • Implement MachineordersProvider to manage its state. • Create widget OperationActionButtons. • Implement pause/resume operation toggle handling in OrdersProgressionPage.
US-3.3	As a Supervisor, I need to finish or cancel a production operation so that the machine is released and the order is closed in the system.	*Business Central (AL):* • Create the functions: – `TryCloseOperation()` in codeunit MESMachineValidation. – `InsertOperationStatus()` and `InsertIdleMachineStatus()` in codeunit MESMachineInsert. – `finishOperation()` and `cancelOperation()` in codeunit MESMachineWrite. • Expose `finishOperation()` and `cancelOperation()` in codeunit MES Web Service.

Continued on next page

ID	User Story	Acceptance Criteria
		Flutter: • Update ErpMachineOrdersService to consume the new endpoints. • Update MachineordersProvider to manage its state. • Update ActionButtonsContainer to handle finish/cancel operation flow. • Implement finish/cancel button state logic in ActionButtonsContainer.
US-3.4	As an Operator or Supervisor, I need a dedicated detail page for an operation so that I can see all real-time information in one place.	*Business Central (AL):* • Create table MES Operation Execution. • Create function `InsertMESOperationExecution()` in codeunit MESMachineInsert. • Create function `fetchOperationLiveData()` in codeunit MESMachineFetch. • Expose `fetchOperationLiveData()` through codeunit MES Web Service. • Create page MES Operation Execution. *Flutter:* • Create model OperationStatusAndProgressModel. • Update ErpMachineOrdersService to consume the new endpoints. • Update MachineordersProvider to manage its state. • Create page OperationDetailPage. • Create widget OperationAppbar. • Create widget CurrentOrderInfoContainer.
US-3.5	As an Operator or Supervisor, I need to see all production cycles declared for an operation so that I can review each declaration's quantity, operator, and timestamp.	*Business Central (AL):* • Create function `fetchProductionCycles()` in codeunit MESMachineFetch. • Expose `fetchProductionCycles()` throught codeunit MES Web Service.

Continued on next page

ID	User Story	Acceptance Criteria
		Flutter: • Create model ProductionCycleModel. • Add `fromJson()` and computed getters to ProductionCycleModel. • Update ErpMachineOrdersService to consume the new endpoints. • Update MachineordersProvider to manage its state. • Create widget ProductionCycle.
US-3.6	As an Operator or Supervisor, I need to see a visual chart of production declarations over time so that I can identify trends and performance patterns.	*Flutter:* • Create widget ProductionChart.
US-3.7	As an Operator or Supervisor, I need to see the Bill of Materials for a production operation so that I know which components are required and how much has already been consumed.	*Business Central (AL):* • Create table MES Component Consumption. • Create function `fetchBom()` in codeunit MESMachineFetch. • Expose `fetchBom()` through codeunit MES Web Service. • Create page MES Component Consumption.

Continued on next page

ID	User Story	Acceptance Criteria
		Flutter: • Create model ComponentConsumption-Model. • Create MesComponentConsumptionService to consume the `fetchBom()` API. • Create MesComponentConsumption-Provider to manage BOM stream and refresh state. • Create widget RequiredComponent. • Implement BOM streaming and refresh flow in MesComponentConsumptionService and MesComponentConsumptionProvider. • Implement component status display and operation-based filtering in RequiredCompo-nent.
US-3.8	As an Operator or Supervisor, I need to report scrapped units with a reason code so that waste is accurately tracked.	*Business Central (AL):* • Create table MES Operation Scrap. • Create function `declareScrap()` in codeu-nit MES Machine Write. • Expose `declareScrap()` through codeunit MES Web Service. • Create MES scrap Code API to fetch scrap codes from the standard Business Central `Scrap Code`. *Flutter:* • Create model MesScrapCode. • Create MesScrapService to consume the scrap code fetch and `declareScrap()` APIs. • Create MesScrapProvider to manage scrap code and scrap declaration state. • Create dialog DeclareScrapDialog.
US-3.9	As a Supervisor, I need to declare production on behalf of an operator so that output is attributed to the correct person.	*Business Central (AL):* • Add field `Declared By` to transactional tables MES Operation Progression and MES Component Consumption.

Continued on next page

ID	User Story	Acceptance Criteria
		Flutter: • Create widget OperatorSelector. • Update DeclareProductionDialog to support supervisor declaration on behalf of an operator. • Connect OperatorSelector to MesUserProvider and pass `onBehalfOfUserId` to Machineorder-sProvider.declareProduction().
US-3.10	As an Operator or Supervisor, I need to scan components into an operation so that consumption is recorded against the BOM.	*Business Central (AL):* • Create the function `insertScans()` in codeunit MES Machine Write. • Create the functions `fetchAllItemBarcodes()` and `resolveBarcode()` in codeunit MES Machine Fetch. • Expose the three function through codeunit MES Web Service. *Flutter:* • Create model ItemBarcodeModel. • Create MesBarcodeService to consume the APIs `fetchAllBarcodes()`, `insertScans()`, and `resolveBarcode()`. • Create MesBarcodeProvider to manage barcode state and API calls. • Create widget ScannerWidget. • Create page BarcodeListPage. • Update widget RequiredComponent to integrate item scanning, search, filtering, and BOM refresh after scan submission. • Implement ondevice camera scanning via mobile_scanner in ScannerWidget

Continued on next page

ID	User Story	Acceptance Criteria
US-3.11	As an Operator or Supervisor, I need to print a label for the active production order so that produced parts are correctly identified.	*Flutter:* • Create page PrintLabelPage. • Update ActionButtonsContainer to integrate the *Print Label* action. • Implement label generation, preview, print flow, and orientation toggle in PrintLabelPage.
US-3.12	As an Operator or Supervisor, I need a label printed for each declared unit so that individual pieces carry traceable information.	*Flutter:* • Create page DeclarationLabelPage. • Implement declaration label generation, preview, orientation toggle, and per-unit PDF export in DeclarationLabelPage.

5.2 Design

5.2.1 Use Case Diagram

This sprint adds production execution actors and interactions centred on the Operator and Supervisor roles. The primary use cases are: Declare Production, Pause Operation, Resume Operation, Finish Operation, Cancel Operation, View BOM Components, View Operation Detail, View Machine History, and Change Password.

The Supervisor role is a superset of Operator: both roles can declare production, pause, resume, and view detail; only the Supervisor may close (finish or cancel) a production order in the intended business flow, although the current implementation enforces this at the UI layer through the `canCloseProductionOrder` flag rather than at the API level. The resulting use case diagram is shown in Figure 5.2.

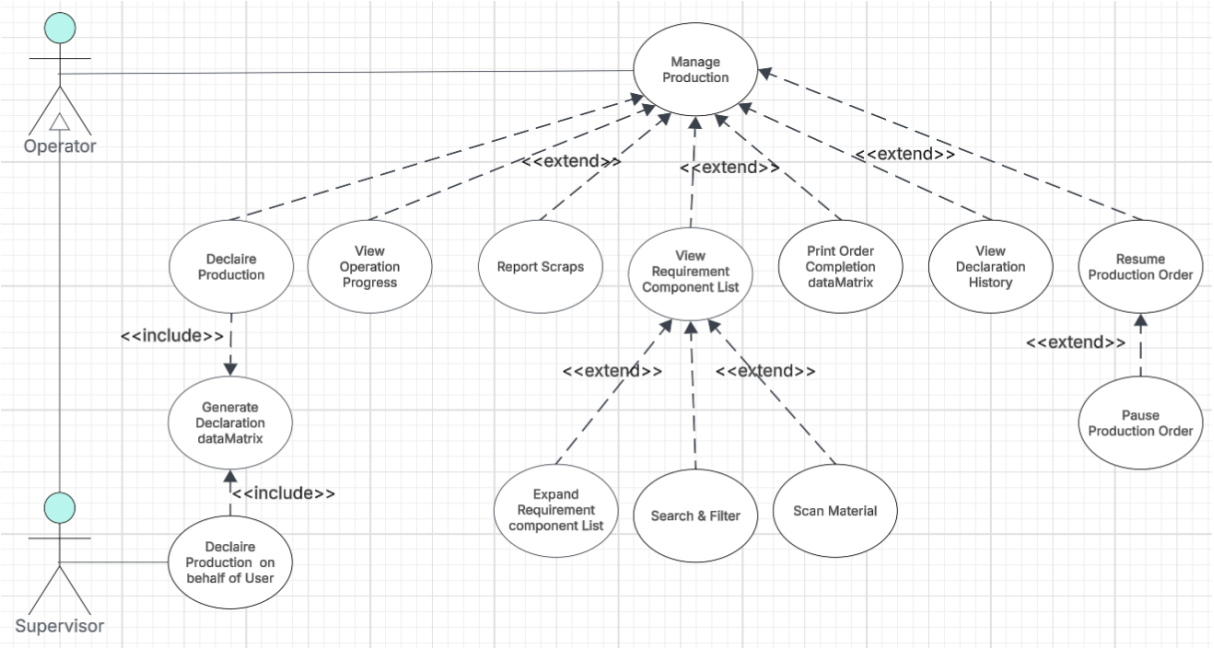

Figure 5.1. UML Use Case Diagram — Sprint 3

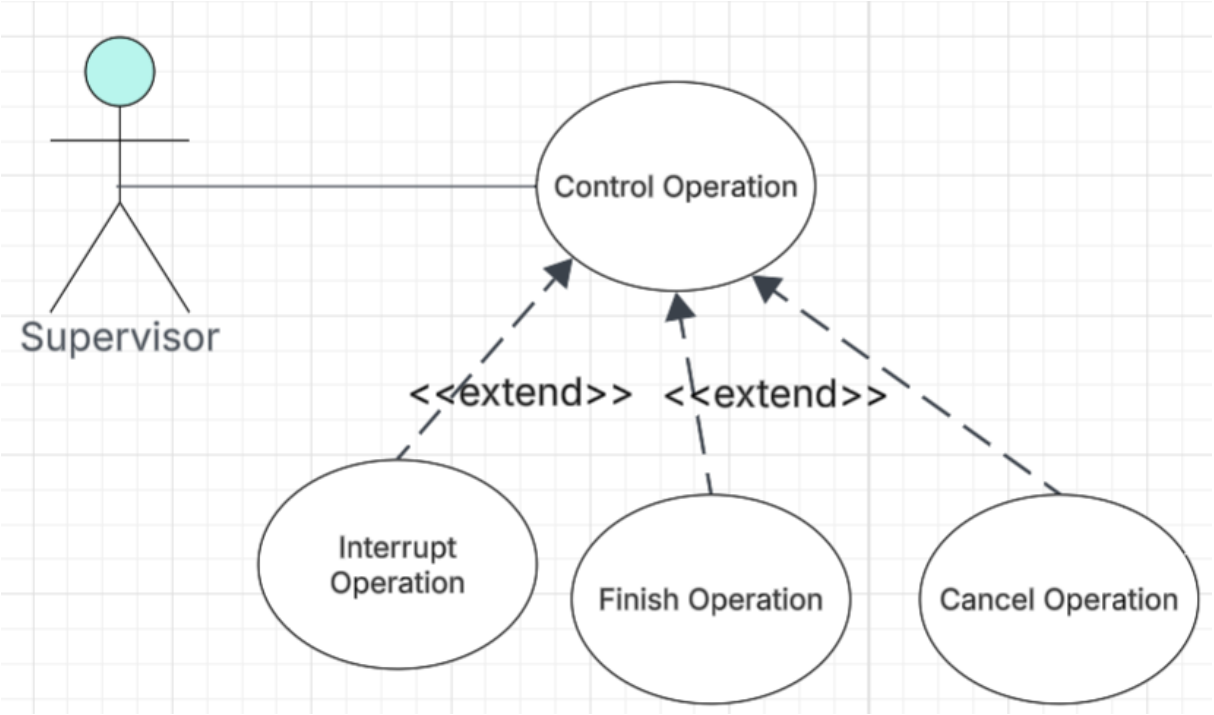

Figure 5.2. UML Use Case Diagram — Sprint 3

5.2.2 Textual Use Case Description

Table 5.2. Textual Use Case Description — Declare Production

Use Case Name	Declare Production
Primary Actor	Operator / Supervisor
Preconditions	• The user is authenticated and navigated to an Operation Detail page. • The operation status is *Running* or *Paused*. • The produced quantity is below the order quantity.
Postconditions	• A new MES Operation Progression record is inserted. • The production chart and cycle table update via the refresh stream. • The progress percentage and produced quantity in the info container update.
Main Scenario	1. The user taps *Declare Production* in the Quick Actions panel. 2. The DeclareProductionDialog opens showing the remaining quantity. 3. The user enters a positive integer not exceeding the remaining quantity. 4. The user taps *Submit*. 5. The system POSTs the declaration to `/MESWebService_declareProduction`. 6. The ERP validates the quantity and inserts the progression record. 7. The dialog closes and the refresh stream triggers a live data update.
Alternative Flow	• *Step 3, invalid quantity*: client-side validation shows an inline error; submission is blocked. • *Step 6, ERP error*: the server error message is displayed in the dialog; the record is not inserted.

Table 5.3. Textual Use Case Description — Finish / Cancel Operation

Use Case Name	Finish / Cancel Operation
Primary Actor	Supervisor
Preconditions	• The user is authenticated with an active session. • The operation is in *Running* or *Paused* state.

Continued on next page

Postconditions	• A *Finished* or *Cancelled* status record is inserted. • The execution end time is stamped. • An Idle machine status record is inserted. • The user is navigated back to the machine page.
Main Scenario	1. The user taps *Finish Production Order* (if complete) or *Cancel Production Order* (if incomplete). 2. A confirmation dialog is displayed; the user confirms. 3. The system POSTs to `/MESWebService_finishOperation` or `cancelOperation`. 4. The ERP validates, inserts the status record, stamps the end time, and inserts an Idle machine status. 5. Flutter calls `Navigator.pop()` to return to the machine page.

5.2.3 Sequence Diagrams

The following sequence diagrams illustrate the communication flow between the Flutter frontend, the Business Central OData layer, and the AL business logic for the production and operation-lifecycle flows introduced in Sprint 3.

Figure 5.3 depicts the production declaration flow. The operator submits a quantity; the Flutter client validates the input locally, then forwards the request to the AL layer, which re-validates the token (SD-02), checks that the declared amount is positive and does not exceed the remaining order quantity, and — on success — inserts a new immutable *MES Operation Progression* record and records the relevant user interaction entries. When a Supervisor declares production on behalf of an operator, the AL layer additionally verifies that the two users share at least one work centre before proceeding.

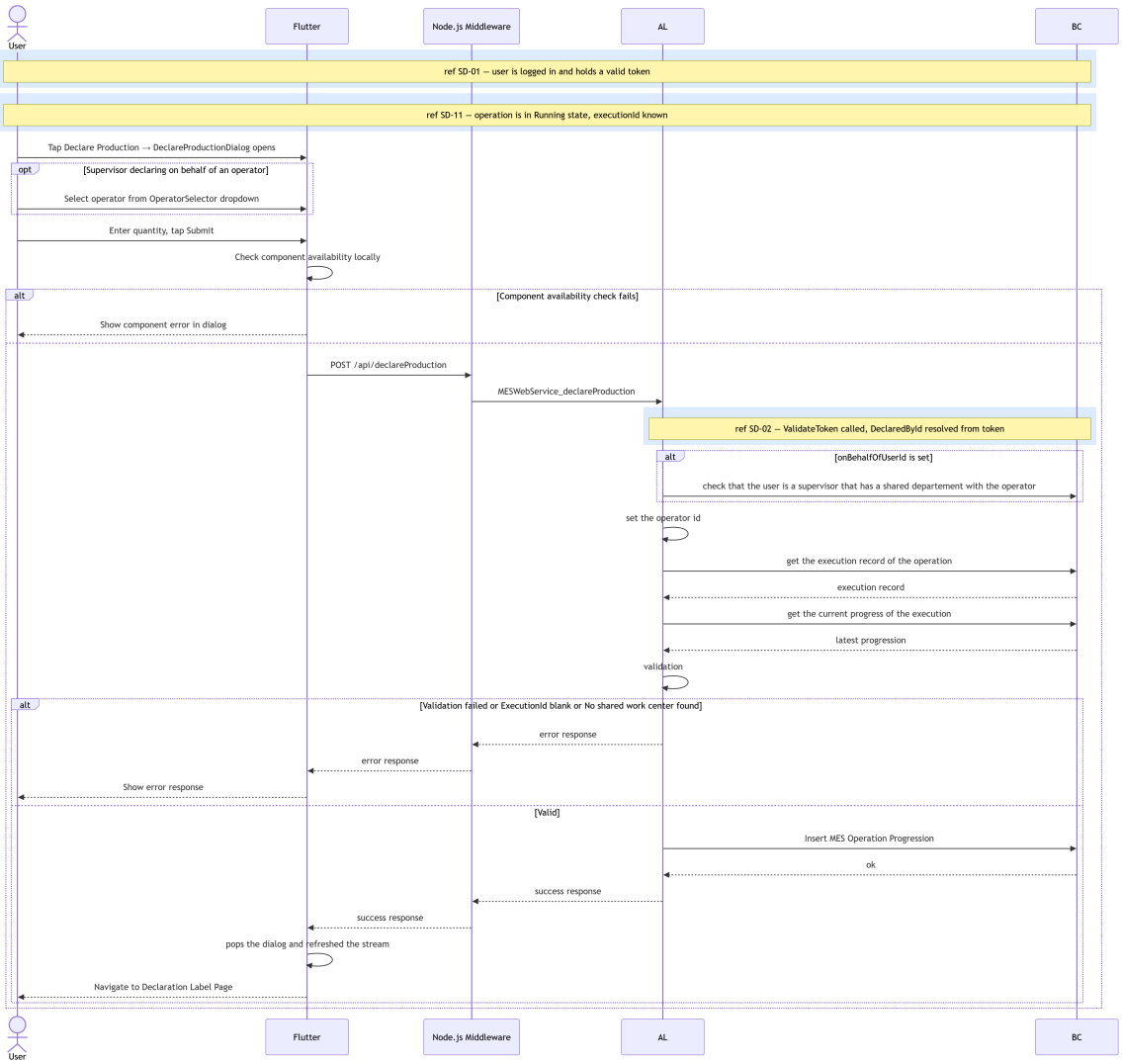

Figure 5.3. SD-12: Declare Production.

Figures 5.4 and 5.5 present the two supplementary data-retrieval flows that run concurrently alongside the main operation detail view. Both are polling streams that re-emit on demand or when a state-change trigger is received from the Flutter client.

Figure 5.4 describes the bill-of-materials retrieval flow. The AL layer resolves the routing link code from the active *ProdOrderRoutingLine*, filters the production order components by that link, aggregates the quantity already scanned per item, and resolves per-unit quantities from the *ItemUnitOfMeasure* table before returning the assembled BOM rows to the Flutter client.

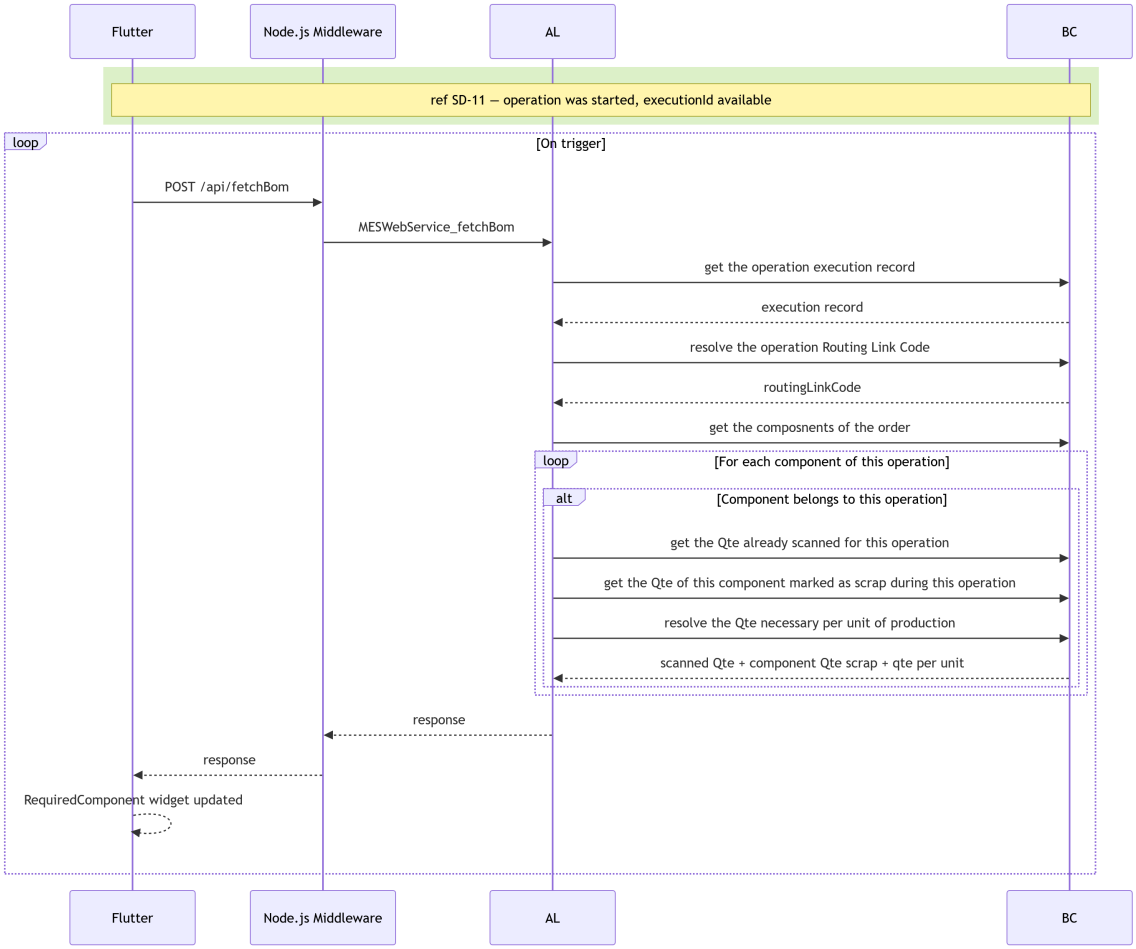

Figure 5.4. SD-21: Fetch Bill of Materials.

Figure 5.5 illustrates the production-cycle retrieval flow. The AL layer fetches the full progression history for the operation, enriches each entry with the name of the user who declared it, and returns the assembled cycle list to the Flutter client.

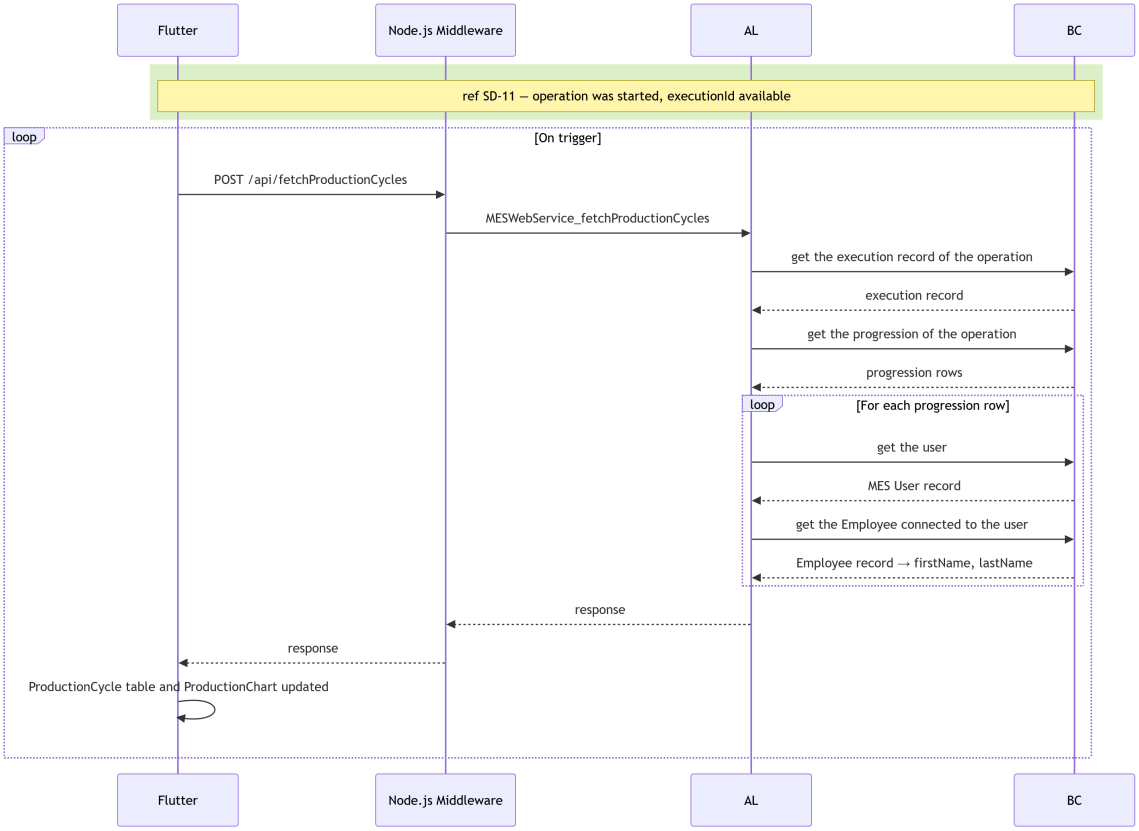

Figure 5.5. SD-22: Fetch Production Cycles.

Figures 5.6 through **??** illustrate the remaining operation lifecycle transition flows: pause, resume, finish, and cancel. All four follow the same structural pattern as the start operation sequence (SD-11): the Flutter frontend POSTs to the corresponding MES Web Service endpoint; the AL backend validates the current state via the `Try*` family of functions; inserts a new immutable status record in Business Central; and returns a structured success or error response to the client.

Figure 5.6 describes the pause flow. The AL layer confirms that the operation is currently in the *Running* state before inserting a *Paused MES Operation State* record and setting the machine status back to *Idle*.

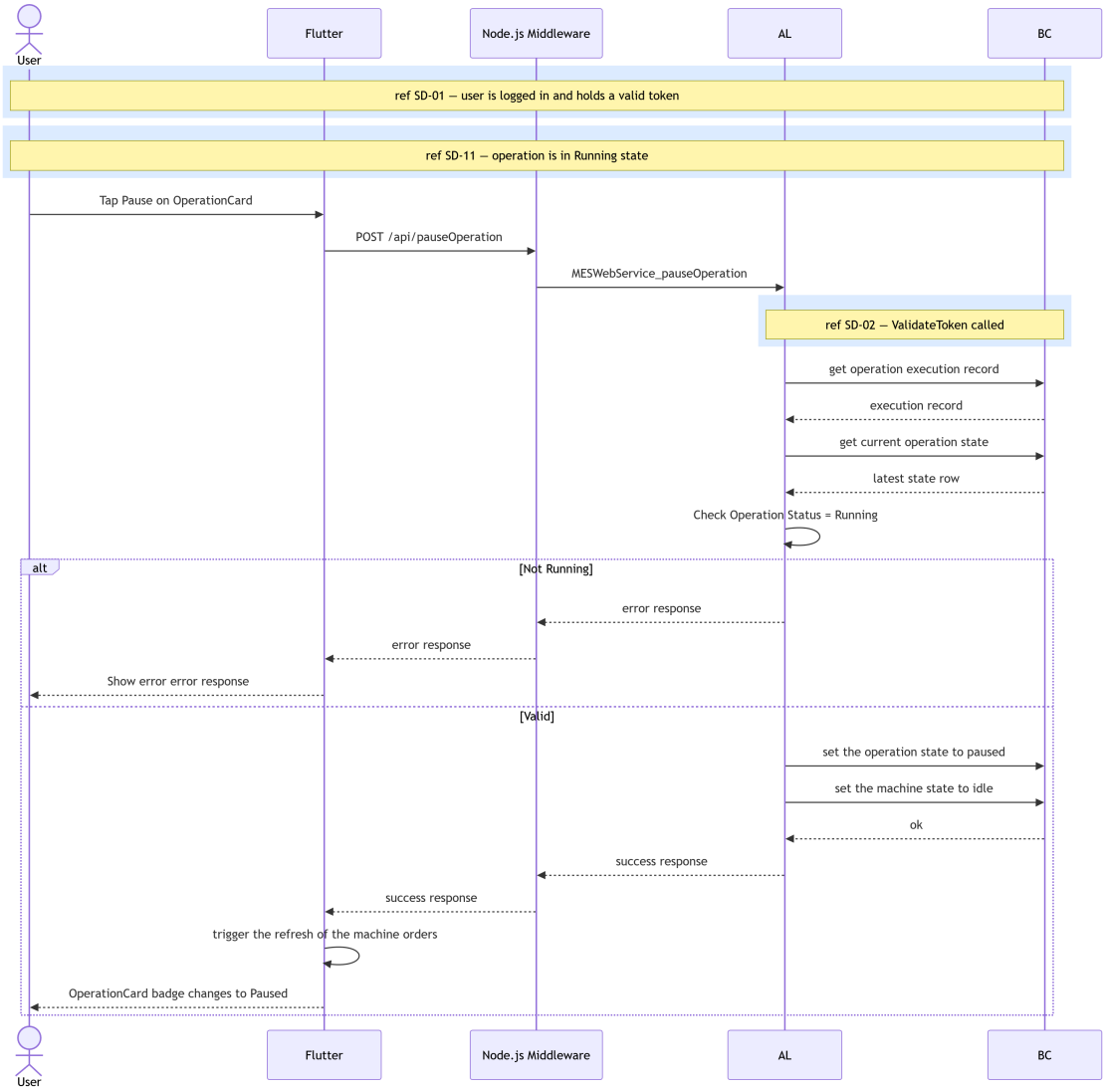

Figure 5.6. SD-13: Pause Operation.

Figure 5.7 presents the resume flow. In addition to confirming that the operation is in the *Paused* state, the AL layer verifies that no other operation is currently running on the same machine before inserting a *Running* state record and marking the machine as *Working*.

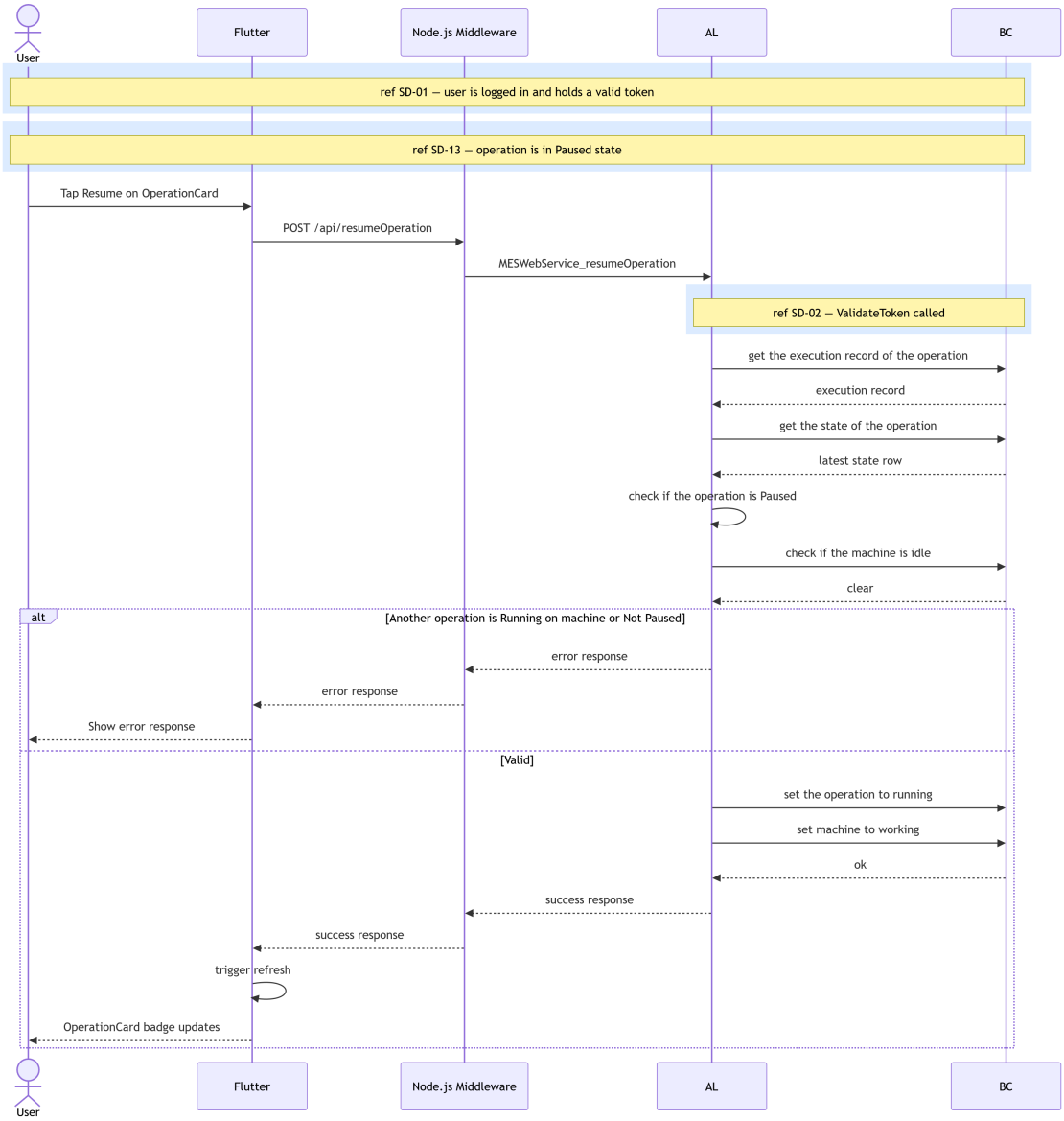

Figure 5.7. SD-14: Resume Operation.

Figure 5.8 covers the finish and cancel flows, which share a single endpoint family. The AL layer rejects the request if the operation is already in a terminal state; otherwise it inserts the appropriate closed-state record, stamps the execution end time, and resets the machine status to *Idle*. The closed operation then appears in the history view rather than the active operations list.

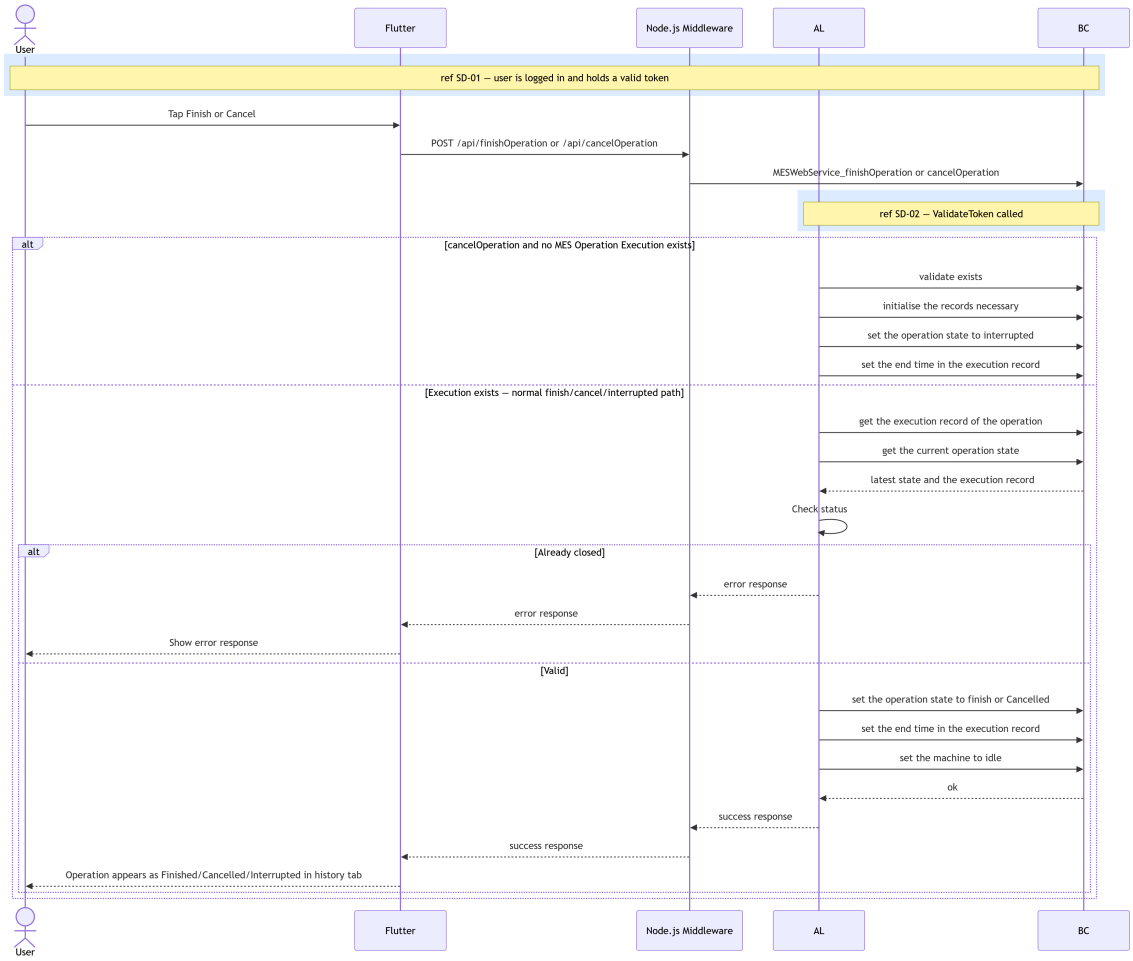

Figure 5.8. SD-15: Finish/Cancel/Interrupt Operation.

Figure 5.9 describes the scrap declaration flow. Before inserting the scrap record, the AL layer verifies that the operation has not already been finished, cancelled, or paused, and that the referenced scrap code exists in Business Central. On success, it inserts a *MES Operation Scrap* entry, records user interaction, and appends a new *MES Operation Progression* row so that the scrap quantity is reflected in the live data stream.

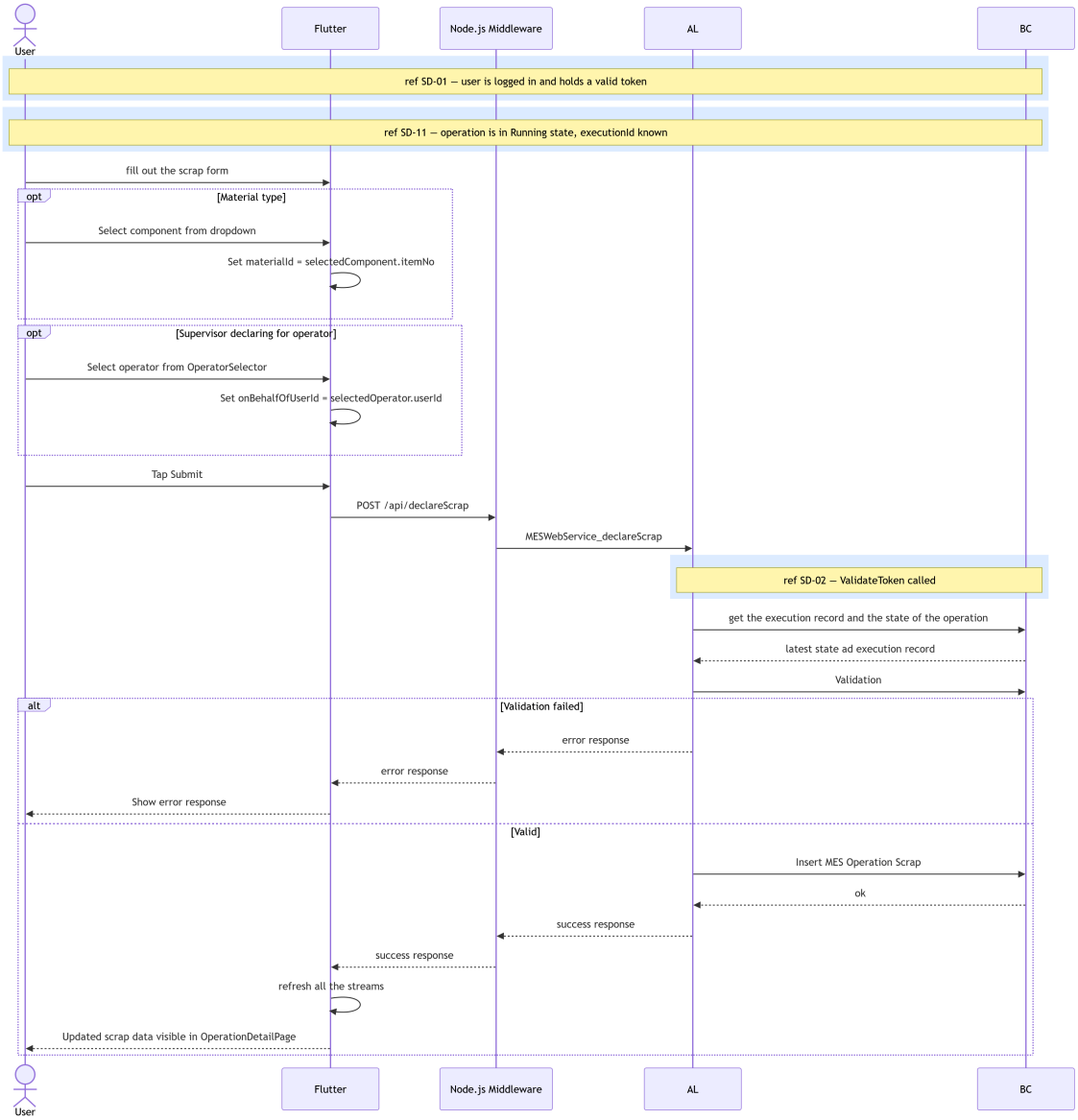

Figure 5.9. SD-16: declare scrap.

Figure 5.10 illustrates the component scanning flow, which operates in two steps. First, the Flutter client resolves each scanned barcode into an item record via a lightweight `resolveBarcode` call and accumulates the results locally. Once the operator confirms the batch, a single `insertScans` request is sent to the AL layer, which inserts one *MES Component Consumption* record per scanned item and records the associated user interaction entries.

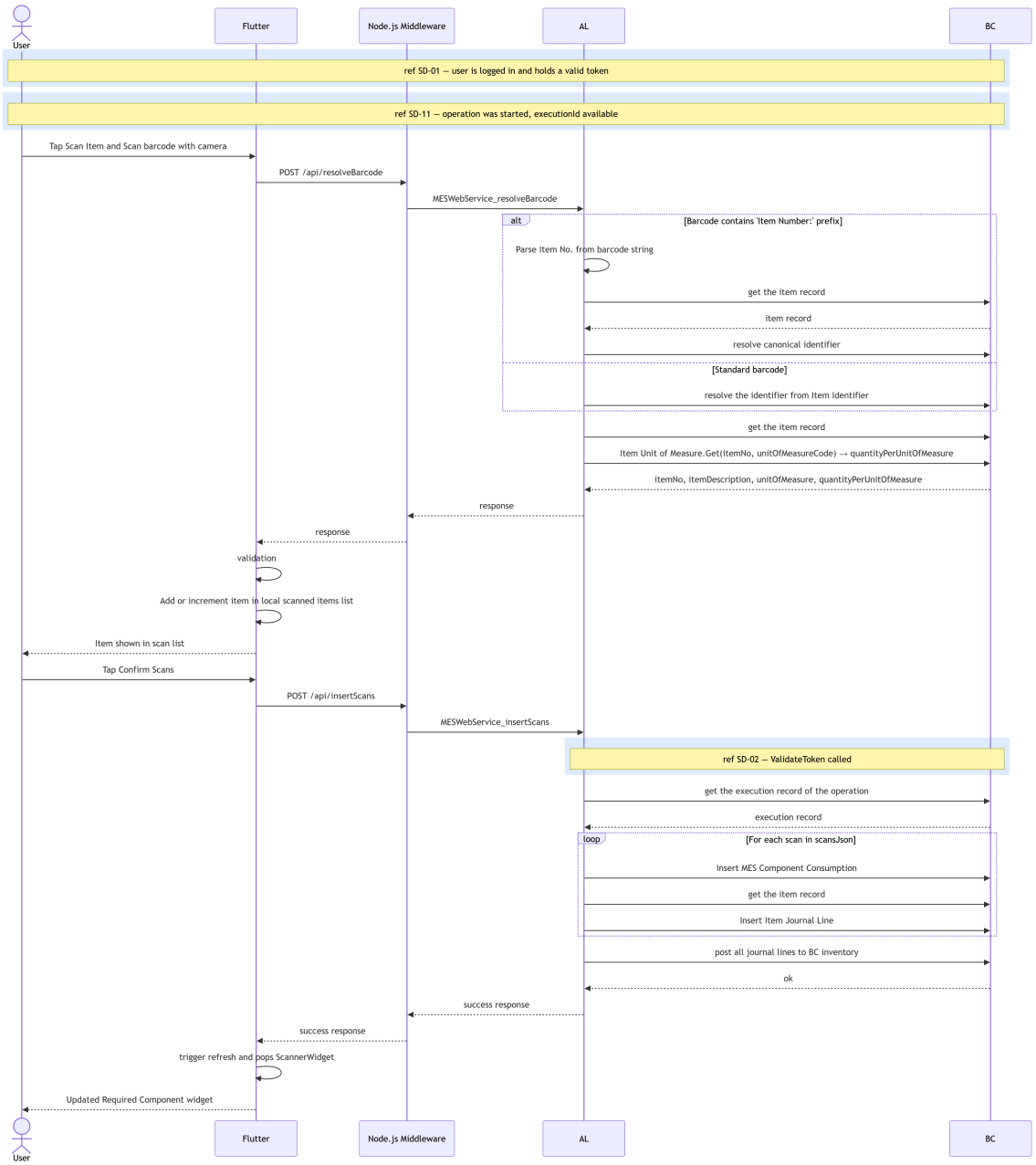

Figure 5.10. SD-17: scan component.

Figure 5.11 presents the live operation data retrieval sequence. When the user navigates to the *OperationDetailPage*, three StreamBuilder instances mount simultaneously: one polling the current operation status and quantities from Business Central (SD-18), one consuming the production-cycle stream (SD-22), and one consuming the BOM stream (SD-21). The Flutter client merges all three data sources and renders the full detail view. If the operation is finished, cancelled, or not yet started, the live-data stream returns an empty response and the client falls back to the history view.

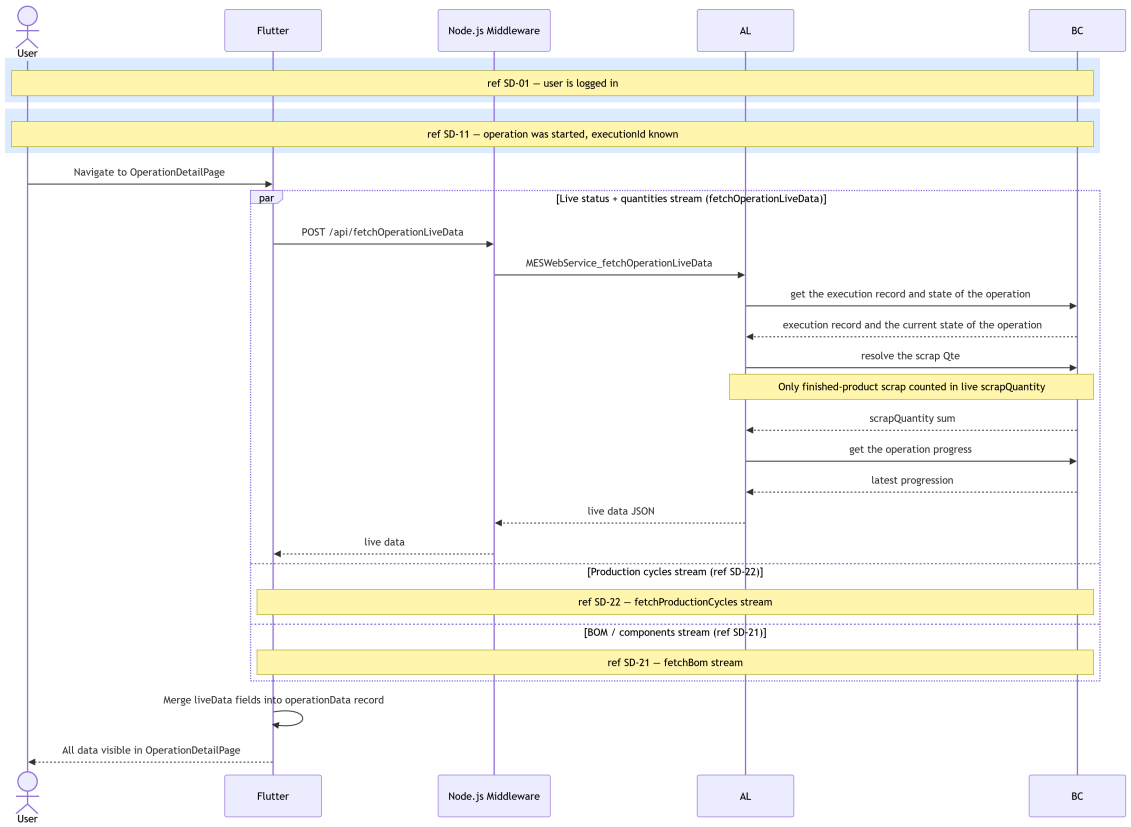

Figure 5.11. SD-18: Fetch Operation Live Data.

5.2.4 Class Diagram

The UML Class Diagram (Figure 5.12) extends the Sprint 2 domain model with three new tables. `MES Operation Execution` (50110) is the central anchor record linking a machine, production order, and operation to its full lifecycle; it holds item description, order quantity, and start/end timestamps. `MES Operation Progression` (50109) is an append-only table of production cycle declarations, each referencing an execution and accumulating `TotalProducedQuantity` from the previous cycle. `MES Component Consumption` (50111) records scrap and barcode scan events, also referencing an execution. Both progression and consumption records are child tables of the execution record, forming a hub-and-spoke traceability model.

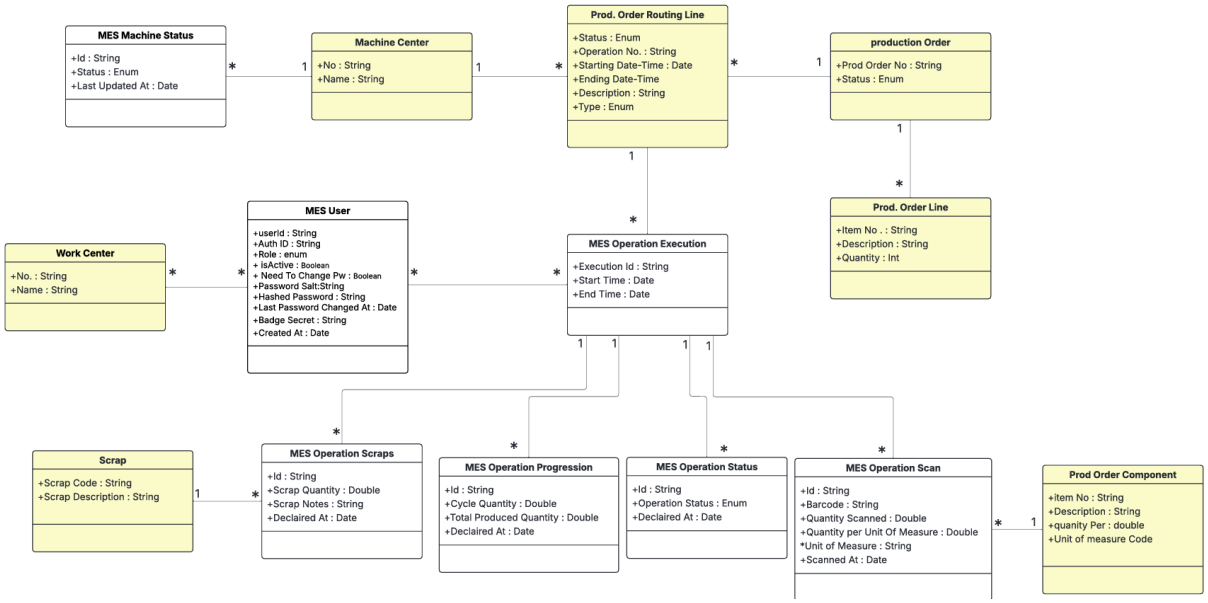

Figure 5.12. UML Class Diagram — Production Execution & Traceability.

5.3 Implementation

5.3.1 Backend Implementation

Tables and Entities

Table 5.4. MES Operation Execution (Table 50110) — Field Dictionary

Name	Type	Description
Execution Id	Code[50]	GUID-based primary key; auto-generated on insert.
Machine No	Code[20]	References Machine Center (no referential constraint enforced in the current implementation).
Prod Order No	Code[20]	Foreign key to Prod. Order Routing Line.
Operation No	Code[10]	Operation identifier from the routing line.
Item No	Code[20]	Copied from Prod. Order Line on execution creation.
Item Description	Text[100]	Copied from Prod. Order Line for denormalized display.
Order Quantity	Decimal	Copied from Prod. Order Line; used for progress calculation.
Start Time	DateTime	Set by `OnInsert` trigger to `CurrentDateTime()`.
End Time	DateTime	Stamped when status transitions to Finished or Cancelled.

Table 5.5. MES Operation Progression (Table 50109) — Field Dictionary

Name	Type	Description
Id	Code[50]	GUID-based primary key; auto-generated on insert.
Execution Id	Code[50]	Foreign key to MES Operation Execution.
Cycle Quantity	Decimal	Units declared in this single production cycle.
Scrap Quantity	Decimal	Units scrapped in this cycle (currently always 0).
Total Produced Quantity	Decimal	Running total: previous total + Cycle Quantity.
Operator Id	Code[50]	BC session UserId of the operator who declared.
Declared At	DateTime	Set by `OnInsert` trigger to `CurrentDateTime()`.

Table 5.6. MES Component Consumption (Table 50111) — Field Dictionary

Name	Type	Description
Id	Code[50]	GUID-based primary key; auto-generated on insert.
Execution Id	Code[50]	Foreign key to MES Operation Execution.
Prod Order No	Code[50]	Foreign key to Prod. Order Component.
Item No	Code[50]	Item identifier of the consumed component.
Barcode	Text[500]	Barcode scanned to register the component.
Unit of Measure	Code[50]	Unit of measure code from the component definition.
Quantity Scanned	Decimal	Raw quantity read from the barcode scan.
Quantity per Unit of Measure	Decimal	Unit-of-measure conversion factor; the effective consumed quantity equals Quantity Scanned multiplied by this value.
Operator Id	Code[50]	BC session UserId of the operator who scanned.
Scanned At	DateTime	Set by `OnInsert` trigger to `CurrentDateTime()`.

REST API and Web Services Implementation

All endpoints are exposed as ODataV4 unbound actions through MES Web Service (codeunit 50126), reachable at:

```
POST <baseUrl>/ODataV4/MESWebService_<ProcedureName>?company=<companyId>
```

Table 5.7 lists all endpoints added in this sprint.

Table 5.7. New API Endpoints — Sprint 3

Endpoint	Key Parameters	Response
`startOperation`	prodOrderNo, operationNo, machineNo	`{value: bool, message?}`
`declareProduction`	machineNo, prodOrderNo, operationNo, input	`{value: bool, message?}`
`pauseOperation`	machineNo, prodOrderNo, operationNo	`{value: bool, message?}`
`resumeOperation`	machineNo, prodOrderNo, operationNo	`{value: bool, message?}`
`finishOperation`	machineNo, prodOrderNo, operationNo	`{value: bool, message?}`
`cancelOperation`	machineNo, prodOrderNo, operationNo	`{value: bool, message?}`
`fetchOperationsStatusAndProgress`	machineNo, fetchFinished	JSON array of operation progress objects
`fetchOperationLiveData`	machineNo, prodOrderNo, operationNo	JSON array (0 or 1 element)
`fetchProductionCycles`	machineNo, prodOrderNo, operationNo	JSON array of cycle records
`fetchBom`	prodOrderNo, operationNo	JSON array of BOM component records

5.3.2 Frontend Implementation

This sprint introduces a key reactive pattern: a shared `StreamController<void>.broadcast()` in MachineordersProvider and MesComponentConsumptionProvider. When any write action (declare, pause, resume, finish, cancel) succeeds, `triggerRefresh()` adds an event to the controller. All live data streams (`fetchOperationLiveDataStream`, `streamProductionCycles`, `streamBom`) are implemented as `async*` generators that yield on both the initial fetch and on each event from the shared broadcast stream, enabling coordinated, push-driven UI updates without polling.

Provider Architecture

Table 5.8. Provider Responsibilities — Sprint 3 Additions

Provider	Responsibility
MachineordersProvider	Owns write operations (start, declare, pause, resume, finish, cancel) and their refresh controller; exposes all live data streams.
MesComponentConsumption Provider	Exposes `getBomStream()` combining service fetch with an independent refresh controller.
AuthProvider	Adds `changePassword()` and `adminSetPassword()`; manages `needsPasswordChange` flag in the in-memory session.

User Interface

Figures 5.13 through 5.16 illustrate the *Operation Detail Page* across its four possible states: Running, Paused, Finished, and Cancelled. Each view adapts its action buttons and status badge to the current state, while the production chart, cycle table, and BOM list remain consistently visible. On desktop, the page uses a two-column layout with order information and production data on the left, and the *Quick Actions* panel on the right; on mobile, these sections stack vertically. Action button availability is state-dependent: *Declare Production* and *Report Reject* are disabled in the Paused, Finished, and Cancelled states; the close button triggers a *Finish Production Order* or *Cancel Production Order* action depending on whether the progress has reached 100%; and *Print Label* is enabled only once the operation is finished.

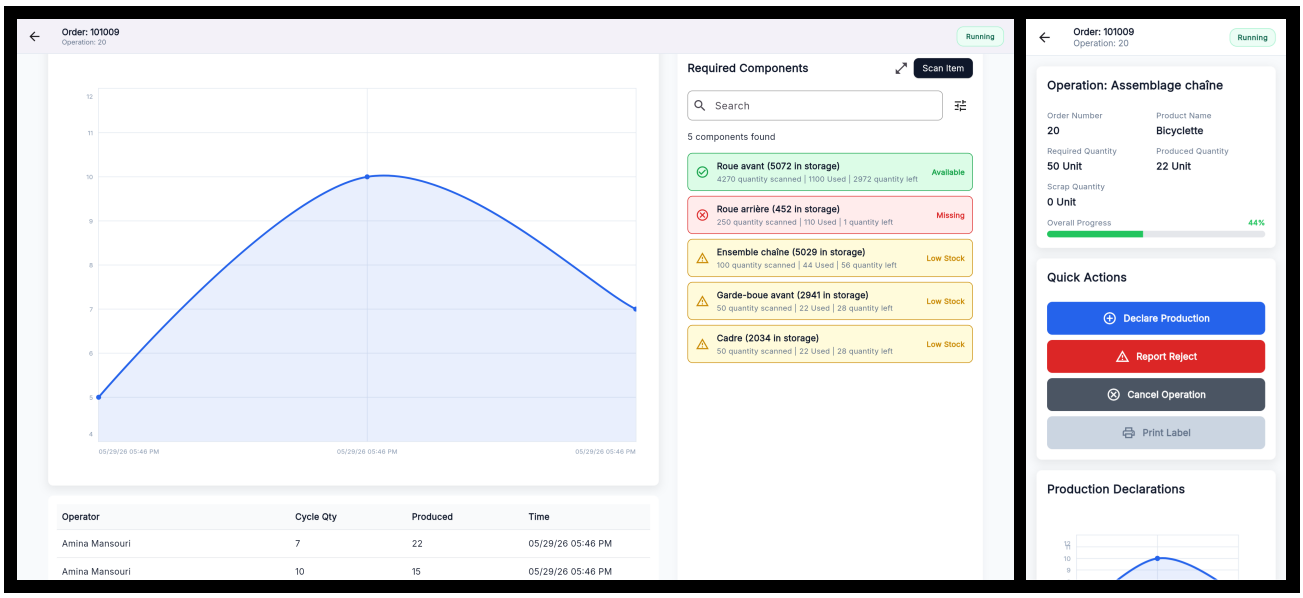

Figure 5.13. *Operation Detail Page* (Running state) — desktop and mobile views.

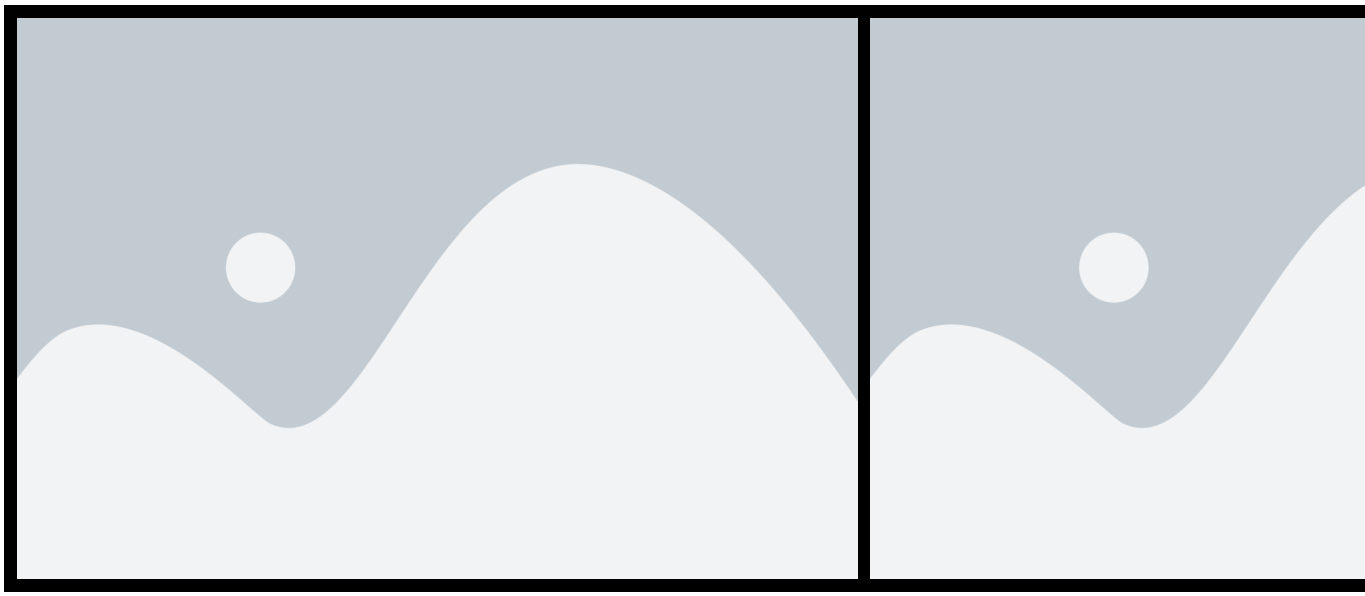

Figure 5.14. *Operation Detail Page* (Finished state) — desktop and mobile views.

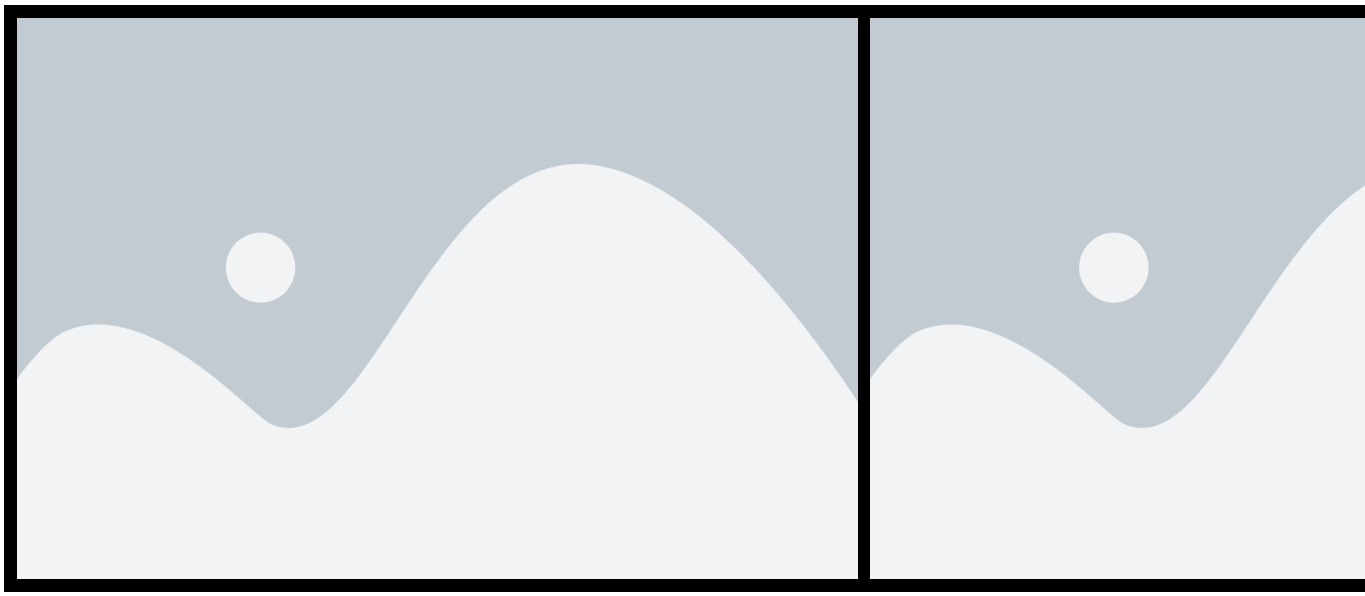

Figure 5.15. *Operation Detail Page* (Paused state) — desktop and mobile views.

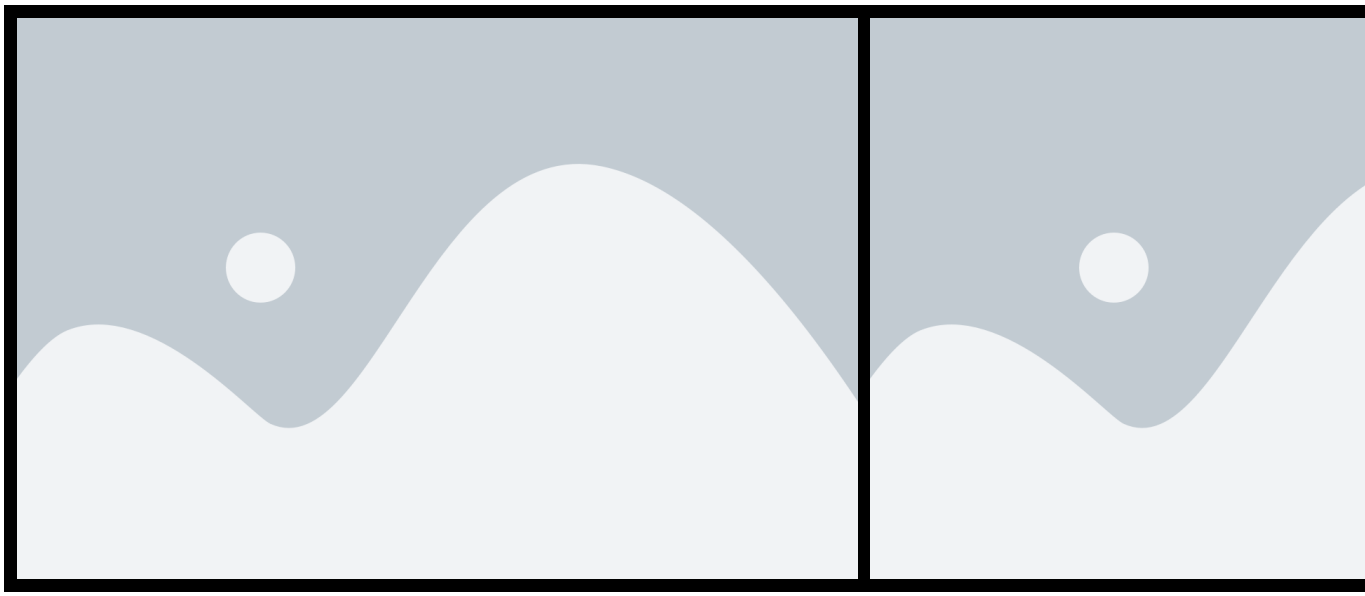

Figure 5.16. *Operation Detail Page* (Cancelled state) — desktop and mobile views.

5.4 Test and Validation

5.4.1 Integration test

For integration testing, we chose Postman, a powerful and user-friendly tool for API evaluation. Figure 5.17 shows a test case for the `/fetchBom` endpoint.

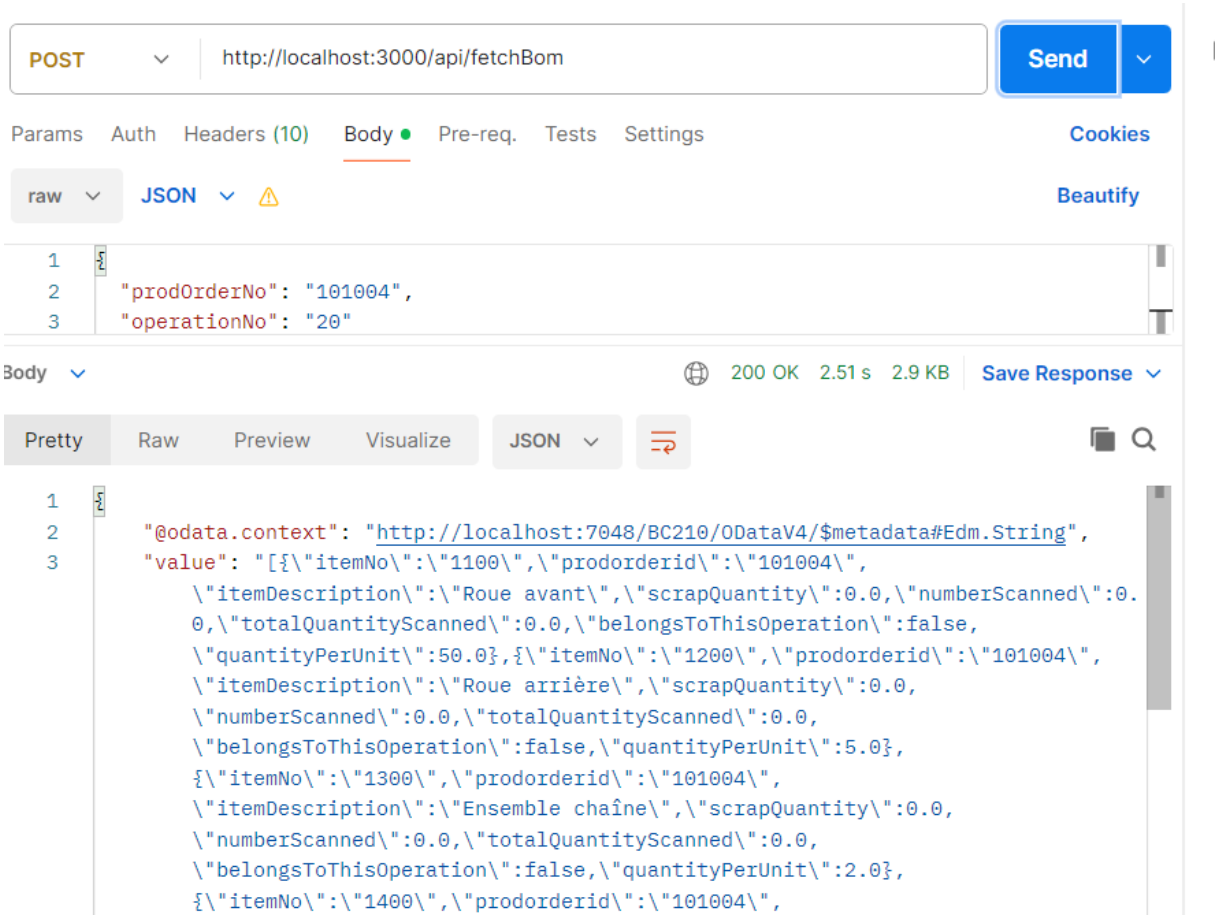

Figure 5.17. Postman endpoint test — fetchBom.

Conclusion

This sprint delivered the complete shop-floor execution loop for the MES system. Operators can now start, pause, resume, and close production operations, declare production output cycle by cycle, track cumulative progress through real-time charts and paginated tables, inspect the Bill of Materials, scan physical components via DataMatrix barcodes, declare scrapped units with reason codes, and print traceable labels for both production orders and individual declared units. Supervisors gain the ability to declare production on behalf of a specific operator, with a full attribution audit trail. The reactive stream architecture ensures that all UI panels update in concert after every write action, providing a consistent and accurate view of the production floor without manual refreshes.

Chapter 6

Sprint 4 Administration, Printing & Scanning

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Introduction

This sprint delivers the administration layer, multi-language support, and in-app onboarding tutorials that complete the *MES* solution. Administrators receive a persistent navigation shell giving access to a a paginated activity log, and a user role management interface. Cross-cutting quality-of-life features include full internationalisation with English, French, and Arabic support, automatic right-to-left layout for Arabic, and guided coach-mark tutorials shown once per screen on first use. This sprint also delivers machine performance dashboard for the Supervisor. No new Business Central tables are introduced in this sprint; all data is derived by querying and aggregating the tables created in Sprints 1 through 3.

6.1 Sprint Backlog

This sprint covers **Domain 4:** ***Administration & Monitoring*** and **Domain 5:** ***UX & Accessibility***.

Table 6.1. Sprint 4 Story Backlog

ID	User Story	Tasks
US-4.1	As an Supervisor, I need a performance dashboard for all machines so that I can monitor production health across the facility.	*Business Central (AL):* • Create function `fetchMachineDashboard()` in codeunit MES Machine Fetch. • Expose the function through the codeunit MES Web Service. *Flutter:* • Create model MachineDashboardModel. • Create LogService to consume the `fetchMachineDashboard` API and LogProvider to manage its state. • Create page MachineDashboardPage. • Implement machine dashboard search and time-range filtering in MachineDashboardPage.
US-4.2	As an Administrator, I need a filterable, paginated log of all system actions so that I can audit operator and machine activity.	*Business Central (AL):* • Create function `fetchActivityLog()` in codeunit MES Machine Fetch. • Add the wrapper of the function to MES Web Service.

Continued on next page

ID	User Story	Tasks
		Flutter: • Create model ActivityLogModel. • Create LogService to consume the `fetchActivityLog()` API and LogProvider to manage its state. • Create page ActivityLogPage. • Implement activity log search, type filtering, time-range filtering, and pagination in ActivityLogPage.
US-4.3	As an Administrator, I need a persistent sidebar so that I can navigate between all admin sections without losing context.	*Flutter:* • Create page AdminPage. • Create widget SidebarItem. • Implement sidebar navigation and profile-page navigation in AdminPage.
US-4.4	As any user, I need the application available in English, French, and Arabic so that I can use it in my preferred language.	*Flutter:* • Integrate the `easy_localization` package. • Create translation JSON files for `en`, `fr`, and `ar`. • Create widget LanguageSelector. • Implement language selection in ProfilePage. • Add the EasyLocalization wrapper in `main()`. • Implement right-to-left layout support through `MaterialApp`.
US-4.5	As a first-time user, I need guided coach-mark tutorials so that I can learn the interface without external documentation.	*Flutter:* • Integrate the `tutorial_coach_mark` package. • Implement tutorial persistence using SharedPreferences. • Create MachineListTutorial, MachineDetailTabsTutorial, OperationDetailTutorial, and AdminDashboardTutorial. • Implement tutorial target styling and localisation.

6.2 Design

6.2.1 Use Case Diagram

The UML Use Case Diagram (Figure 6.1) covers the actors and interactions introduced in this sprint. The Administrator gains the interaction *View Activity Log*.The Supervisor gains the interaction *View Machine Dashboard*. All three actors (Operator, Supervisor, Administrator) share the *Select Language* and *View Tutorial* use cases, which are cross-cutting accessibility features available on every screen.

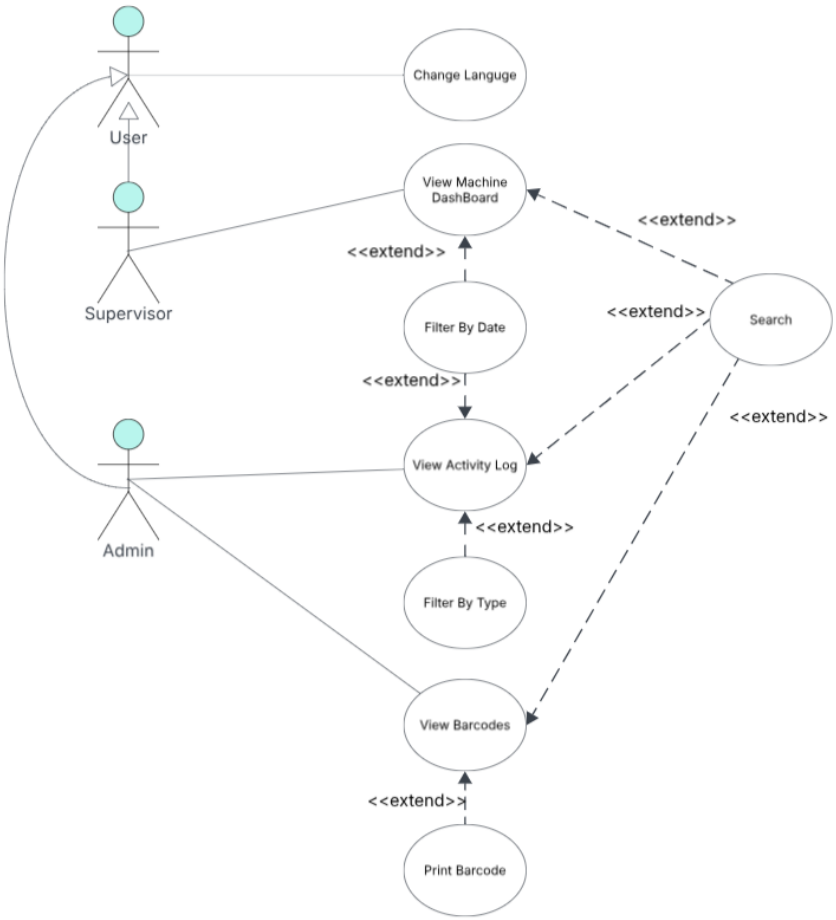

Figure 6.1. UML Use Case Diagram — Administration & UX.

6.2.2 Textual Use Case Description

The following table (Table 6.2) provides the detailed textual description of the *View Machine Dashboard* use case, which represents the most significant new administrator workflow in this sprint.

Table 6.2. Textual Use Case Description — View Machine Dashboard

Use Case Name	View Machine Dashboard
Primary Actor	Supervisor
Preconditions	• The user is authenticated with the *Supervisor* role. • At least one MES Operation Execution record exists.
Postconditions	• A grid of machine cards is displayed, each showing uptime, production totals, and scrap for the selected time window. • Changing the time-range selector refreshes all card metrics.
Main Scenario	1. The Supervisor logs in and is routed to machineList. 2. The Supervisor selects *Machine Dashboard* in the topBar. 3. The system POSTs to `/MESWebService_fetchMachineDashboard` with the default `hoursBack = 24`. 4. The backend aggregates execution and progression records and returns per-machine metrics. 5. The Supervisor changes the time range to 48 h via the AppBar dropdown; the page re-fetches and updates.

6.2.3 Sequence Diagrams

The following sequence diagrams illustrate the communication flow between the Flutter frontend, the Business Central OData layer, and the AL business logic for the administrator analytics flows introduced in Sprint 4.

Figure 6.2 depicts the machine dashboard retrieval flow. The AL layer iterates over all *Machine Center* records and, for each machine, computes the total operation count, cumulative produced and scrap quantities, and uptime percentage from the corresponding Business Central records before assembling the per-machine KPI rows returned to the Flutter client.

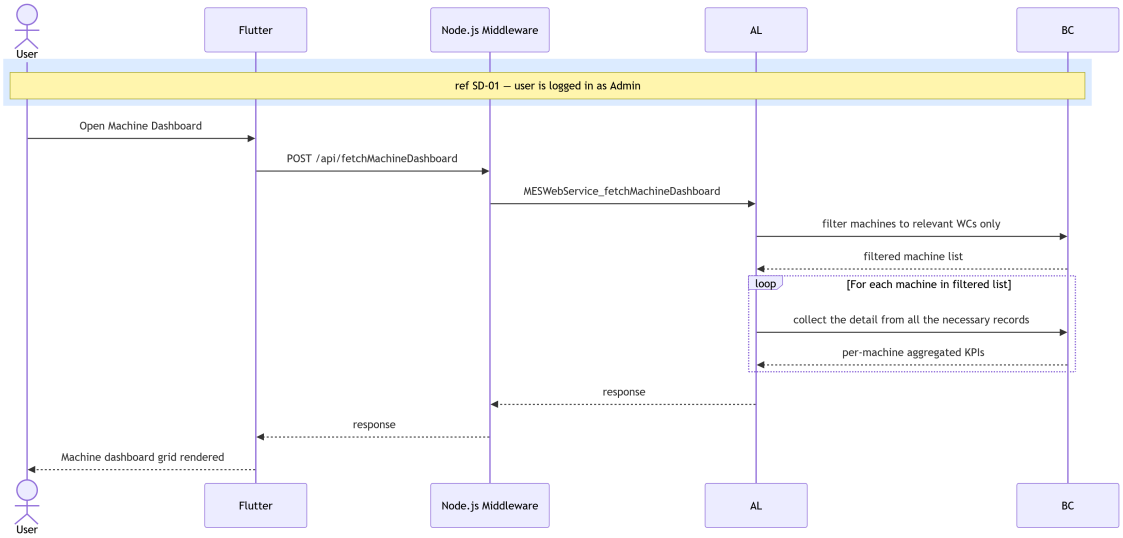

Figure 6.2. SD-19: Fetch Machine Dashboard.

Figure 6.3 presents the activity log retrieval flow. The AL layer queries four event tables in parallel; *MES Operation State*, *MES Operation Progression*, *MES Operation Scrap*, and *MES Component Consumption* , filtering each by the requested time window. For every event row it joins the associated *MES User* and *MES Operation Execution* records to enrich the entry with operator identity and execution context. The resulting flat event list is returned to the Flutter client, which renders it as a chronological activity table.

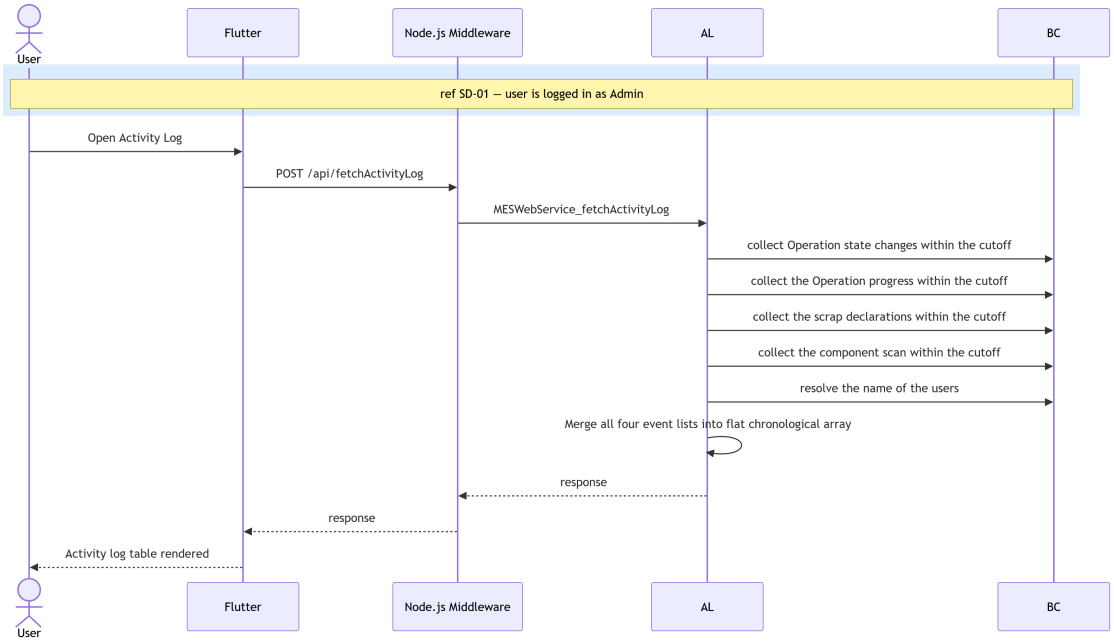

Figure 6.3. SD-20: Fetch Activity Log.

6.2.4 Class Diagram

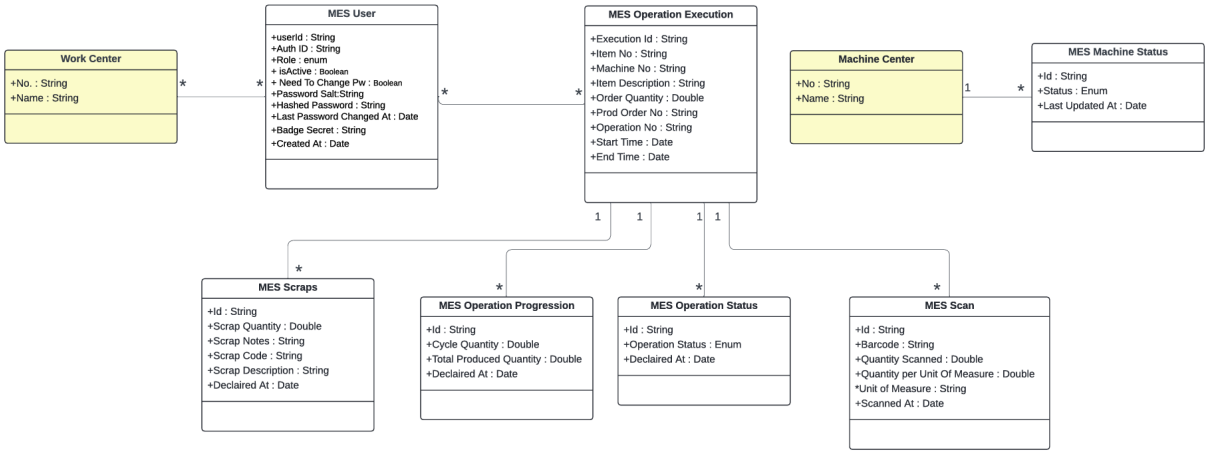

Figure 6.4. UML Class Diagram.

6.3 Implementation

6.3.1 Backend Implementation

No new Business Central tables are created in this sprint. All backend work consists of new query and aggregation procedures added to the existing MES Web Service codeunit (50126), reading from the tables defined in Sprints 1 through 3.

Dashboard Aggregation

`fetchMachineDashboard()` accepts a `hoursBack` integer and an optional `workCenterNoJson` (JSON array of work centre numbers). It iterates all `Machine Center` records, and for each machine it filters `MES Operation Execution` records whose `Start Time` falls within the requested window. For each matched execution it sums the `Cycle Quantity` fields from linked `MES Operation Progression` records to obtain `totalProduced`, and counts records marked as scrap to obtain `totalScrap`. Running minutes are derived from the difference between `End Time` (or the current datetime for open executions) and `Start Time`. Uptime percentage is computed as `runningMinutes / (hoursBack × 60) × 100`, capped at 100.

Activity Log Aggregation

`fetchActivityLog()` queries four tables in descending timestamp order, filtered to the `hoursBack` window: `MES Operation State` (type `status_change`), `MES Operation Progression` (type `production`), `MES Operation Scrap` (type `scrap`), and `MES Component Consumption` (type `scan`). For each record the operator's first and last name are resolved from the `Employee` table via the `Operator Id` foreign key. The results are merged into a single chronological list before being returned as a JSON array.

API Endpoints Added in Sprint 4

Table 6.3. New API Endpoints — Sprint 4

Endpoint	Key Parameters	Response
`fetchMachineDashboard`	hoursBack, workCenterNoJson	JSON array of `MachineDashboardModel` objects
`fetchActivityLog`	hoursBack	JSON array of `ActivityLogModel` objects

6.3.2 Frontend Implementation

Shared HTTP Infrastructure

All services in this sprint (and retroactively migrated services from earlier sprints) use a shared HttpClient and HttpResponseParser layer, replacing the inline `http.post` calls used in Sprint 1.

`HttpClient.post()`: centralised `http.post` wrapper that injects JSON content-type headers and encodes the request body. All services call this method rather than the raw `http` package directly.

`HttpResponseParser`: provides four static methods. `parseList()` decodes Business Central OData list responses (nested value string). `parseObject()` decodes single-object responses. `parseWriteResult()` decodes write-result responses and throws with the backend error message on failure. `_assertSuccess()` validates HTTP status codes and throws a structured exception for non-2xx responses.

Provider Architecture

Table 6.4. Provider Responsibilities — Sprint 4 Additions

Provider	Responsibility
LogProvider	Owns dashboard and activity log fetch calls; exposes `selectedHours`, `hourOptions`, and `setHours()` shared between the two admin pages.
MesBarcodeProvider	Exposes `fetchAllBarcodes()`, `insertScans()`, and `resolveBarcode()`; resolves session token automatically from AuthProvider.

User Interface

Figures 6.5 and 6.6 illustrate the two pages. The *Machine Dashboard Page* renders one card per machine, each displaying a circular uptime indicator and four metric rows (operations, produced, scrap, running minutes); the AppBar hosts a time-range dropdown that re-fetches data on change.

The *Activity Log Page* presents a paginated list of all system events ordered newest-first; each row shows a type icon, operator name, machine, action text (expandable for long entries), and timestamp; type and time-range dropdowns in the AppBar narrow the view.

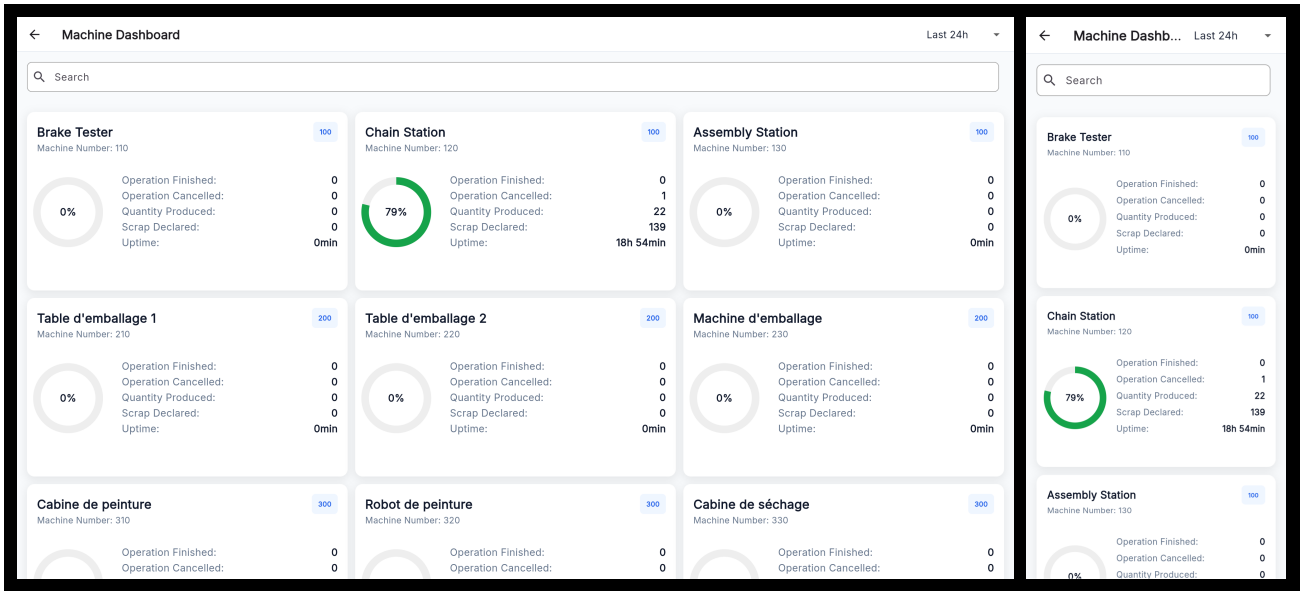

Figure 6.5. *Machine Dashboard Page* — desktop and mobile views.

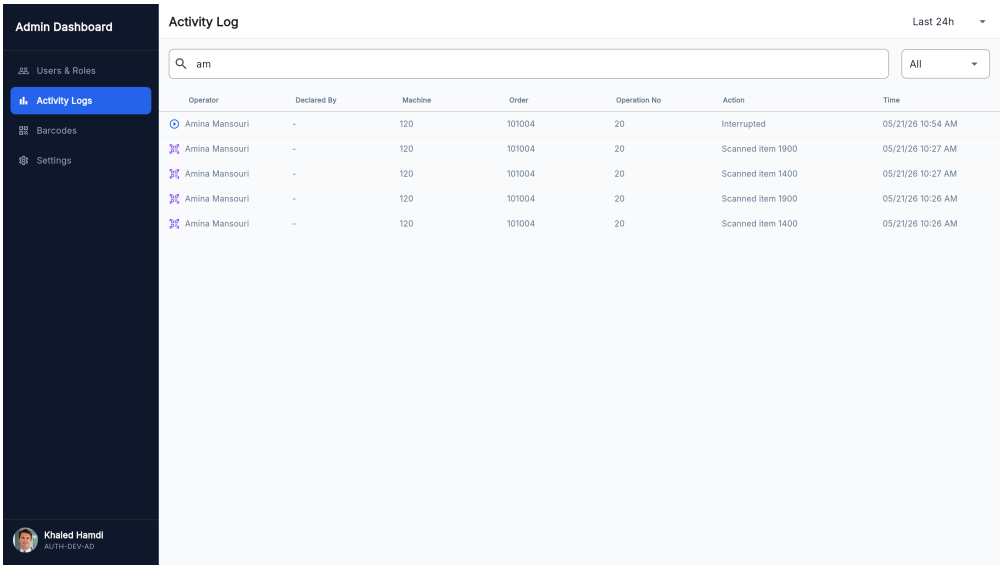

Figure 6.6. *Activity Log Page* — desktop and mobile views

6.4 Test and validation

6.4.1 Integration test

For integration testing, we chose Postman, a powerful and user-friendly tool for API evaluation. Figure 6.7 shows a test case for the `/activityLog` endpoint.

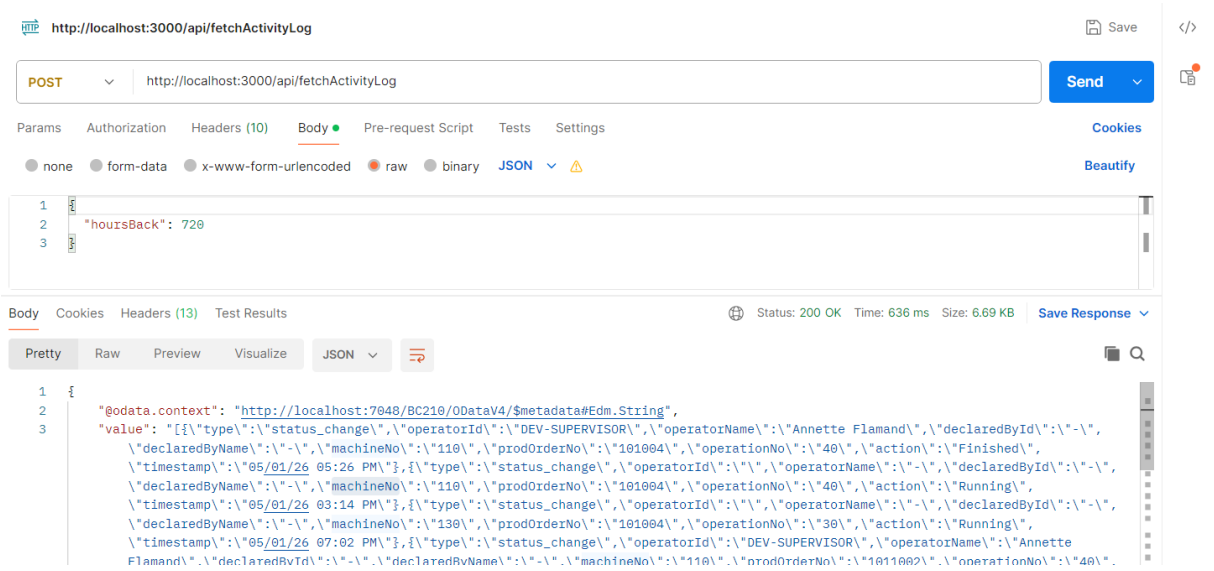

Figure 6.7. Postman endpoint test — fetchActivityLog.

Figure 6.8 shows a test case for the `/MachineDashboard` endpoint.

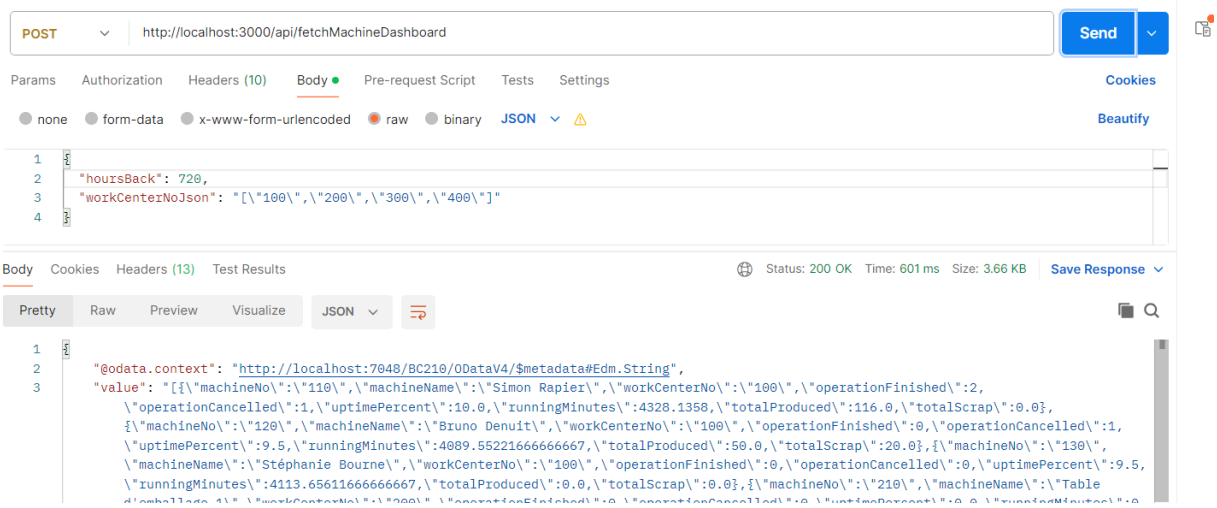

Figure 6.8. Postman endpoint test — fetchMachineDashboard.

Conclusion

This sprint completed the *MES* solution by delivering the administration layer and cross-cutting quality-of-life features. Administrators can now monitor the production health of the entire facility through a configurable machine performance dashboard, audit all operator and machine actions through a paginated activity log, and manage users from a persistent navigation shell without losing context between sections. All users benefit from full internationalisation covering English, French, and Arabic with automatic right-to-left layout, and from guided coach-mark tutorials that ease onboarding for first-time users.

Chapter 7

Sprint 5 AI Assistant & Analytical Intelligence.

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Introduction

This sprint delivers the conversational intelligence layer of the *MES* solution. Building upon the production execution and administration infrastructure established in the previous sprints, this sprint introduces an embedded natural-language chat assistant that allows operators and supervisors to query machine status, production progress, scrap data, Bill of Materials, and operation history without navigating the full interface. The assistant resolves ambiguous machine references, answers fleet-wide questions through parallel fan-out queries, proactively surfaces operational anomalies, and provides contextual navigation shortcuts directly within the chat reply.

The implementation spans three new components: a Node.js reverse-proxy middleware layer that handles *SSPI/Windows* authentication toward Business Central and routes AI chat traffic to a dedicated Python agent; the Python FastAPI AI agent itself, which classifies user intent, executes tool chains against the existing BC web services, enriches results with pure-Python analytics, and synthesises natural-language replies using a configurable multi-provider *LLM* client; and the Flutter chat UI, integrated directly into the machine list page as a side panel or dialog.

7.1 Sprint Backlog

This sprint covers **Domain 5:** textbf*AI Chat Assistant & Analytical Intelligence*.

Table 7.1. Sprint 5 Story Backlog

ID	User Story	Tasks
US-5.1	As an Operator or Supervisor, I need a natural-language chat assistant so that I can query machine status, production progress, and operational data without navigating the full UI.	*Node.js Middleware* • Implement Express application (`index.js`) with global middleware stack (JSON body parser, CORS, request logger). • Implement `src/config.js` loading all environment variables and computing BC URL constants (`odataBase`, `webServiceBase`, `companyParam`). • Implement `src/proxy.js`: `buildTargetUrl()`, `buildPsScript()`, `runPowerShell()`, and `forwardRequest()` to proxy Flutter requests to BC via PowerShell *SSPI*. • Register all 44 AL web service endpoints in the `webServiceEndpoints` set. • Implement `src/cors.js` CORS options. Continued on next page

ID	User Story	Tasks
		Python AI Agent • Implement FastAPI application (`main.py`) with `/health` and `POST /chat` endpoints and lifespan context manager. • Implement `agent/models.py`: `ChatRequest`, `ChatResponse`, `UserContext`, `ConversationTurn`, `ToolResult`, `RedirectAction`, `ActionType`. • Implement `agent/config.py` with all environment variable constants (LLM provider, model, timeouts, middleware URL). • Implement `agent/llm_client.py`: `LLMClient` abstraction and `build_llm_client()` factory supporting Ollama, OpenAI, HuggingFace, OpenAI-compatible, and Anthropic providers. • Implement `tools/mes_tools.py`: all 17 BC web service tool classes, `TOOL_MAP` registry, `MACHINE_SCOPED_TOOLS` set, `_post()` HTTP helper, and `_extract_value()` response normaliser. *Flutter* • Create model AiChatResponse, ConversationTurn, and AiRedirectAction. • Create AiChatService consuming `POST /api/ai/chat`. • Create AiChatProvider managing conversation history, loading state, and last response. • Create page AiChatPage rendering as side overlay (width $\geq$ 600 dp) or dialog (mobile). • Integrate chat FAB and overlay toggle in Machinelistpage.

Continued on next page

ID	User Story	Tasks
US-5.2	As an Operator or Supervisor, I need the assistant to resolve ambiguous machine references so that I do not need to know exact machine codes.	*Python AI Agent* • Implement `agent/resolver.py`: `MachineEntry`, `ResolveResult`, `build_machine_index()`, and `resolve_machine()` with five-tier cascade (exact ID, normalised exact, digit suffix, token overlap, edit distance). • Implement text normalisation helpers `_norm()`, `_digits_only()`, `_tokenise()`, `_edit_distance()`, `_token_overlap()`. • Implement `format_candidates_for_llm()` disambiguation message builder. • Implement `_substitute_machine_no()` in `agent/orchestrator.py` to replace `__RESOLVE__` slot placeholders after resolution. • Integrate resolution into `Orchestrator.run()` when `chain.unresolved_machine_ref` is set.
US-5.3	As an Operator or Supervisor, I need the assistant to answer fleet-wide questions so that I get a consolidated view without visiting each machine individually.	*Python AI Agent* • Implement `agent/composite.py`: `is_composite()` detection, `CompositeResult` dataclass, `run_composite()` with status filters (running/idle/overdue/scrap), fan-out capped at 12 machines in batches of 4. • Implement `build_composite_context()` context builder. • Integrate composite path in `Orchestrator.run()` and `Orchestrator._run_composite()`. • Add `is_composite` flag detection to both LLM and regex classifiers.

Continued on next page

ID	User Story	Tasks
US-5.4	As an Operator or Supervisor, I need the assistant to provide contextual navigation shortcuts so that I can open a machine, operation, or dashboard directly from the chat response.	*Python AI Agent* • Define `ActionType` enum and `RedirectAction` schema in `agent/models.py`. • Implement `Orchestrator._parse_actions()` to extract `actions` JSON block from LLM reply (up to 4 actions). • Document supported action types and payload shapes in `prompts/system_prompts.py` (`SYNTHESIS_USER_TEMPLATE`). *Flutter* • Implement `AiChatPage._handleAction()` routing: `redirect_machine` → MachineMainPage, `redirect_machine_list` → `popUntil`, `redirect_operation` → OperationDetailPage. • Render action buttons only on the last assistant message.

Continued on next page

ID	User Story	Tasks
US-5.5	As a Supervisor, I need the assistant to proactively highlight anomalies so that I can act before problems escalate.	*Python AI Agent* • Implement `agent/data_analysis.py`: `enrich_deadlines()` (hours until deadline, overdue flag, risk level), `analyse_scrap()` (scrap rate, severity, spike detection at $2.5\times$ baseline), `analyse_velocity()` (units per hour, projected end, velocity status), `summarise_machines()`, and `summarise_dashboard()`. • Integrate enrichment into `Orchestrator._enrich()` for all result key types (machines, orders, live_data, cycles, activity, dashboard, history, bom, etc.). • Implement `agent/intent.py` deterministic regex classifier: `Intent` enum, `ToolStep`, `ToolChain` dataclasses, extraction helpers (`_extract_machine_no`, `_extract_order_no`, `_extract_hours`), and `classify()` priority chain. • Implement `agent/llm_intent.py`: `LLMIntentIdentifier` with `_llm_classify()`, validation helpers, `_plan_to_tool_chain()`, and `ThinkingBlock` reasoning trace. • Implement `agent/orchestrator.py`: full 7-stage pipeline (`run()`, `_execute_step()`, `_fetch_all_machines()`, `_enrich()`, `_build_context()`, `_synthesise()`, `_parse_actions()`). • Implement `prompts/system_prompts.py`: `SYNTHESIS_SYSTEM` and `SYNTHESIS_USER_TEMPLATE` with anomaly alert rules and action block instructions. • Wire all 17 tools to the `fetchSupervisorOverview` and `fetchDelayReport` BC endpoints used in anomaly detection.

7.2 Design

7.2.1 Use Case Diagram

The UML Use Case Diagram (Figure **??**) shows the two primary actors (Operator and Supervisor) and their interactions with the AI chat assistant. The Supervisor role has access to anomaly-oriented queries (overdue orders, scrap spikes, idle operators) in addition to the machine and operation queries shared with Operator. Both roles benefit from the fuzzy machine resolution and redirect action use cases.

Figure 7.1. UML Use Case Diagram — Sprint 5

7.2.2 Textual Use Case Description

The following table (Table 7.2) provides the detailed textual description of the *Query via Chat Assistant* use case, which represents the central interaction delivered in this sprint.

Table 7.2. Textual Use Case Description — Query via Chat Assistant

Use Case Name	Query via Chat Assistant
Primary Actors	Operator, Supervisor
Preconditions	• The user is authenticated and holds a valid session token. • The Node.js middleware and Python AI agent are running and reachable. • At least one work centre is assigned to the authenticated user.
Postconditions	• A natural-language answer is displayed in the chat panel. • Up to 4 contextual navigation action buttons may appear below the reply. • The conversation turn is appended to the session history for multi-turn context.
Main Scenario	1. The user opens the chat panel from the *Machine List Page*. 2. The user types a question in natural language (e.g., "What is MC-001 producing?"). 3. The Flutter client sends a `POST /api/ai/chat` request to the Node.js middleware with the message, user context, and conversation history. 4. The middleware forwards the request to the Python AI agent. 5. The AI agent classifies intent via the *LLM* classifier (with regex fallback) and builds a `ToolChain`. 6. If the machine reference is ambiguous, the agent resolves it using the five-tier fuzzy resolver. 7. The agent executes each `ToolStep` in sequence, calling the corresponding BC web service via the Node.js middleware. 8. The agent enriches the raw results with pure-Python analytics (deadline enrichment, scrap analysis, velocity computation). 9. The agent synthesises a natural-language answer via a second *LLM* call using the enriched context block. 10. Any contextual redirect actions are extracted from the reply and returned as structured `RedirectAction` objects. 11. The Flutter client displays the answer and optional action buttons.

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Alternative Flows	• *Ambiguous machine name:* the agent returns a disambiguation message listing up to 3 candidates; the user clarifies in the next turn. • *Fleet-wide query:* the agent triggers the composite fan-out path, querying up to 12 machines in parallel batches of 4. • *LLM classifier failure:* the deterministic regex classifier is used as a fallback; the `ThinkingBlock` records `fallback_used = true`. • *General question:* no tool calls are made; the *LLM* answers from general manufacturing knowledge.

7.2.3 Sequence Diagrams

The following sequence diagrams illustrate the communication flow between the Flutter frontend, the Node.js middleware, the Python AI agent, and the Business Central OData layer.

Figure **??** depicts the standard single-machine query flow (e.g., "What is MC-001 producing?"). The Flutter client posts the message and user context to the middleware; the middleware forwards to the AI agent; the agent classifies intent, builds a `ToolChain` with two steps (`get_ongoing_operations` then `get_operation_live_data` via slot reference), executes each step sequentially against BC, enriches the results, and synthesises a final reply.

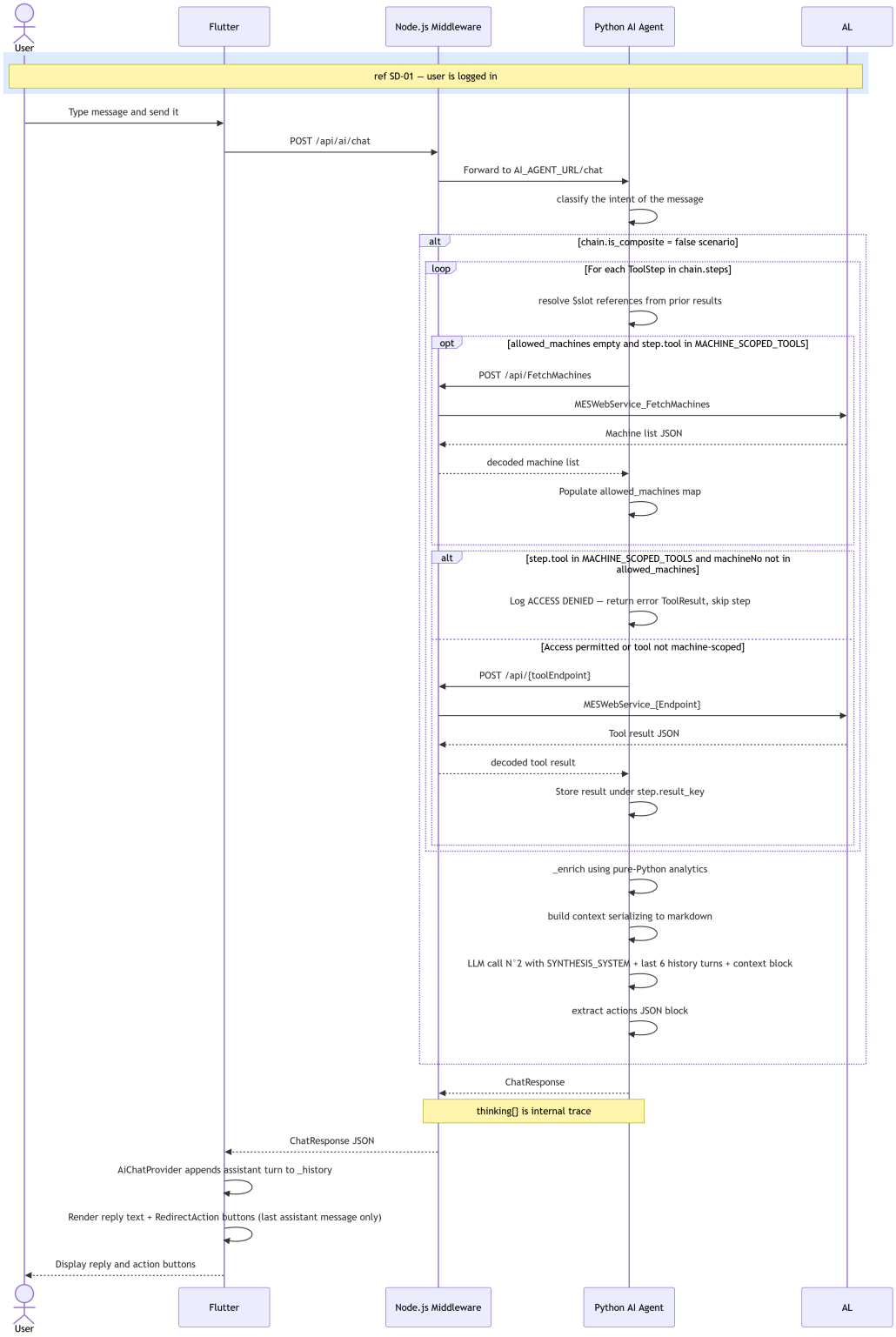

Figure 7.2. SD-30: AI Chat — Main Flow.

Figure 7.3 illustrates the two-tier intent classification flow, showing the LLM-based primary classifier, the JSON plan validation step, and the fallback path to the regex classifier. The *ThinkingBlock* capturing the classification trace is also shown.

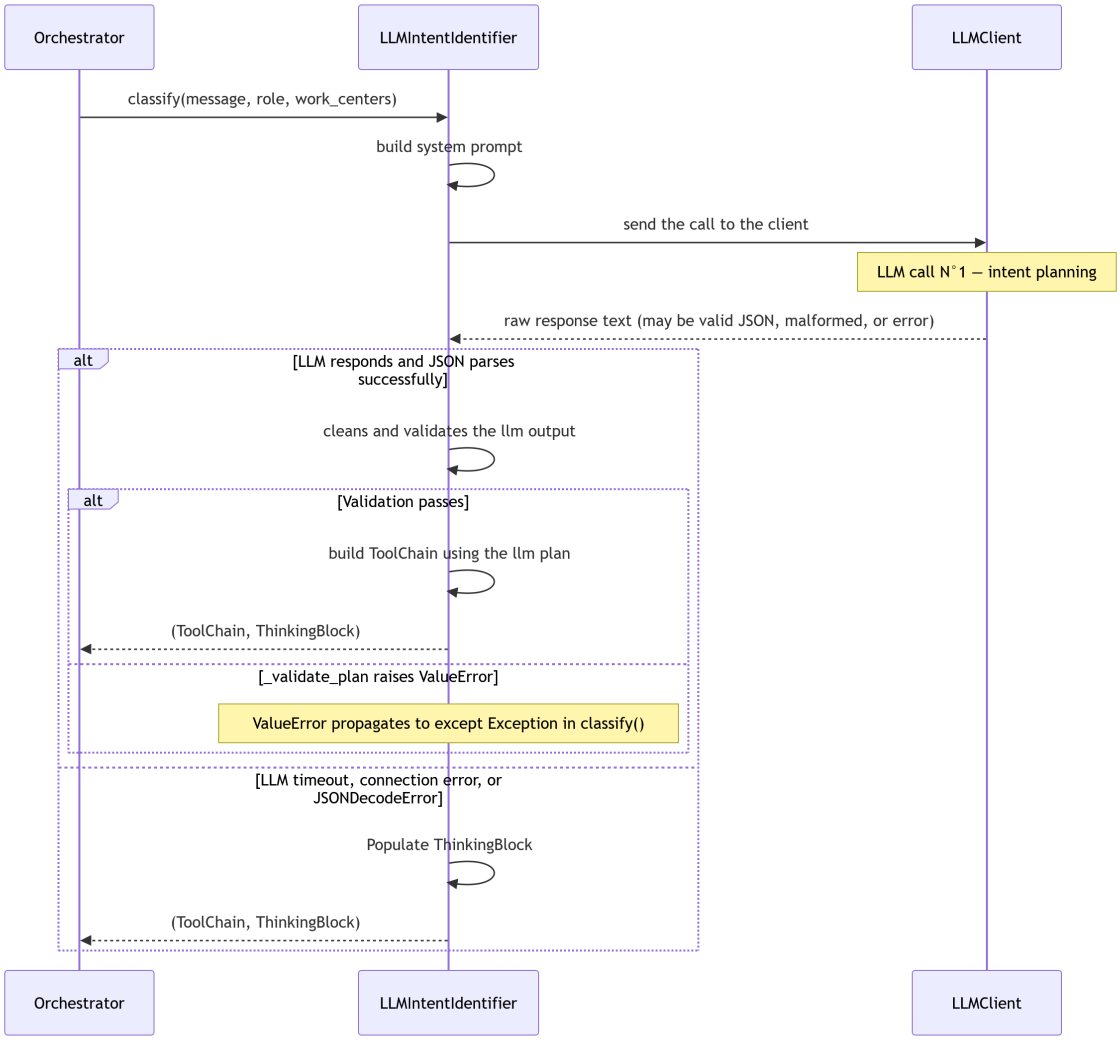

Figure 7.3. SD-31: Intent Classification — LLM with Regex Fallback.

7.2.4 System Architecture Diagram

Figure 7.4 shows the physical architecture of the AI assistant layer, illustrating how the Flutter client, Node.js middleware, Python AI agent, and Business Central server interact. The middleware plays a dual role: it proxies all regular MES API calls using PowerShell *SSPI* authentication, and it also routes AI chat requests to the Python agent which calls back into the same middleware for its data fetching needs.

Figure 7.4. AI Assistant Layer — Physical Architecture.

7.2.5 Class Diagram

The UML Class Diagram (Figure **??**) covers the main structural components of the AI agent: the `MESAgentOrchestrator` central controller, the `ToolChain` and `ToolStep` planning structures produced by the classifiers, the `ResolveResult` returned by the fuzzy resolver, and the `CompositeResult` produced by the fan-out executor. The 17 tool classes in `tools/mes_tools.py` are represented as a single `BaseTool` with a `TOOL_MAP` registry.

Figure 7.5. UML Class Diagram — AI Assistant & Analytical Intelligence.

7.3 Implementation

7.3.1 Node.js Middleware

Overview

The Node.js middleware (`node-middleware/`) is an Express 5 application that acts as the single network bridge between the Flutter client and Business Central.

The middleware exposes two distinct route families: `/api/<endpoint>` (proxied to BC) and `/api/ai/*` (forwarded to the Python agent).

AI Proxy Layer

The `src/aiProxy.js` Express Router, mounted at `/api/ai`, forwards `POST /chat` requests to the Python agent at `AI_AGENT_URL/chat` using the native `fetch` API. It validates that both `message` and `user_context` are present before forwarding, enforces a `AI_TIMEOUT_MS` abort timeout, and passes the Python agent's response through unchanged. A `GET /api/ai/health` endpoint aggregates the Node process uptime with the Python agent's own health status for monitoring purposes.

Configuration

Table 7.3 lists the environment variables consumed by the middleware.

Table 7.3. Node.js Middleware — Environment Variables

Variable	Default	Description
BC_HOST	—	Base URL of the BC server (no trailing slash)
BC_INSTANCE	—	BC server instance name
BC_COMPANY	—	Company GUID
PROXY_PORT	3000	Listening port
ALLOWED_ORIGINS	""	Comma-separated CORS origins
BC_TIMEOUT_MS	30 000	BC request timeout in milliseconds
AI_AGENT_URL	http://localhost:8000	Python agent base URL
AI_TIMEOUT_MS	60 000	AI request abort timeout in milliseconds

7.3.2 Python AI Agent

The agent makes exactly two *LLM* calls per user message . **Call 1 — Intent classification.** The LLMIntentIdentifier sends the user message, role, and work centres to the LLM with a structured system prompt listing all 17 available tools and their required arguments. The LLM responds with a JSON plan containing the intent label, a list of ToolStep objects (each with a tool name, arguments — which may be slot references to previous steps — and a result key), and a reasoning string. The plan is validated against the tool catalogue before being converted to a ToolChain dataclass. If the LLM call fails for any reason, the deterministic regex classifier in agent/intent.py is used as a fallback. The entire classification reasoning is stored in a ThinkingBlock and returned in ChatResponse.thinking for debugging; it is never shown to the user.

Call 2 — Synthesis. After all tool steps have been executed and their results enriched, the Orchestrator._synthesise() method builds a context block containing the enriched data in markdown format (truncated at LLM_MAX_CONTEXT_CHARS characters) and calls the LLM with the SYNTHESIS_SYSTEM and SYNTHESIS_USER_TEMPLATE prompts. The system prompt instructs the model to answer only from the pre-fetched data, to highlight anomalies prominently, to respond in the same language as the user, and to optionally append a actions JSON block containing up to 4 RedirectAction objects.

Intent Classification

LLM classifier. The LLMIntentIdentifier constructs a system prompt containing the full TOOL_CATALOGUE and the list of valid intent labels. The user message is injected via _IDENTIFIER_USER_TEMPLATE along with the user's role and work centres. The raw LLM response is stripped of JSON fences, parsed, validated against _VALID_TOOLS and _INTENT_MAP,

and converted to a `ToolChain` via `_plan_to_tool_chain()`.

Regex classifier (fallback). The `classify()` function in `agent/intent.py` evaluates the user message against 10 keyword pattern variables in strict priority order. Table 7.4 lists the 12 intent values and their associated tool chains.

Table 7.4. Intent Values and Associated Tool Chains

Intent	Primary Tool(s)	Trigger Keywords
`machine_status`	get_ongoing_operations	machine, mc-N, status
`machine_orders`	get_machine_orders	order, queue, po-N
`machine_live`	get_ongoing_operations + get_operation_live_data	live, real-time, progress
`machine_history`	get_operations_history	history, finished, past
`machine_dashboard`	get_ongoing_operations + get_machine_dashboard	dashboard, kpi, uptime
`department_overview`	list_machines (fan-out)	department, all machines
`activity_log`	get_activity_log	activity, log, happened
`scrap_analysis`	get_activity_log	scrap, reject, defect
`operation_bom`	get_bom	bom, component, material
`production_cycles`	get_production_cycles	cycle, declaration, shift
`operation_live`	get_operation_live_data	live + order number
`general`	(none)	No tool trigger matched

Tool Catalogue

The 17 Business Central web service wrappers in `tools/mes_tools.py` correspond directly to the OData functions exposed by MES Web Service (Codeunit 50126). Table 7.5 lists all tools, their AL endpoint mappings, and access scope.

Table 7.5. AI Agent Tool Catalogue

Tool name	AL Endpoint	Scope
`list_machines`	FetchMachines	Work-centre scoped
`get_machine_orders`	getMachineOrders	Machine scoped

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Tool name	AL Endpoint	Scope
get_ongoing_operations	fetchOngoingOperationsState	Machine scoped
get_operation_live_data	fetchOperationLiveData	Machine scoped
get_production_cycles	fetchProductionCycles	Machine scoped
get_operations_history	fetchOperationsHistory	Machine scoped
get_activity_log	fetchActivityLog	Work-centre scoped
get_machine_dashboard	fetchMachineDashboard	Work-centre scoped
get_production_orders	fetchProductionOrders	Machine scoped
get_work_center_summary	fetchWorkCenterSummary	Work-centre scoped
get_operator_summary	fetchOperatorSummary	Work-centre scoped
get_my_data	fetchMyData	User scoped
get_scrap_summary	fetchScrapSummary	Machine scoped
get_bom	fetchBom	Machine scoped
get_delay_report	fetchDelayReport	Work-centre scoped
get_consumption_summary	fetchConsumptionSummary	Machine scoped
get_supervisor_overview	fetchSupervisorOverview	Work-centre scoped

Machine-scoped tools are subject to access control: the orchestrator verifies that the resolved `machineNo` is present in the `allowed_machines` map (populated by `list_machines` calls for the authenticated user's work centres) before executing the step.

Data Enrichment

The `Orchestrator._enrich()` method runs pure-Python analytics over the raw tool results before they are passed to the synthesis LLM. Table 7.6 summarises the enrichment applied to each result key.

Table 7.6. Data Enrichment Applied per Result Key

Result key	Enrichment
machines	`summarise_machines()` — utilisation percentage, working/idle counts, interpretation string

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Result key	Enrichment
`orders`	`enrich_deadlines()` — hours until deadline, overdue flag, risk level (*at_risk* threshold: < 4 h)
`live_data`	`analyse_scrap()` + `analyse_velocity()` — scrap severity, spike detection, units per hour, projected end
`cycles`	`analyse_scrap()` — scrap trend from cycle history
`activity`	`analyse_scrap(activity=data)` — recent scrap event count
`dashboard`	`summarise_dashboard()` — fleet totals, overall scrap severity
`bom`	Missing components list (items with zero scanned quantity)

Scrap spike detection. The `analyse_scrap()` function computes the scrap rate for the most recent quarter of declared cycles and compares it to the rate for the earlier three-quarters. If the recent rate exceeds the baseline by a factor of 2.5 or is above the critical threshold of 15%, a spike is flagged and explicitly mentioned in the synthesis prompt.

Velocity projection. The `analyse_velocity()` function derives the actual production rate (units per hour) from the last 6 declared cycles and compares it to the required rate (remaining quantity divided by time until the planned end). The result is one of *ahead*, *on_pace*, or *behind*, along with a projected completion timestamp.

7.3.3 Flutter Frontend

Chat Page and Provider

The AiChatPage widget adapts its rendering to the available screen width: on screens wider than 600 dp it is displayed as a `Positioned` overlay panel (400 dp wide, full height) anchored to the right side of the MachineListPage; on mobile it opens as a `Dialog`. The chat input field and conversation history list are shared between both modes.

The AiChatProvider maintains the in-session conversation history as a `List<ConversationTurn>`. On each `sendMessage()` call it appends the user turn, calls `AiChatService.sendMessage()` with the history (excluding the most recently appended turn, which the server receives as the primary message), appends the assistant reply, and stores the last AiChatResponse for action button rendering.

Redirect Actions

After each assistant reply, the AiChatPage inspects `lastResponse.actions`. If non-empty and the message is the last in the conversation, up to 4 ElevatedButton widgets are rendered below the reply bubble. The `_handleAction()` method routes each action type:

- `redirect_machine` → `Navigator.push` to MachineMainPage with the payload `machineNo`.
- `redirect_operation` → `Navigator.push` to OperationDetailPage with `machineNo`, `prodOrderNo`, and `operationNo`.
- `redirect_machine_list` → `Navigator.popUntil` to the root route (machine list).
- `redirect_machine_dashboard` → `Navigator.push` to the admin machine dashboard.
- `redirect_history` → `Navigator.push` to MachineHistoryPage with `machineNo`.

User Interface

The *AI Chat Panel*, shown in Figure 7.6, is accessible from the *Machine List Page* via a floating action button. On wide screens it opens as a non-modal side panel; on mobile it opens as a full-screen dialog. The panel displays the conversation history, a text input field at the bottom, and optional action buttons below the last assistant message. A *Clear* button resets the conversation history.

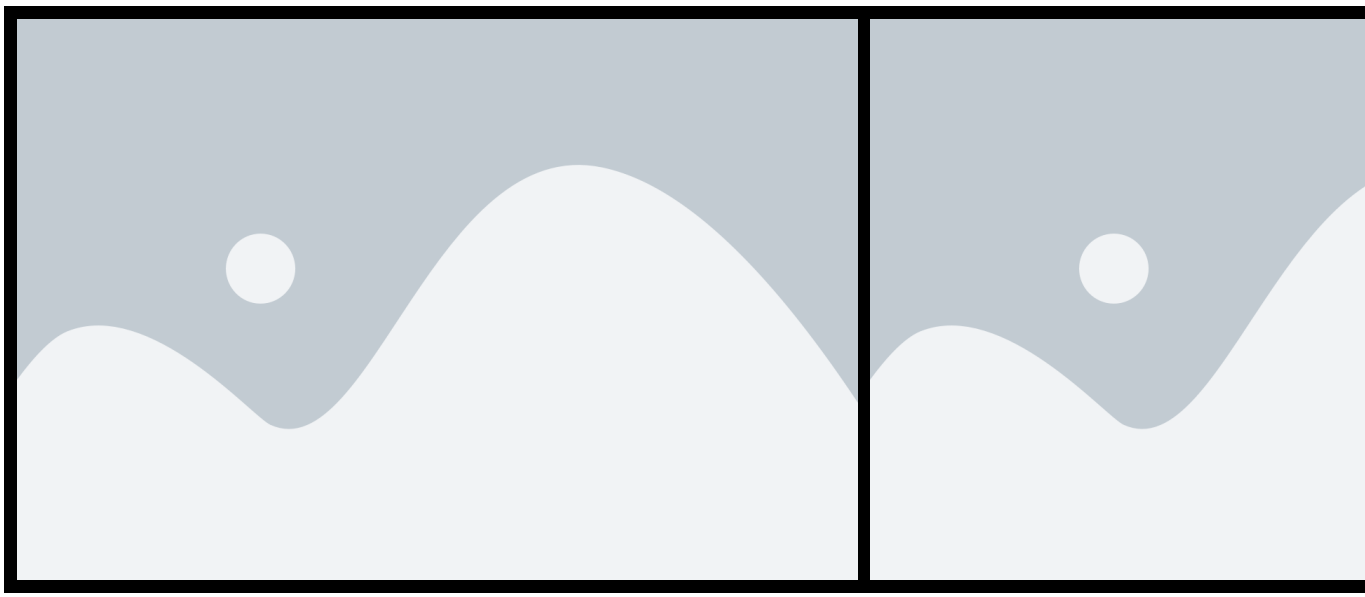

Figure 7.6. *AI Chat Panel* — desktop side panel and mobile dialog views.

7.3.4 Integration Tests

For integration testing we used Postman for the Node.js middleware endpoints and the FastAPI built-in `/docs` interface for the Python agent.

Figure 7.7 shows a test of the `POST /api/ai/chat` endpoint with a single-machine live data query. The response contains the synthesised text, the `thinking` block with the classified intent and tool steps, and one `redirect_machine` action.

Figure 7.7. Postman endpoint test — `POST /api/ai/chat`.

Figure 7.8 shows a test of the `GET /api/ai/health` endpoint confirming that both the Node.js proxy and the Python agent are reachable.

Figure 7.8. Postman endpoint test — `GET /api/ai/health`.

Conclusion

This sprint delivered the conversational intelligence layer of the *MES* solution.

The AI agent implements a two-*LLM*-call pipeline: a first call classifies user intent and produces an executable `ToolChain` plan , and a second call synthesises the final natural-language reply from the enriched tool results. Pure-Python analytics (deadline enrichment, scrap spike detection, velocity projection) pre-compute key facts so the synthesis *LLM* receives answers rather than raw numbers.

Operators and supervisors can now ask questions in natural language, receive machine and production data scoped strictly to their work centres, navigate directly to relevant screens via action buttons embedded in the reply, and obtain proactive alerts for overdue orders, scrap anomalies, and paused operations — all without leaving the chat interface.

General Conclusion

This report has presented the full lifecycle of the MES (*Manufacturing Execution System*) project developed at **Mavision** over the course of five sprints. Starting from a blank slate, the team designed and implemented a complete shop-floor execution system natively integrated with Microsoft Dynamics 365 Business Central. Sprint 1 established the security and identity foundation: a token-based authentication system, role-based access control, and a full administrator interface for user and account management. Sprint 2 delivered real-time machine monitoring, production order management, and a complete operation history trail. Sprint 3 completed the production execution loop with cycle declarations, pause, resume, and close flows, Bill of Materials visibility, component scanning, scrap tracking, and label printing. Sprint 4 added the administration layer with a machine performance dashboard, a paginated activity log, multi-language support covering English, French, and Arabic, and in-app onboarding tutorials. Sprint 5 delivered the AI intelligence layer: a natural-language chat assistant embedded in the Flutter frontend, a Node.js reverse-proxy middleware bridging the Flutter application to both Business Central and a Python AI agent, and the agent itself, providing intent classification, multi-provider LLM synthesis, fuzzy machine resolution, composite fan-out queries, and contextual redirect actions. Across all five sprints, the append-only data model ensures a tamper-evident audit trail from every machine state change through to every production cycle declaration. The reactive stream architecture in the Flutter frontend provides operators and supervisors with a consistent, real-time view of the production floor without manual refreshes. The project demonstrates that a tightly integrated MES solution can be built on top of Microsoft Dynamics 365 Business Central using OData V4 unbound actions, with a Node.js middleware layer handling Windows authentication and AI routing. The extensible layered architecture positions the system for future enhancements such as statistical reporting, alert management, and additional language support.

Bibliography

[1] Mavision, *Official Website*, https://www.mavision.tn, visited April 2025.